Chapter 3

Analysis

3.1 General Use cases Diagram

EPS is an academic escrow simulation for goods and services. Each agreement has one Buyer, one distinct Seller and one full VND amount, funded through a Bank or Payment Gateway mock. The overview is divided into focused views for readability.

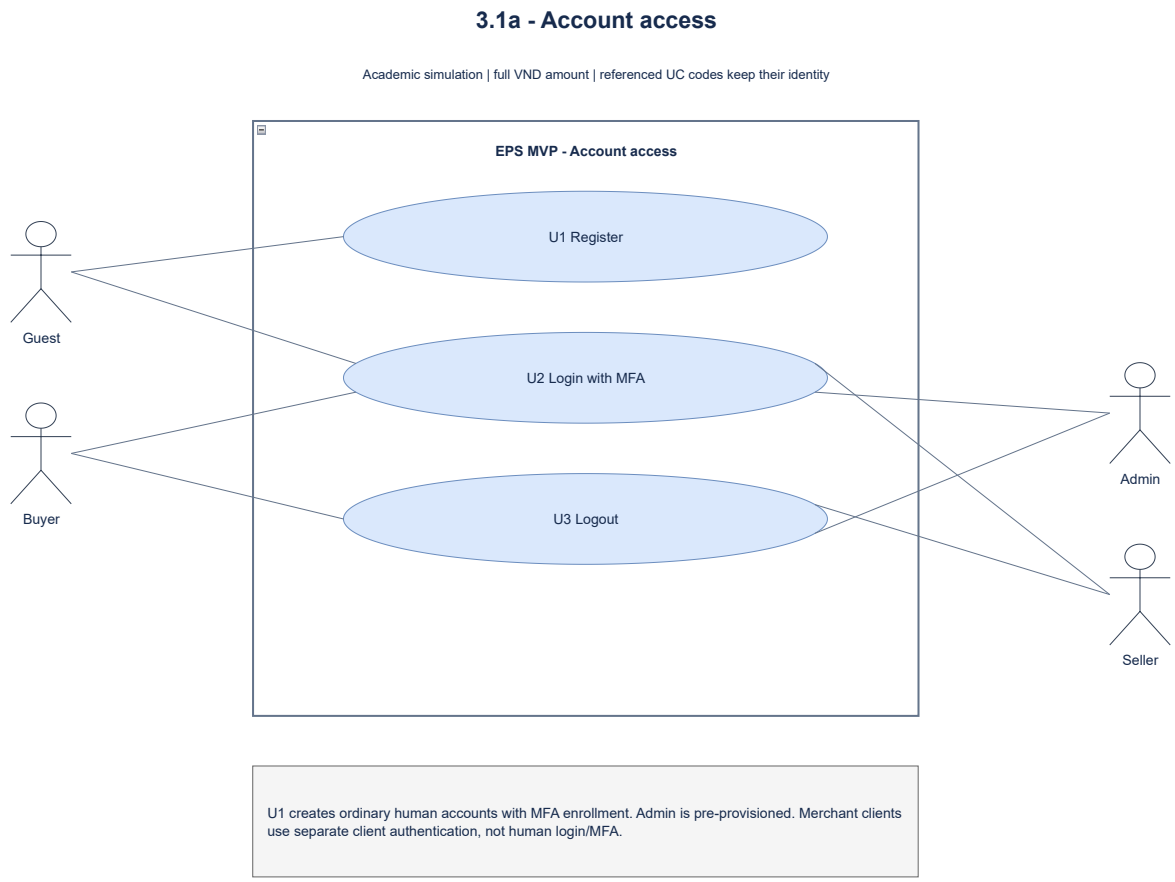


Figure 3.1: Account access

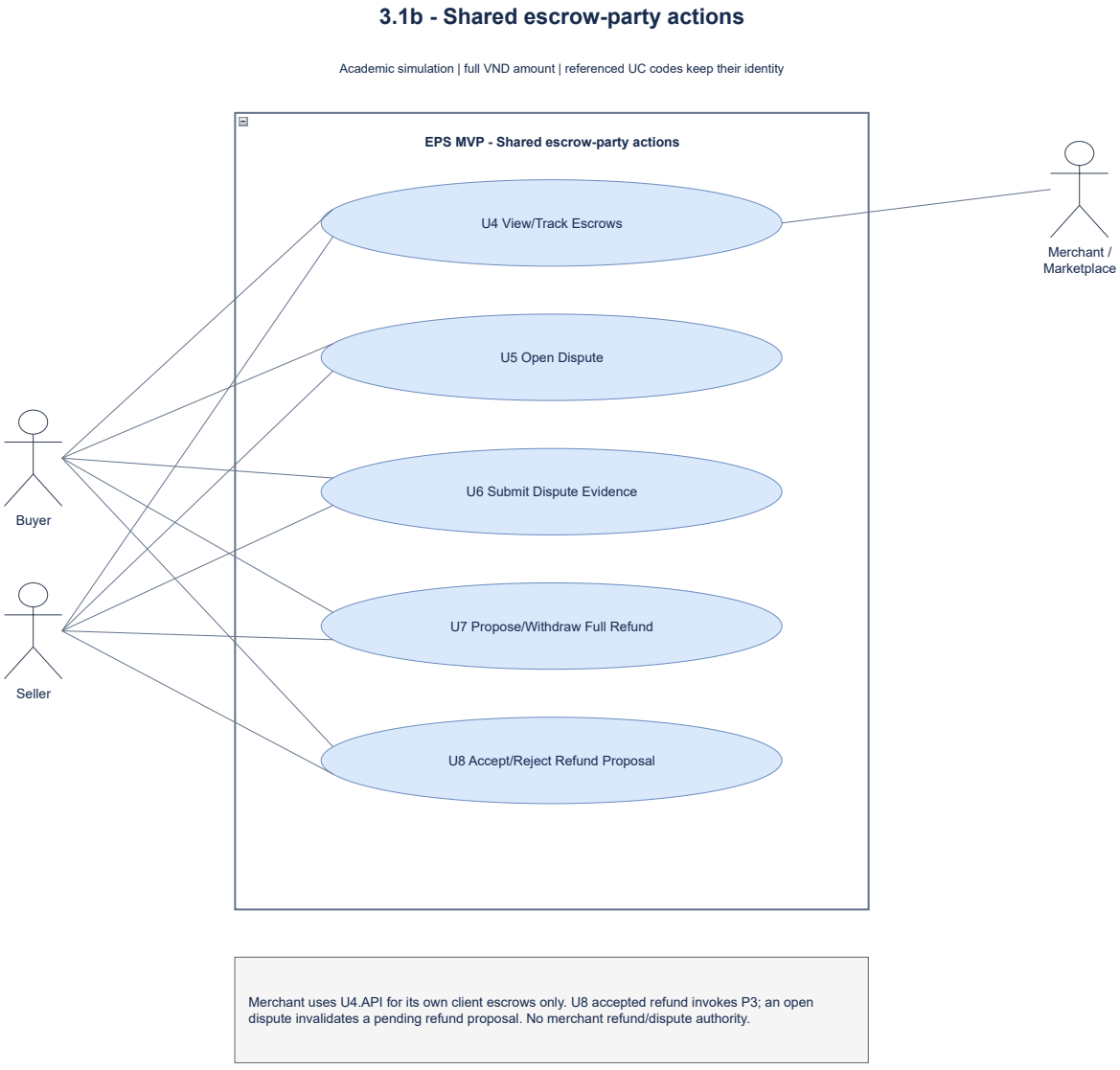


Figure 3.2: Shared escrow-party actions

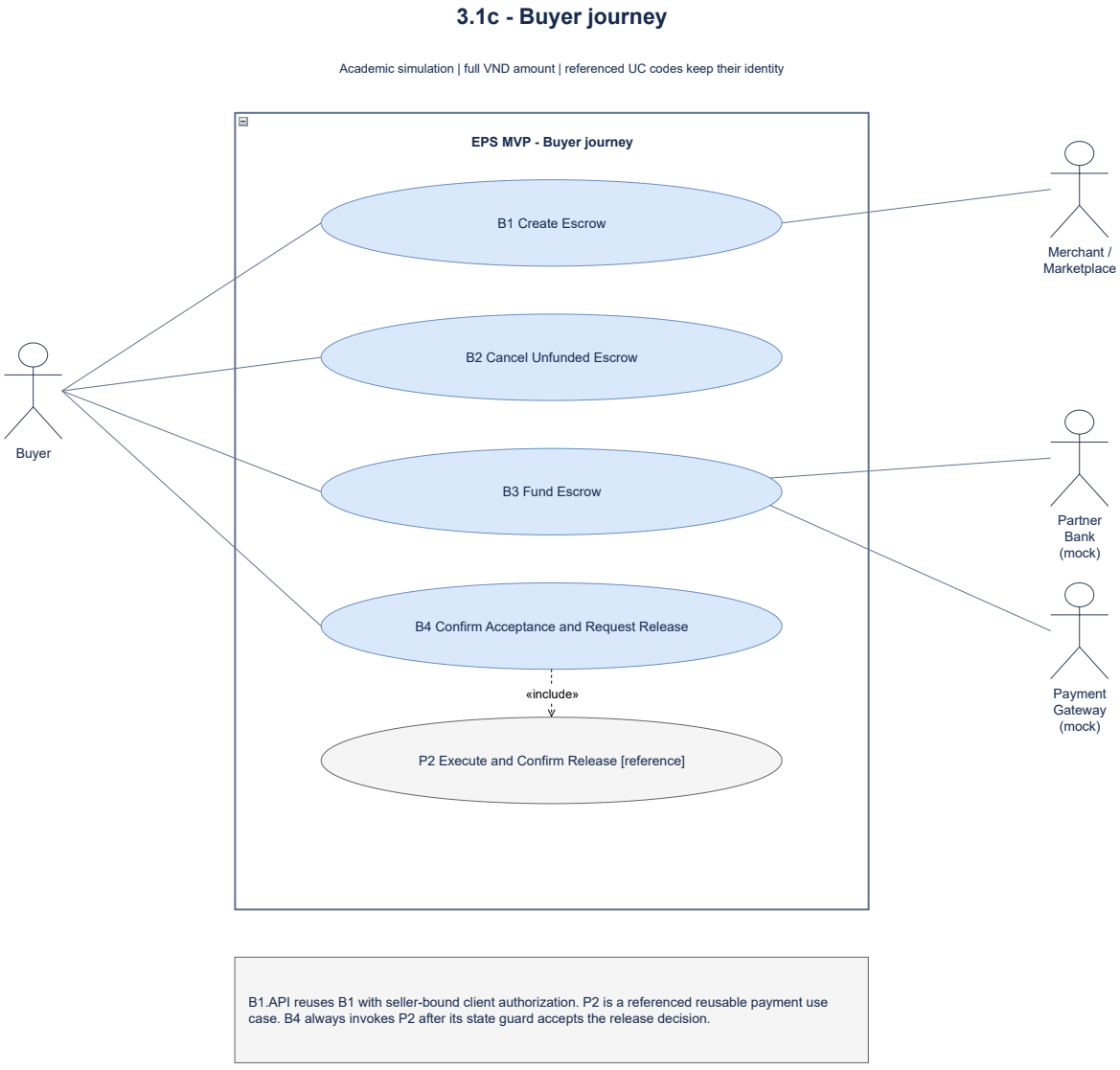


Figure 3.3: Buyer journey

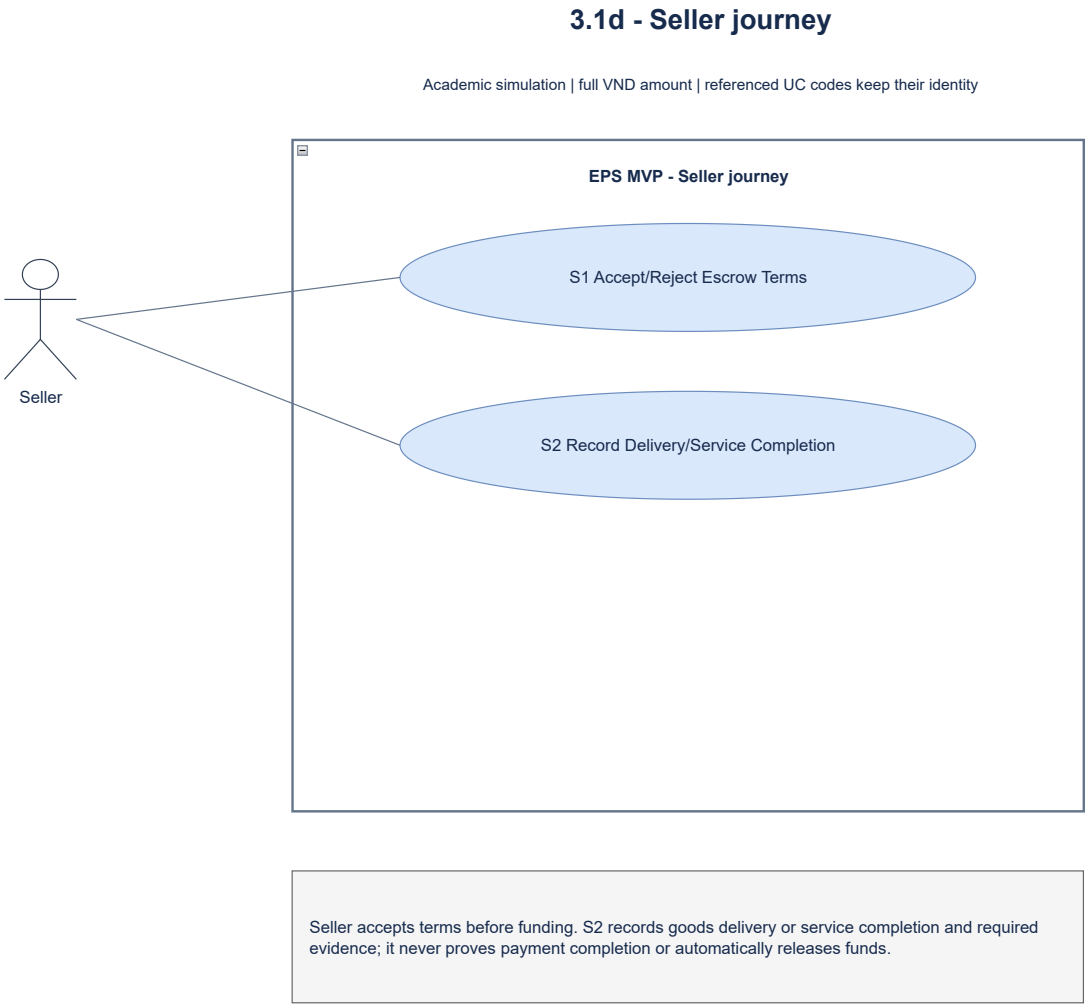


Figure 3.4: Seller journey

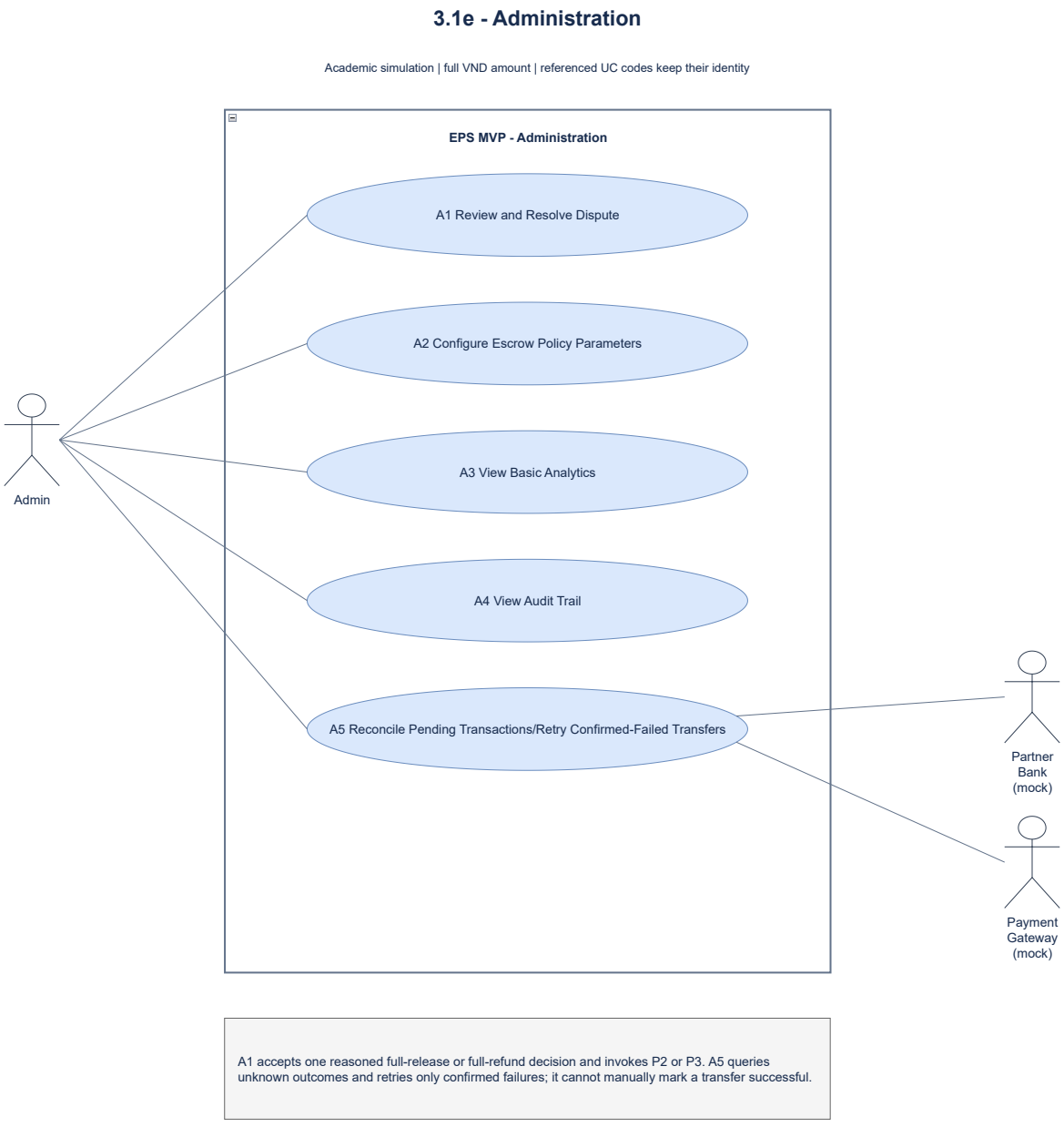


Figure 3.5: Administration

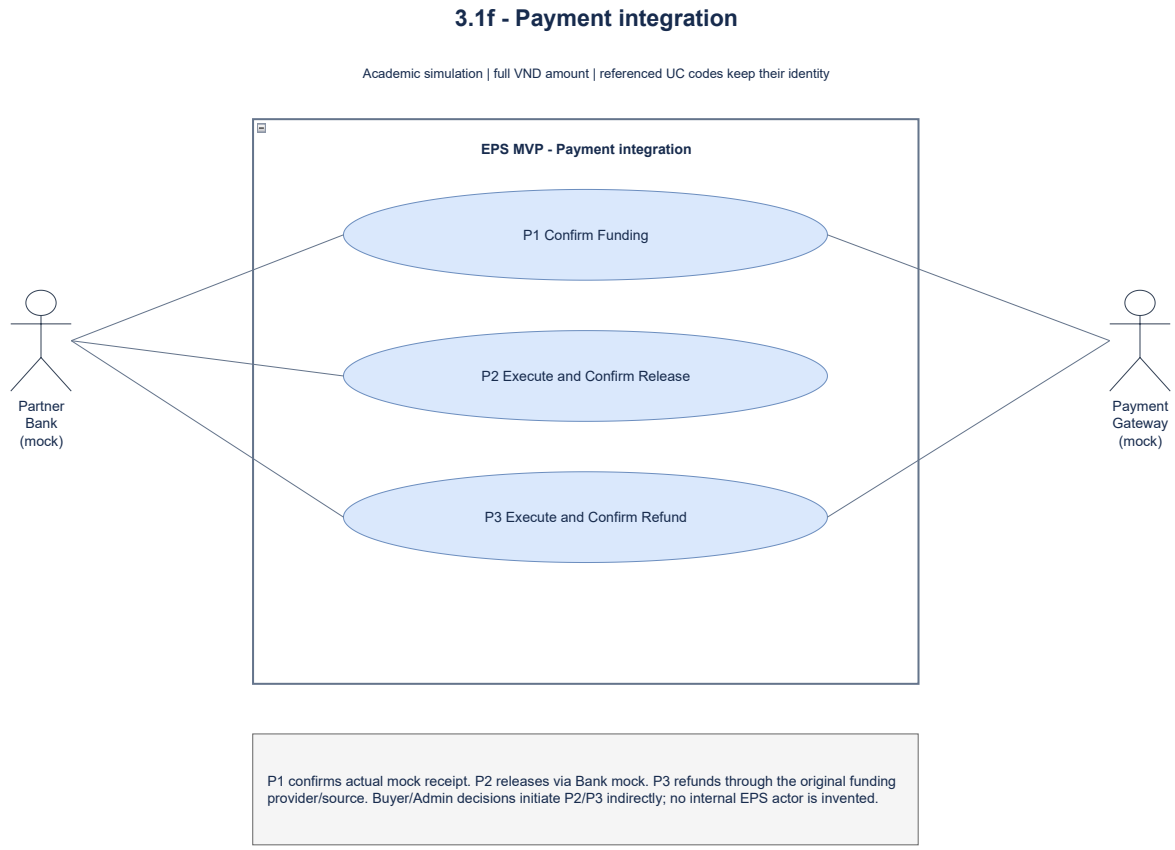


Figure 3.6: Payment integration

- **Guest**: registers an ordinary account and logs in with MFA.
- **Buyer**: creates and funds an escrow, accepts delivery and requests release.
- **Seller**: accepts the agreement and records delivery or service completion. Both participants may request a refund or open a dispute.
- **Admin**: resolves disputes, updates policy, views analytics/audit and reconciles pending transfers.
- **Merchant/Marketplace**: creates and queries only its own escrows through a separately authenticated client.
- **Partner Bank**: simulates funding, Seller release and refunds for Bank-funded escrows.
- **Payment Gateway**: simulates funding and refunds to the original Gateway source.

Buyer and Seller are roles in each transaction. Release uses the Bank mock; refund returns through the original funding provider. Real money, partial payments, milestones and automatic release are outside this MVP.

3.2 Use case Details

3.2.1 User's Use cases Details

U1: Register

Use Case Code	U1	**Use Case Name**	Register
Actor	Guest		
Usage	Create an ordinary user account.		
Activation Event	Guest selects Register.		
Preconditions	The login identifier is not already registered.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Guest	Enter account details and password.
	2	EPS	Validate the details and start MFA enrollment; do not grant Admin or Merchant privileges.
	3	Guest	Complete the MFA enrollment challenge.
	4	EPS	Create the account after successful verification and direct the user to Login.
	5	EPS	For invalid details, a duplicate identifier or failed MFA, reject registration without activating an account.
Postconditions	A valid ordinary account is created; the user must log in separately.		

U2: Login with MFA

Use Case Code	U2	**Use Case Name**	Login with MFA
Actor	Buyer; Seller; Admin		
Usage	Start an authenticated user session.		
Activation Event	User selects Login.		
Preconditions	An active user or pre-provisioned Admin account exists.		
Main Event Flow	**Index**	**Done by**	**Action**

U2 (continued)			
	1	User	Submit the login identifier and password.
	2	EPS	Verify credentials and request the enrolled MFA factor.
	3	User	Submit the MFA response.
	4	EPS	Verify MFA, create the session and show the permitted workspace.
	5	EPS	Reject invalid credentials or MFA without creating a session. Merchant client credentials cannot be used as a human login.
Postconditions	A successful login creates a session with the account's existing permissions.		

U3: Logout

Use Case Code	U3	**Use Case Name**	Logout
Actor	Buyer; Seller; Admin		
Usage	End the current user session.		
Activation Event	User selects Logout.		
Preconditions	A current or previously expired session is available.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	User	Select Logout.
	2	EPS	Invalidate the current session and show Login. An already expired session returns the same result.
Postconditions	The previous session cannot access protected functions.		

U4: View/Track Escrows

Use Case Code	U4	**Use Case Name**	View/Track Escrows
Actor	Buyer; Seller; Merchant/Marketplace (U4.API variant)		
Usage	View escrows and their current progress.		
Activation Event	User opens the escrow list or an escrow detail.		
Preconditions	The user is authenticated and is a participant in the requested escrow.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	User	Request the list or select an escrow.
	2	EPS	Check access and retrieve the authorised record.
	3	EPS	Show terms, lifecycle, payment status, delivery, refund proposal and dispute status.
	4	EPS	Show unresolved operations as pending with a query reference; viewing does not move money.
	5	EPS	Reject access to another user's escrow without disclosing its data.

U4 (continued)			
	API	Merchant	U4.API: Authenticate the Merchant client and return only its owned escrow status. Pending results include a query reference. Reject other clients' escrows and exclude private dispute evidence and Admin audit data.
Postconditions	The authorised current status is displayed; the escrow is unchanged.		

U5: Open Dispute

Use Case Code	U5	**Use Case Name**	Open Dispute
Actor	Buyer; Seller		
Usage	Request Admin resolution of a held escrow.		
Activation Event	Buyer or Seller selects Open Dispute.		
Preconditions	The escrow is HELD or DELIVERED; no dispute has previously been opened and no payout is reserved.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer/Seller	Enter the dispute reason.
	2	EPS	Recheck participant access and dispute eligibility.
	3	EPS	Open the dispute and invalidate any pending refund proposal in one state change.
	4	EPS	Save the case and allow both parties to submit evidence; block delivery and ordinary release.
	5	EPS	Reject a stale or concurrent request if another payout or dispute decision has already been accepted.
Postconditions	One OPEN dispute exists; the full amount remains held.		

U6: Submit Dispute Evidence

Use Case Code	U6	**Use Case Name**	Submit Dispute Evidence
Actor	Buyer; Seller		
Usage	Add evidence to an open dispute.		
Activation Event	A participant selects Submit Evidence.		
Preconditions	The participant belongs to the escrow and the dispute remains OPEN.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer/Seller	Submit evidence and a short explanation.
	2	EPS	Validate the submission and recheck that no final decision has been recorded.
	3	EPS	Save the evidence with its author and submission time.
	4	EPS	Reject unauthorised submissions or evidence received after the final decision.
Postconditions	Accepted evidence is available for dispute review.		

U7: Propose/Withdraw Full Refund

Use Case Code	U7	**Use Case Name**	Propose/Withdraw Full Refund
Actor	Buyer; Seller		
Usage	Propose or withdraw a full refund.		
Activation Event	A participant selects Propose Refund or Withdraw Proposal.		
Preconditions	The escrow is HELD or DELIVERED, with no open dispute or reserved payout.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer/Seller	Submit a full-refund reason, or select their pending proposal for withdrawal.
	2	EPS	For a new proposal, check that no other proposal is pending; save it and notify the counterpart.
	3	EPS	While the proposal is pending, block new delivery recording and ordinary Buyer release.
	4	EPS	Allow only the proposer to withdraw before approval. Restore otherwise eligible actions without deleting saved delivery.
	5	EPS	Reject stale requests; repeated requests return the recorded result.
Postconditions	The proposal is PENDING or WITHDRAWN; no money is transferred.		

U8: Accept/Reject Refund Proposal

Use Case Code	U8	**Use Case Name**	Accept/Reject Refund Proposal
Actor	Buyer; Seller		
Usage	Accept or reject the counterpart's full-refund proposal.		
Activation Event	The counterpart responds to a proposal.		
Preconditions	The proposal is PENDING and the escrow has no open dispute or reserved payout.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer/Seller	Review the proposal and choose Accept or Reject.
	2	EPS	Recheck that the responder is the counterpart and the proposal remains eligible.
	3	EPS	On acceptance, approve the proposal and reserve the full refund as REFUND_ PENDING; invoke P3.
	4	EPS	On rejection, mark REJECTED and restore otherwise eligible actions without reverting delivery.
	5	EPS	Reject self-approval or a stale response. A repeated response returns the recorded result.
Postconditions	Approval starts P3; rejection makes no payment. Only confirmed provider success produces REFUNDED.		

3.2.2 Buyer's Use cases Details

B1: Create Escrow

Use Case Code	B1	**Use Case Name**	Create Escrow
Actor	Buyer; Merchant/Marketplace (B1.API variant)		
Usage	Create an escrow agreement.		
Activation Event	Buyer selects Create Escrow.		
Preconditions	The Buyer is authenticated; the Buyer and Seller are distinct existing users.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer	Enter the Seller, GOODS/SERVICES, full VND amount, deliverable and acceptance criteria.
	2	EPS	Validate the agreement and capture the current evidence policy and Seller destination.
	3	EPS	Save the escrow as AWAITING_SELLER and notify the Seller.
	4	EPS	Reject invalid terms or self-trading. An identical repeated creation returns the existing escrow.
	API	Merchant	B1.API: Authenticate the Merchant client, use its bound Seller and validate the same B1 terms. Save client ownership and return the escrow reference and AWAITING_SELLER. Reject another Seller or conflicting request reuse; identical repeats return the existing escrow.
Postconditions	The agreement awaits Seller acceptance; no funding has started.		

B2: Cancel Unfunded Escrow

Use Case Code	B2	**Use Case Name**	Cancel Unfunded Escrow
Actor	Buyer		
Usage	Cancel an escrow before funding.		
Activation Event	Buyer selects Cancel.		
Preconditions	The escrow is AWAITING_SELLER or AWAITING_FUNDING; no money was received and no funding result is unresolved.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer	Confirm cancellation of their escrow.
	2	EPS	Recheck ownership and funding status.
	3	EPS	Set CANCELLED and record the reason.
	4	EPS	Reject cancellation while funding is active or unknown. A repeated cancellation returns the existing result.
Postconditions	The cancelled escrow cannot accept funding or payout.		

B3: Fund Escrow

Use Case Code	B3	**Use Case Name**	Fund Escrow
Actor	Buyer; Partner Bank; Payment Gateway		
Usage	Fund the full accepted agreement through a mock provider.		
Activation Event	Buyer selects Fund Escrow.		
Preconditions	Seller has accepted; the escrow is AWAITING_FUNDING with no unresolved funding attempt.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer	Review the terms, accept them and choose Bank or Gateway.
	2	EPS	Save a funding reference and set FUNDING_ PENDING.
	3	Bank/Gateway	Process the full VND funding request and return a result through P1.
	4	EPS	Show the confirmed result, or keep pending if the result is uncertain.
	5	EPS	Only confirmed failure with no receipt returns AWAITING_FUNDING, allowing the Buyer to retry or cancel.
Postconditions	Only P1 confirmation establishes HELD. A timeout does not permit another funding attempt.		

B4: Confirm Acceptance and Request Release

Use Case Code	B4	**Use Case Name**	Confirm Acceptance and Request Release
Actor	Buyer; Partner Bank		
Usage	Accept delivery and request full release to the Seller.		
Activation Event	Buyer selects Confirm Acceptance.		
Preconditions	The escrow is DELIVERED and its evidence policy is satisfied; no refund proposal, open dispute or payout is pending.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Buyer	Review delivery and confirm acceptance.
	2	EPS	Recheck access, evidence and escrow eligibility.
	3	EPS	Reserve the full amount as RELEASE_ PENDING and invoke P2 through the Bank mock.
	4	EPS	Show the pending or confirmed result.
	5	EPS	Reject a conflicting request; a repeated acceptance returns the saved release decision.
Postconditions	Release is requested once. RELEASED requires confirmed Bank settlement.		

3.2.3 Seller's Use cases Details

S1: Accept/Reject Escrow Terms

Use Case Code	S1	**Use Case Name**	Accept/Reject Escrow Terms
Actor	Seller		
Usage	Accept or reject a proposed escrow agreement.		
Activation Event	Seller opens the proposed terms.		
Preconditions	The escrow is AWAITING_SELLER and the user is its Seller.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Seller	Review the terms and choose Accept or Reject.
	2	EPS	Recheck Seller access and the current state.
	3	EPS	On acceptance, lock the terms and set AWAITING_FUNDING; on rejection, set CANCELLED with the reason.
	4	EPS	Reject a stale change or return the previously recorded response.
Postconditions	Accepted terms cannot be edited; funding is permitted only after acceptance.		

S2: Record Delivery/Service Completion

Use Case Code	S2	**Use Case Name**	Record Delivery/Service Completion
Actor	Seller		
Usage	Record goods delivery or service completion.		
Activation Event	Seller selects Record Delivery/Completion.		
Preconditions	The escrow is HELD, with no pending refund proposal, open dispute or reserved payout.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Seller	Submit delivery information or the completed work and required evidence.
	2	EPS	Check Seller access and the escrow's captured evidence policy.
	3	EPS	Save completion, set DELIVERED and notify the Buyer.
	4	EPS	Reject missing required evidence or an ineligible state; do not release money automatically.
Postconditions	Delivery is recorded; the full amount remains held for Buyer acceptance or another eligible outcome.		

3.2.4 Admin's Use cases Details

A1: Review and Resolve Dispute

Use Case Code	A1	**Use Case Name**	Review and Resolve Dispute
Actor	Admin		
Usage	Resolve a dispute with one full payout decision.		
Activation Event	Admin selects an open dispute.		
Preconditions	Admin is authenticated; the dispute is OPEN, funds are held and no payout is reserved.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Review the agreement, evidence and payment status.
	2	Admin	Choose full release or full refund and enter a reason.
	3	EPS	Recheck the case; save RESOLVED and reserve the chosen payout in one state change.
	4	EPS	Invoke P2 for release or P3 for refund. This decision does not require Buyer acceptance.
	5	EPS	Show the pending or confirmed settlement; reject a conflicting second decision.
Postconditions	The decision and payout direction are final; only provider confirmation completes the payment.		

A2: Configure Escrow Policy Parameters

Use Case Code	A2	**Use Case Name**	Configure Escrow Policy Parameters
Actor	Admin		
Usage	Configure delivery-evidence requirements for new escrows.		
Activation Event	Admin opens Escrow Policy.		
Preconditions	Admin is authenticated with policy permission.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Review and edit the evidence-required flag for GOODS and SERVICES.
	2	EPS	Validate the flags and save a new policy version.
	3	EPS	Apply it only to escrows created afterwards; keep existing snapshots unchanged.
	4	EPS	Record the change in the audit trail. Automatic release and milestones are not configurable.
Postconditions	A valid policy version is available for new agreements.		

A3: View Basic Analytics

Use Case Code	A3	**Use Case Name**	View Basic Analytics
Actor	Admin		
Usage	View basic statistics for simulated escrows.		
Activation Event	Admin opens Analytics.		
Preconditions	Admin is authenticated with analytics permission.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Select the period and transaction type.
	2	EPS	Calculate counts and VND amounts by lifecycle state.
	3	EPS	Show the observation time and simulated-data label; separate pending amounts from final outcomes.
Postconditions	Statistics are displayed without changing escrow records.		

A4: View Audit Trail

Use Case Code	A4	**Use Case Name**	View Audit Trail
Actor	Admin		
Usage	Inspect recorded system and escrow events.		
Activation Event	Admin opens Audit Trail.		
Preconditions	Admin is authenticated with audit permission.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Filter events by escrow, period or event type.
	2	EPS	Return attributable events and correlation references, excluding authentication secrets.
	3	EPS	Display the audit trail as read-only; show an empty result when no events match.
Postconditions	Audit records remain unchanged.		

A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers

Use Case Code	A5	**Use Case Name**	Reconcile Pending Transactions/Retry Confirmed-Failed Transfers
Actor	Admin; Partner Bank; Payment Gateway		
Usage	Reconcile uncertain operations and retry confirmed-failed payouts.		
Activation Event	Admin selects an unresolved or failed payment attempt.		
Preconditions	Admin is authenticated; the original provider and operation references are available.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	Admin	Request the status of the original operation.
	2	EPS	Authenticate and match the provider result, then apply P1, P2 or P3 once.
	3	EPS	For an unknown result, retain pending and allow queries only.

A5 (continued)			
	4	EPS	For confirmed failed funding with no receipt, return AWAITING_FUNDING; only the Buyer may fund again.
	5	Admin	For a provider-confirmed failed payout with no money moved, request a retry.
	6	EPS	Create one linked attempt with the same amount, direction and beneficiary. Never permit manual success or an opposite payout.
Postconditions	The status is reconciled, remains pending, or has one authorised same-direction retry.		

3.2.5 Payment Integration's Use cases Details

P1: Confirm Funding

Use Case Code	P1	**Use Case Name**	Confirm Funding
Actor	Partner Bank; Payment Gateway		
Usage	Confirm whether the full funding amount was received.		
Activation Event	Bank/Gateway sends a callback or answers a status query.		
Preconditions	A saved funding attempt exists.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	EPS	Authenticate the provider and match escrow, operation, attempt, full amount and VND.
	2	EPS	On confirmed receipt, record the held balance and set HELD once.
	3	EPS	On confirmed failure with no receipt, return AWAITING_FUNDING.
	4	EPS	For pending or unknown results, retain FUNDING_PENDING.
	5	EPS	Return the stored result for duplicates. Reject mismatched or contradictory results and flag them for reconciliation.
Postconditions	Funds are held only after a valid receipt; a duplicate never creates another credit.		

P2: Execute and Confirm Release

Use Case Code	P2	**Use Case Name**	Execute and Confirm Release
Actor	Partner Bank		
Usage	Execute and confirm one full release to the Seller.		
Activation Event	B4/A1 authorises release, or A5 authorises an eligible retry.		
Preconditions	A full release decision and fixed Seller destination are saved; a retry requires confirmed failure with no money moved.		
Main Event Flow	**Index**	**Done by**	**Action**

P2 (continued)			
	1	EPS	Send the full VND amount to the Bank mock using the saved release reference.
	2	Bank	Execute the mock transfer and return a callback or query result.
	3	EPS	Authenticate and match the attempt, amount, currency and destination.
	4	EPS	On success, set RELEASED once. On failure, keep RELEASE_PENDING and its reservation for an eligible A5 retry; on uncertainty, query only.
	5	EPS	Reuse duplicate results and reject mismatches. Do not accept a new dispute or opposite payout after reservation.
Postconditions	The Seller is paid once on confirmed success; otherwise the same release remains pending.		

P3: Execute and Confirm Refund

Use Case Code	P3	**Use Case Name**	Execute and Confirm Refund
Actor	Partner Bank; Payment Gateway		
Usage	Execute and confirm one full refund to the original source.		
Activation Event	U8/A1 authorises refund, or A5 authorises an eligible retry.		
Preconditions	A full refund decision and original funding source are saved; a retry requires confirmed failure with no money moved.		
Main Event Flow	**Index**	**Done by**	**Action**
	1	EPS	Send the full VND refund through the original funding provider using the saved refund reference.
	2	Bank/Gateway	Execute the mock refund and return a callback or query result.
	3	EPS	Authenticate and match the attempt, amount, currency and original source.
	4	EPS	On success, set REFUNDED once. On failure, keep REFUND_PENDING and its reservation for an eligible A5 retry; on uncertainty, query only.
	5	EPS	Reuse duplicate results and reject mismatches. Do not accept a new dispute or opposite payout after reservation.
Postconditions	The original source is refunded once on confirmed success; otherwise the same refund remains pending.		

3.3 Activity Diagram

3.3.1 User's Use cases

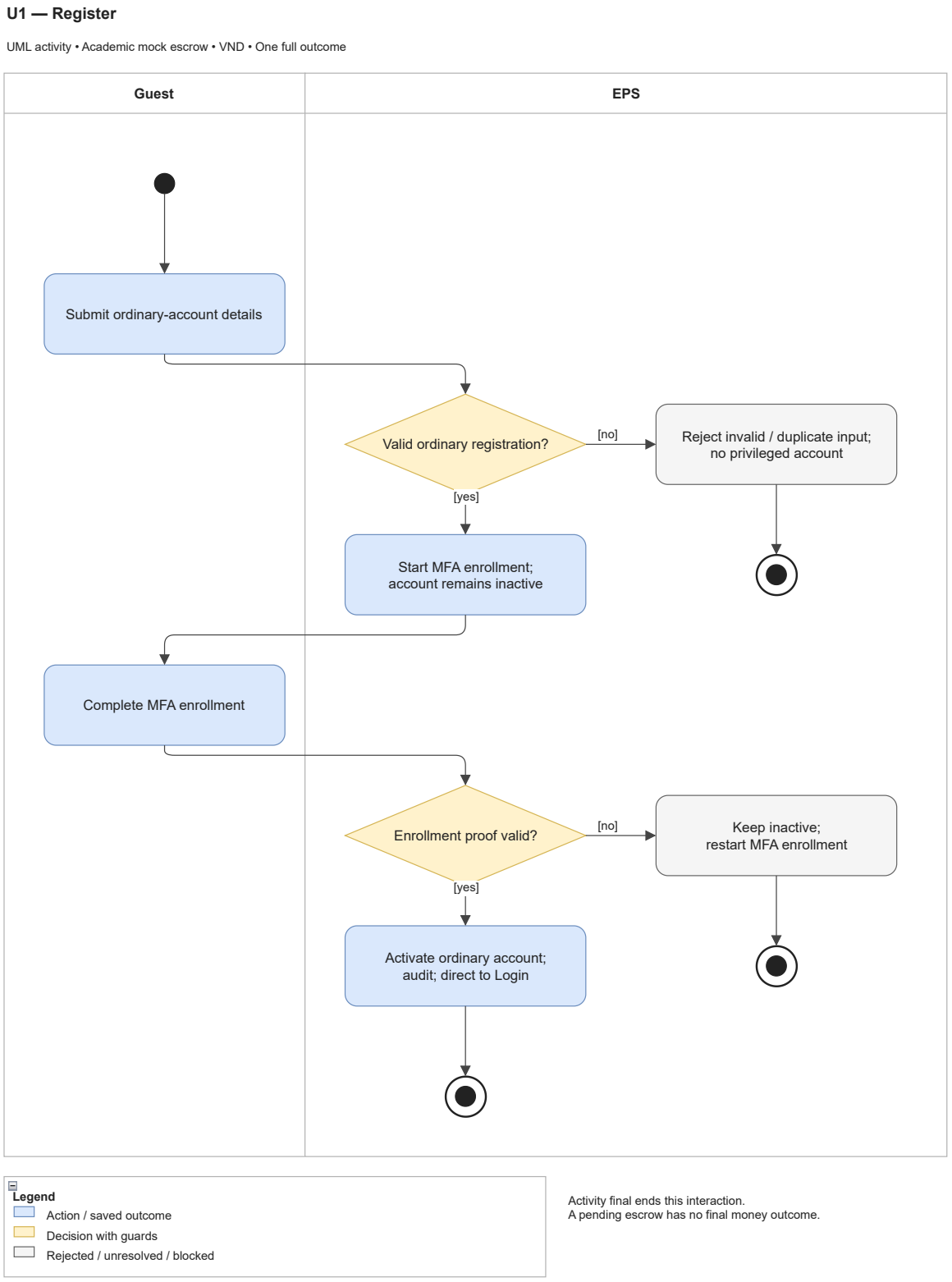


Figure 3.7: U1: Register

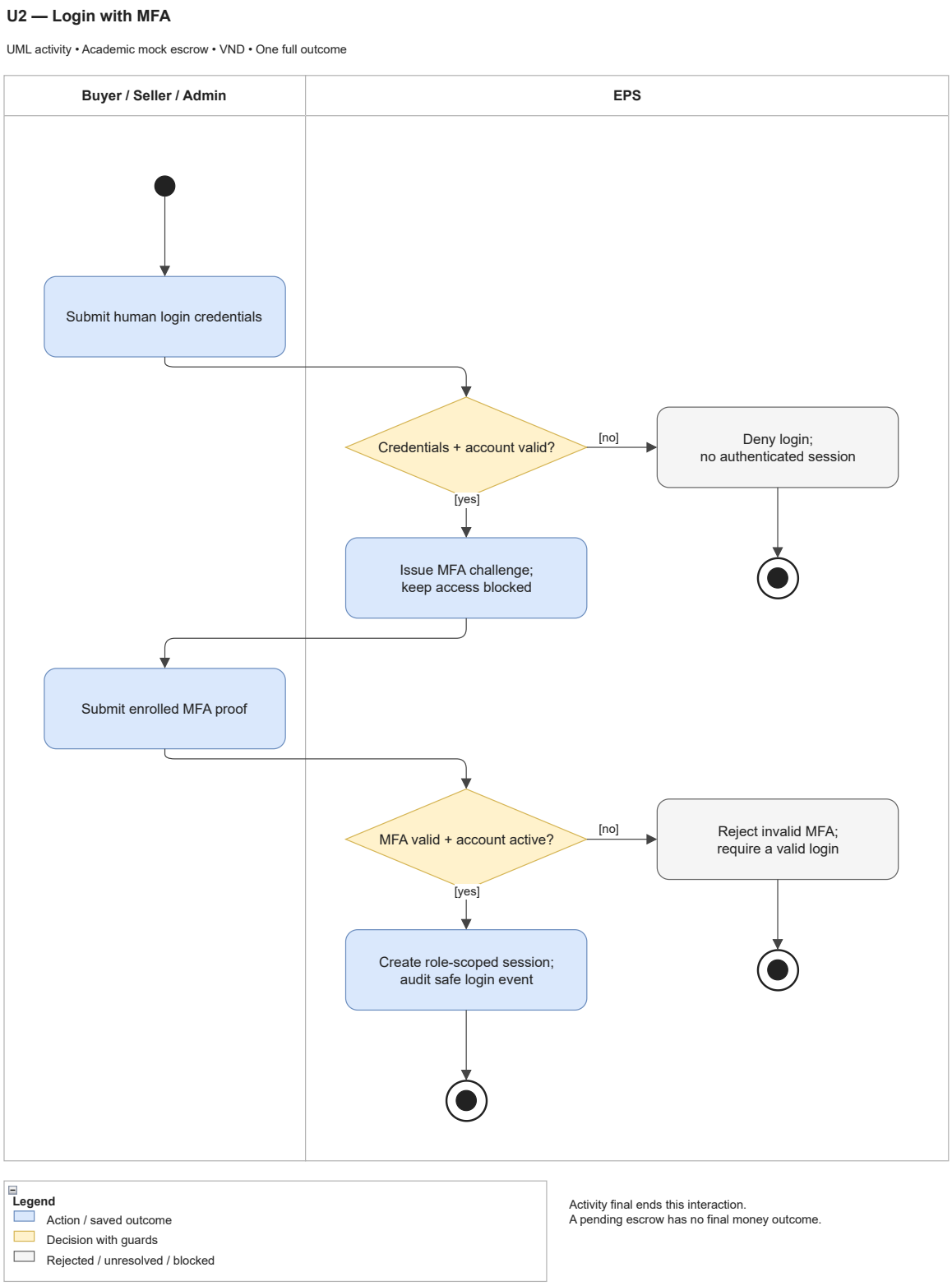


Figure 3.8: U2: Login with MFA

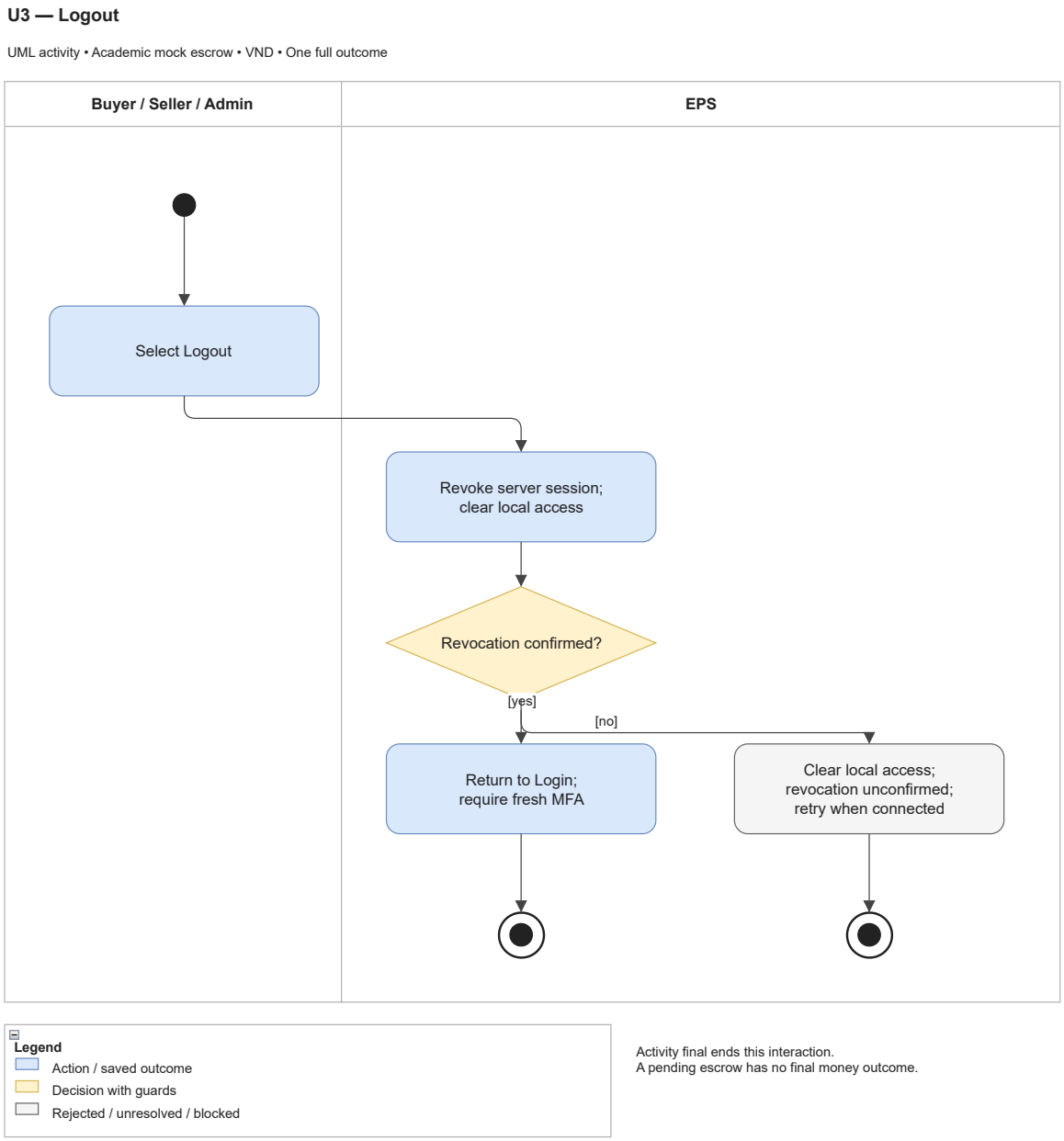


Figure 3.9: U3: Logout

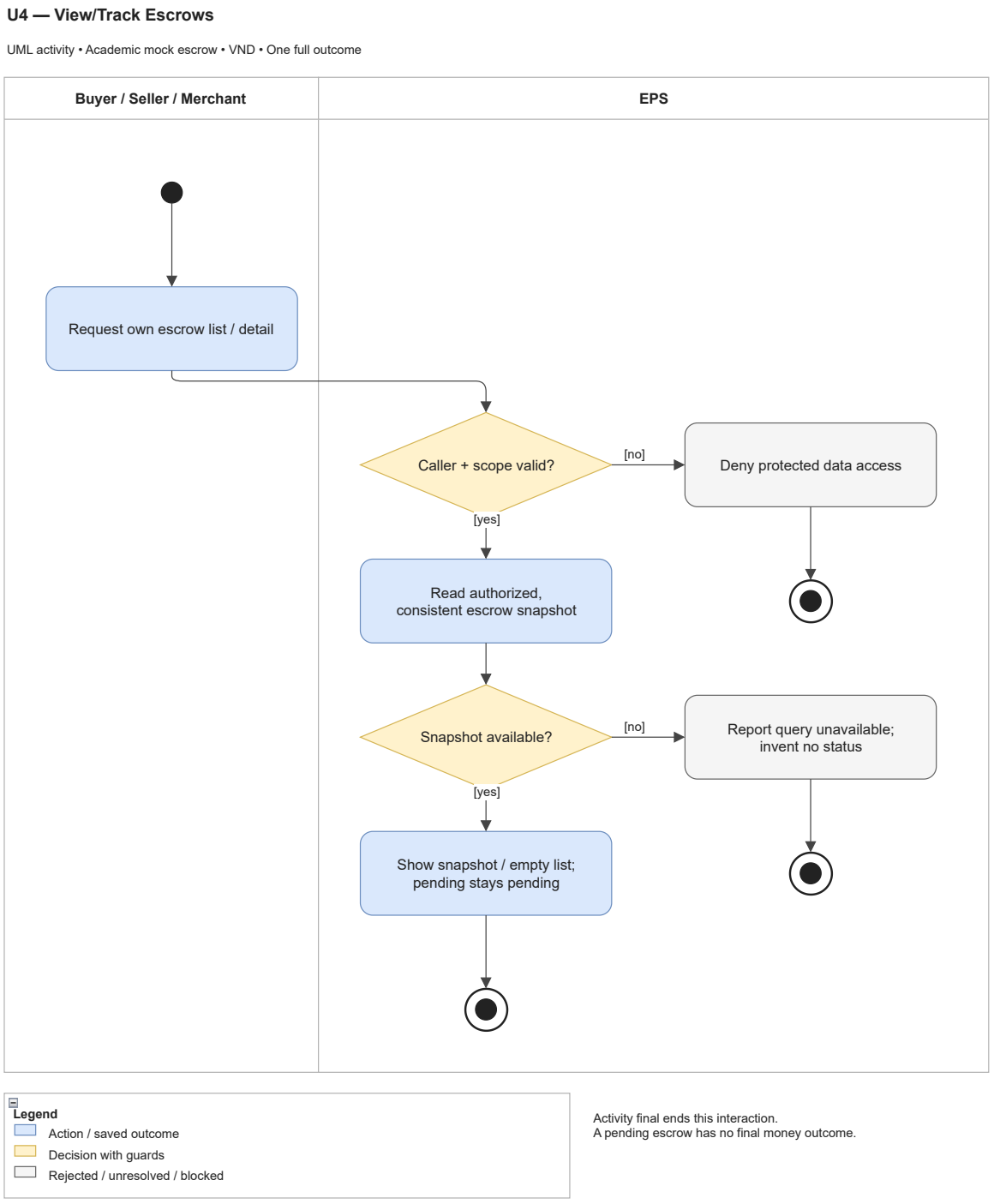


Figure 3.10: U4: View/Track Escrows

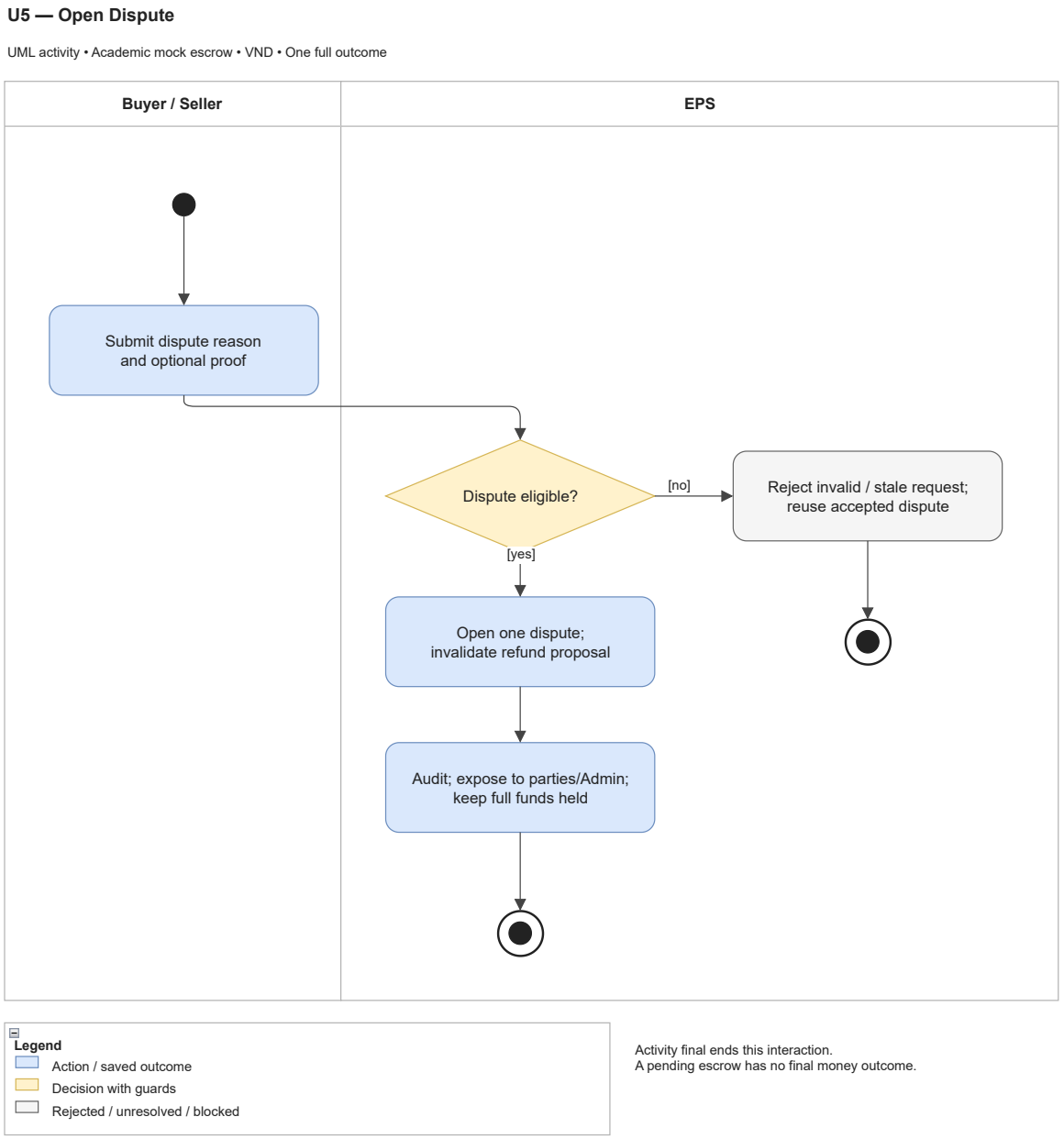


Figure 3.11: U5: Open Dispute

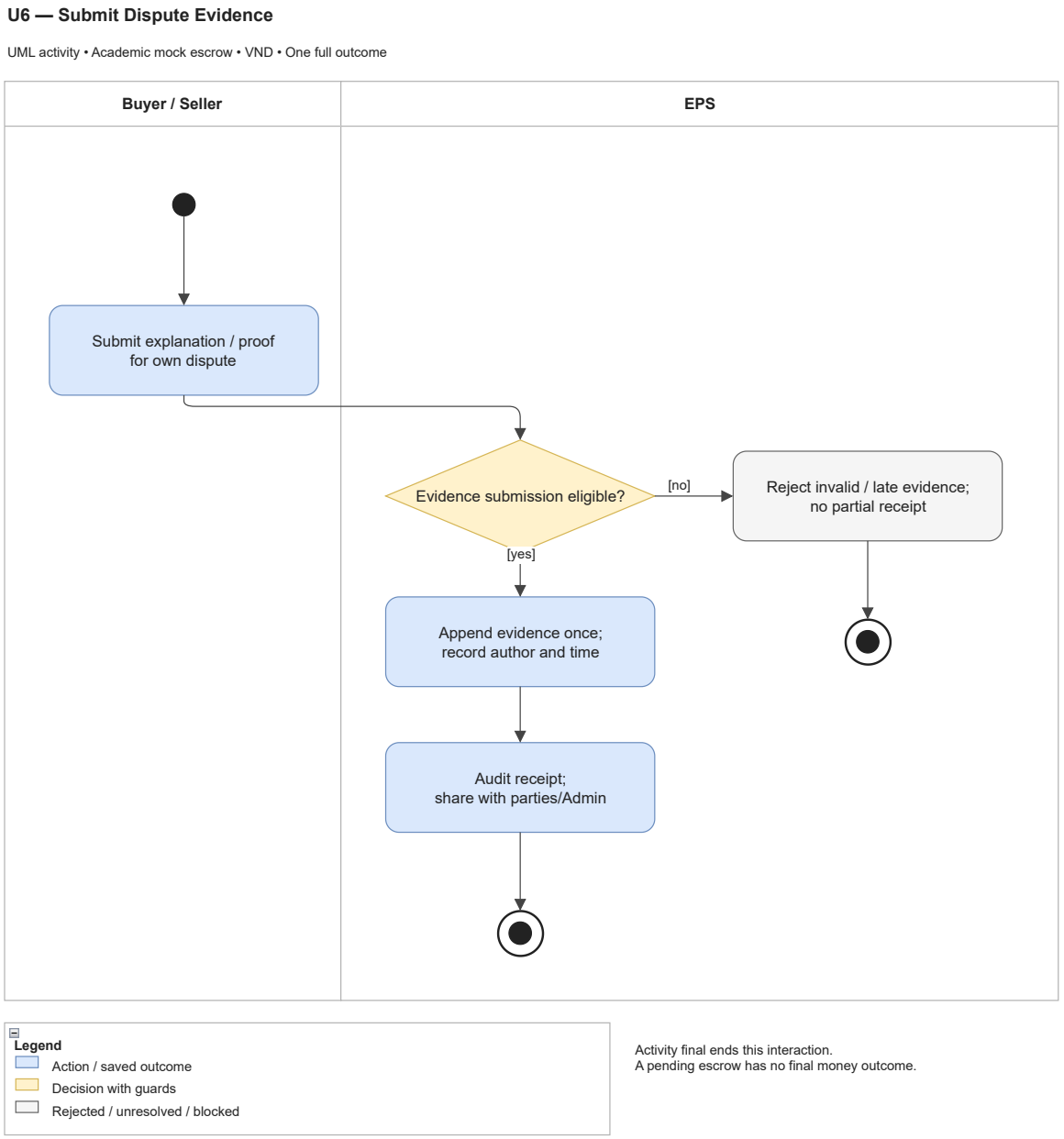


Figure 3.12: U6: Submit Dispute Evidence

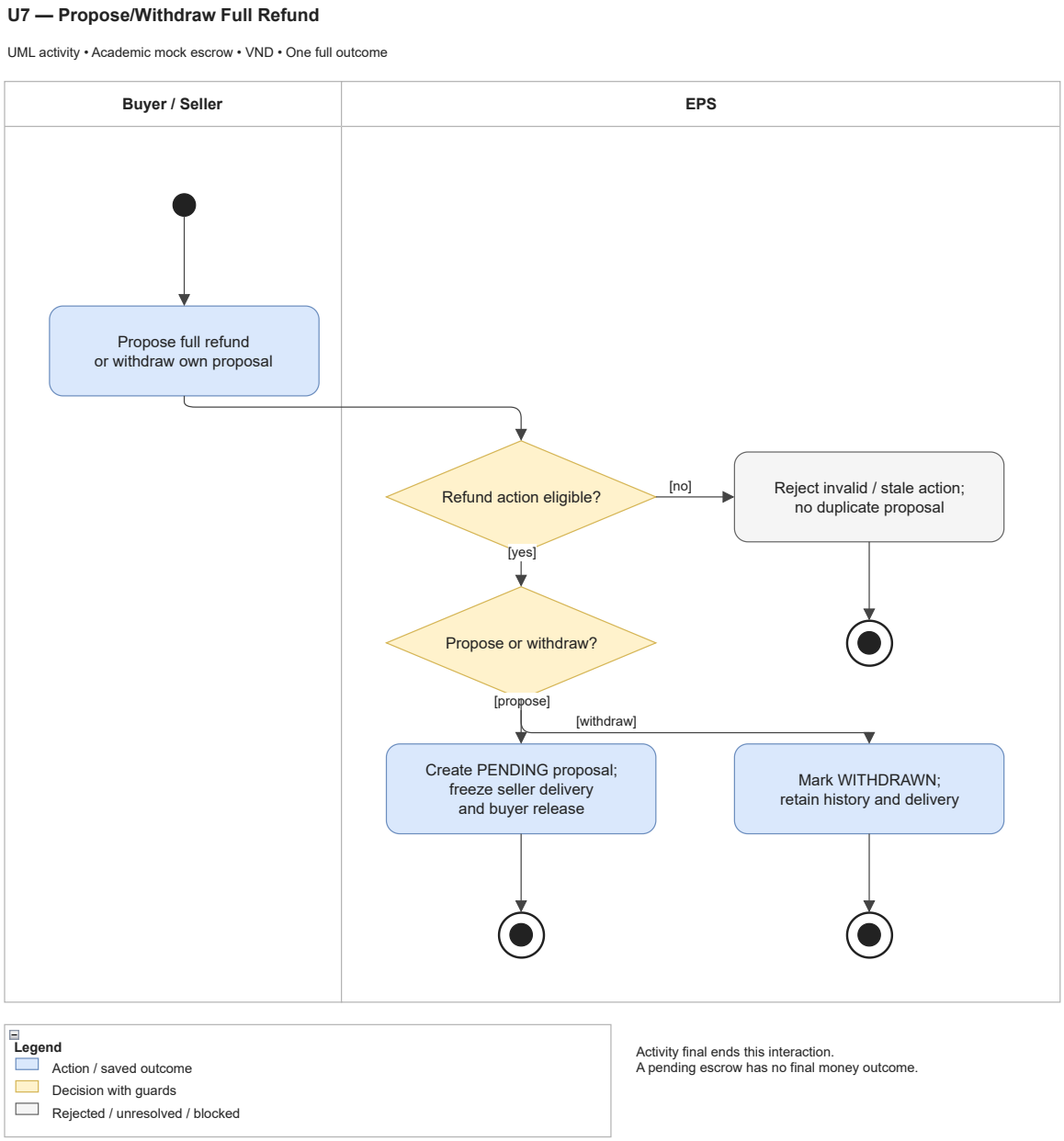


Figure 3.13: U7: Propose/Withdraw Full Refund

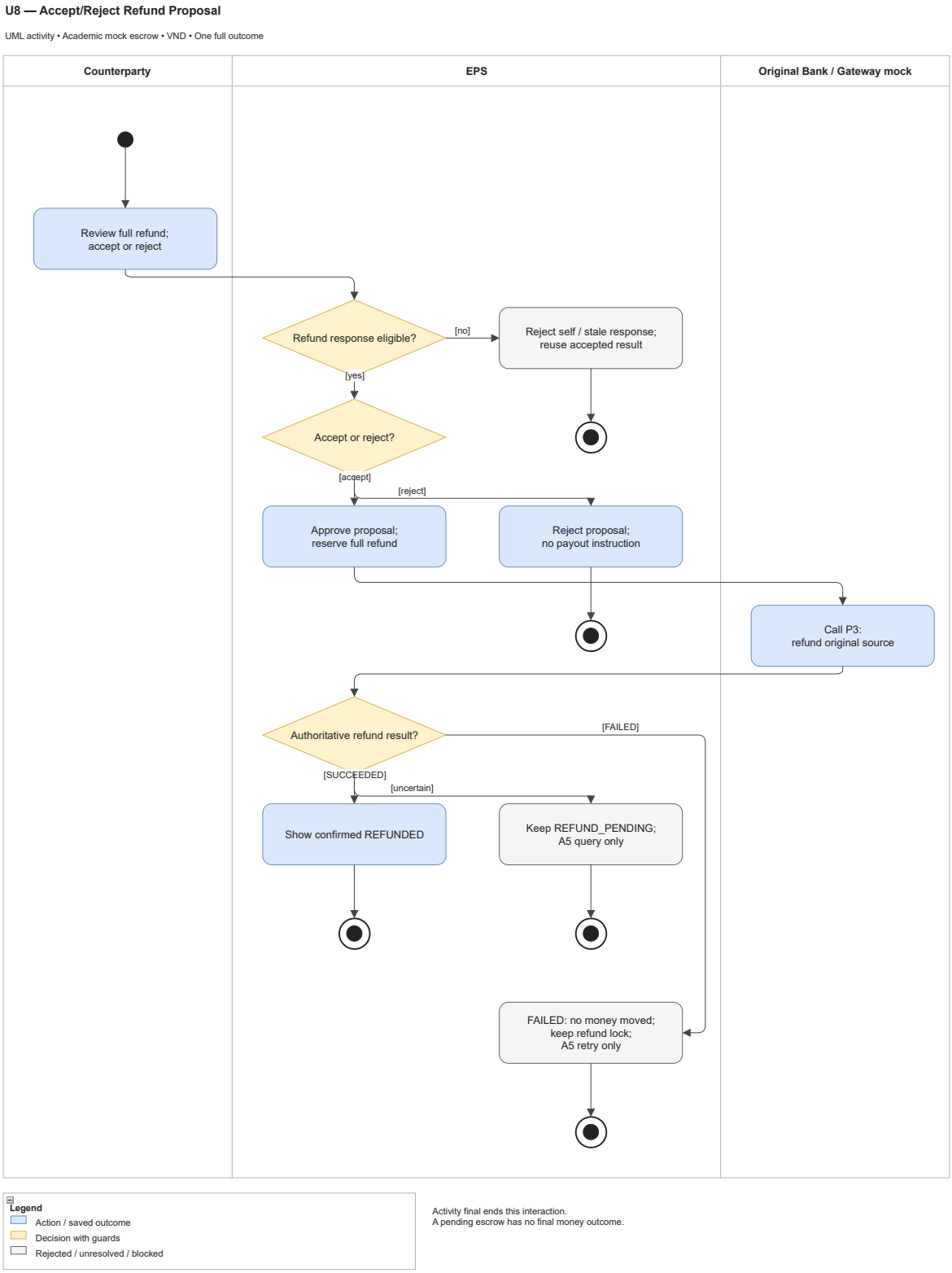


Figure 3.14: U8: Accept/Reject Refund Proposal

3.3.2 Buyer's Use cases



Figure 3.15: B1: Create Escrow

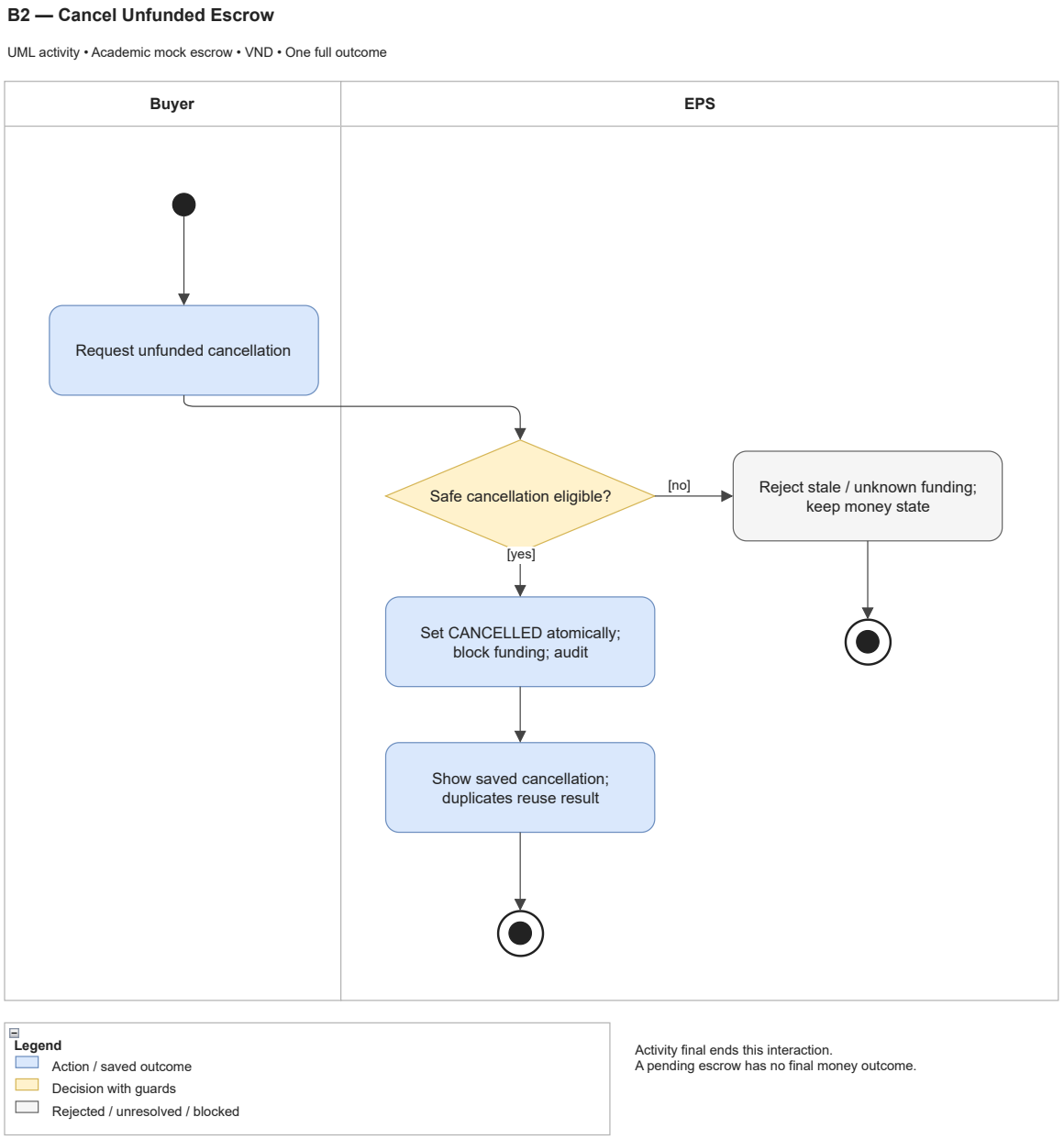


Figure 3.16: B2: Cancel Unfunded Escrow

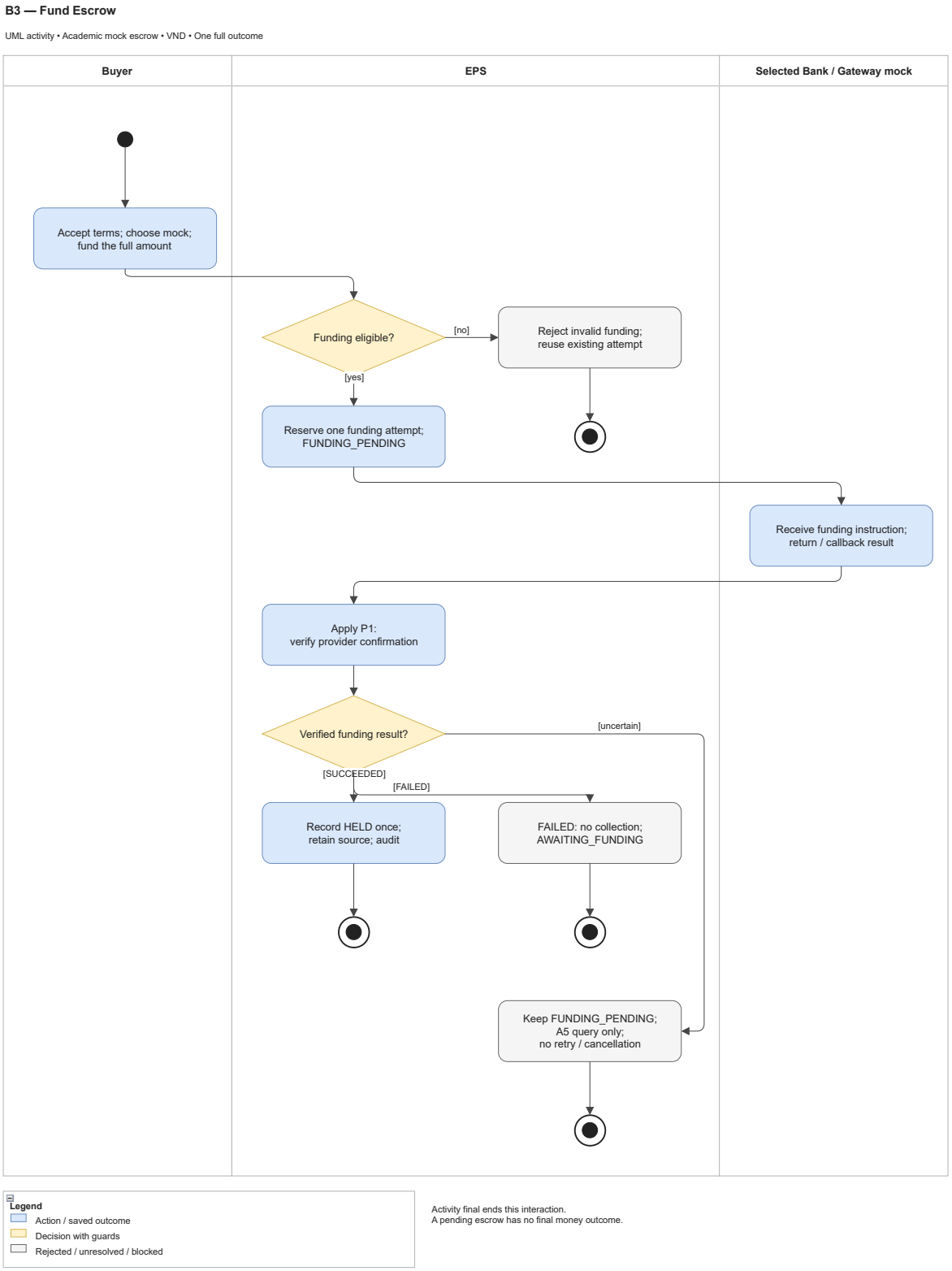


Figure 3.17: B3: Fund Escrow

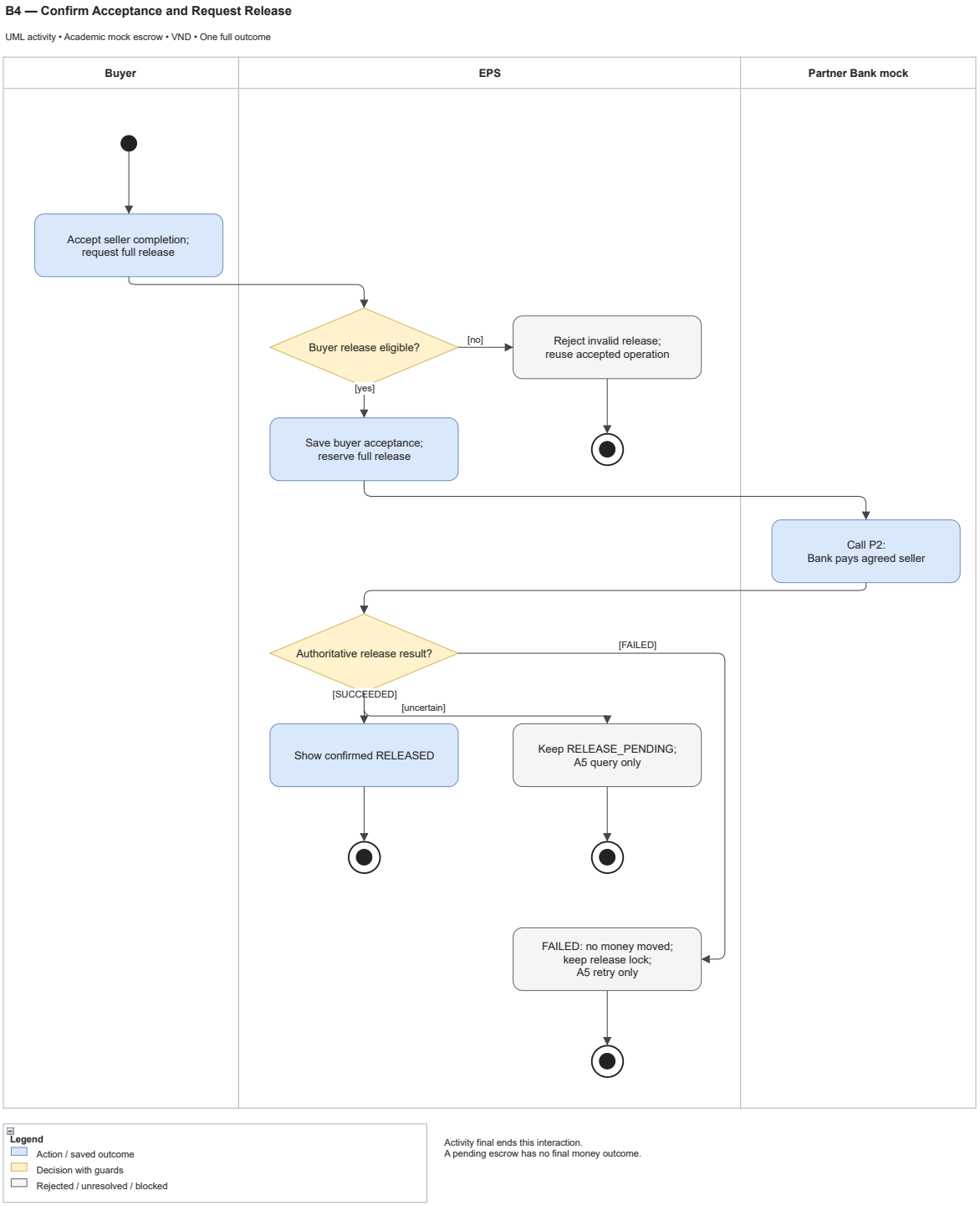


Figure 3.18: B4: Confirm Acceptance and Request Release

3.3.3 Seller's Use cases

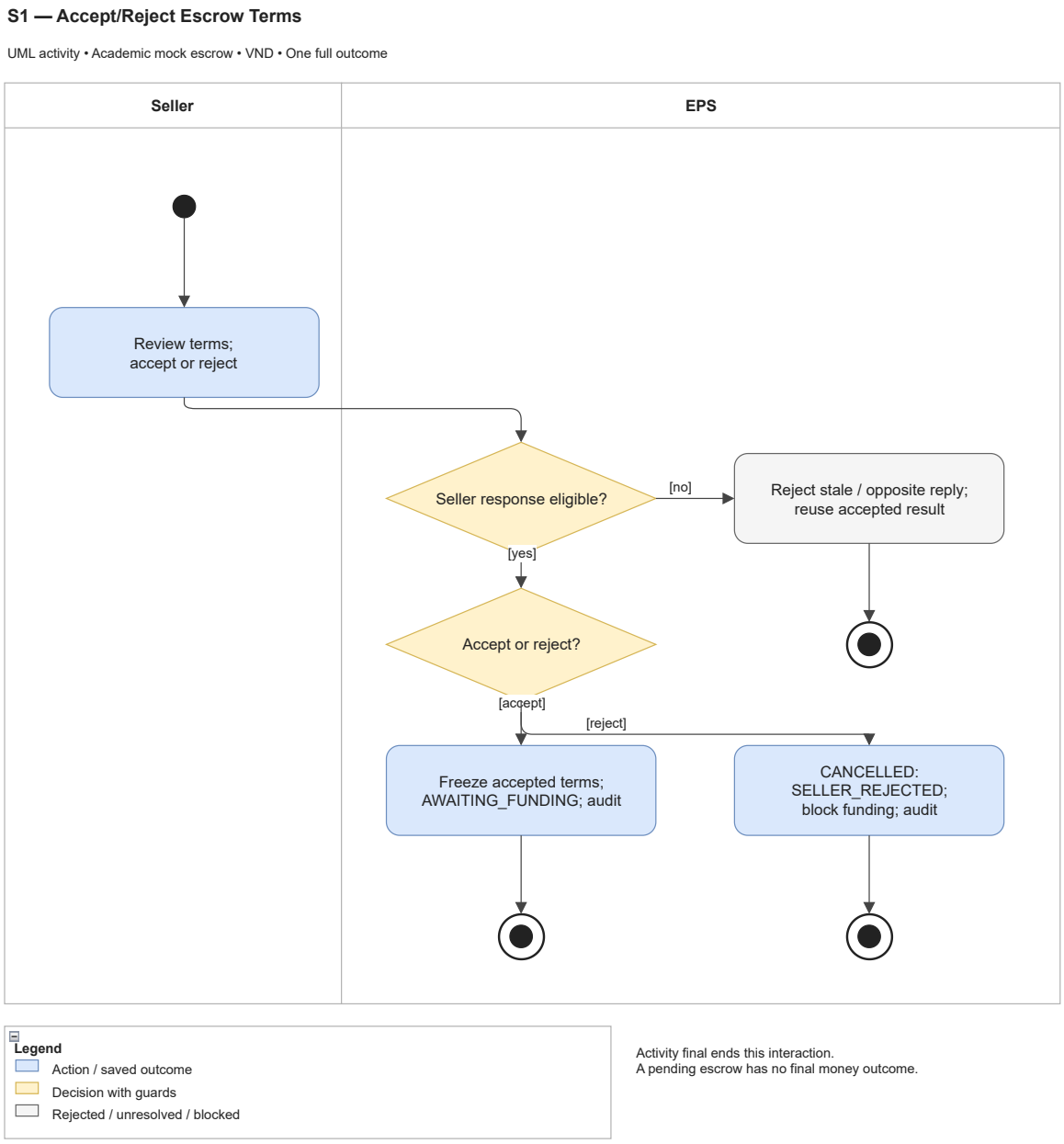


Figure 3.19: S1: Accept/Reject Escrow Terms

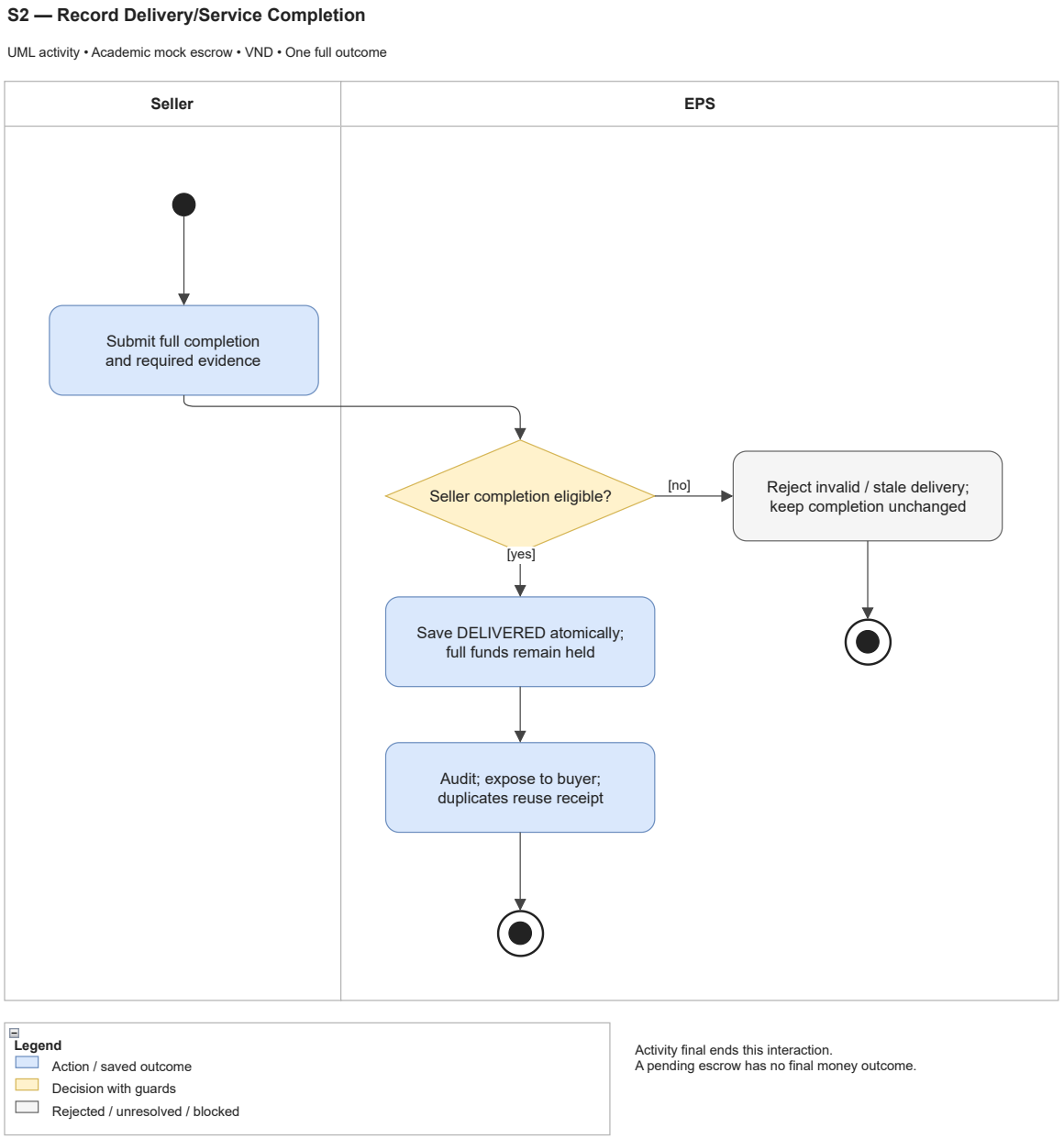


Figure 3.20: S2: Record Delivery/Service Completion

3.3.4 Admin's Use cases

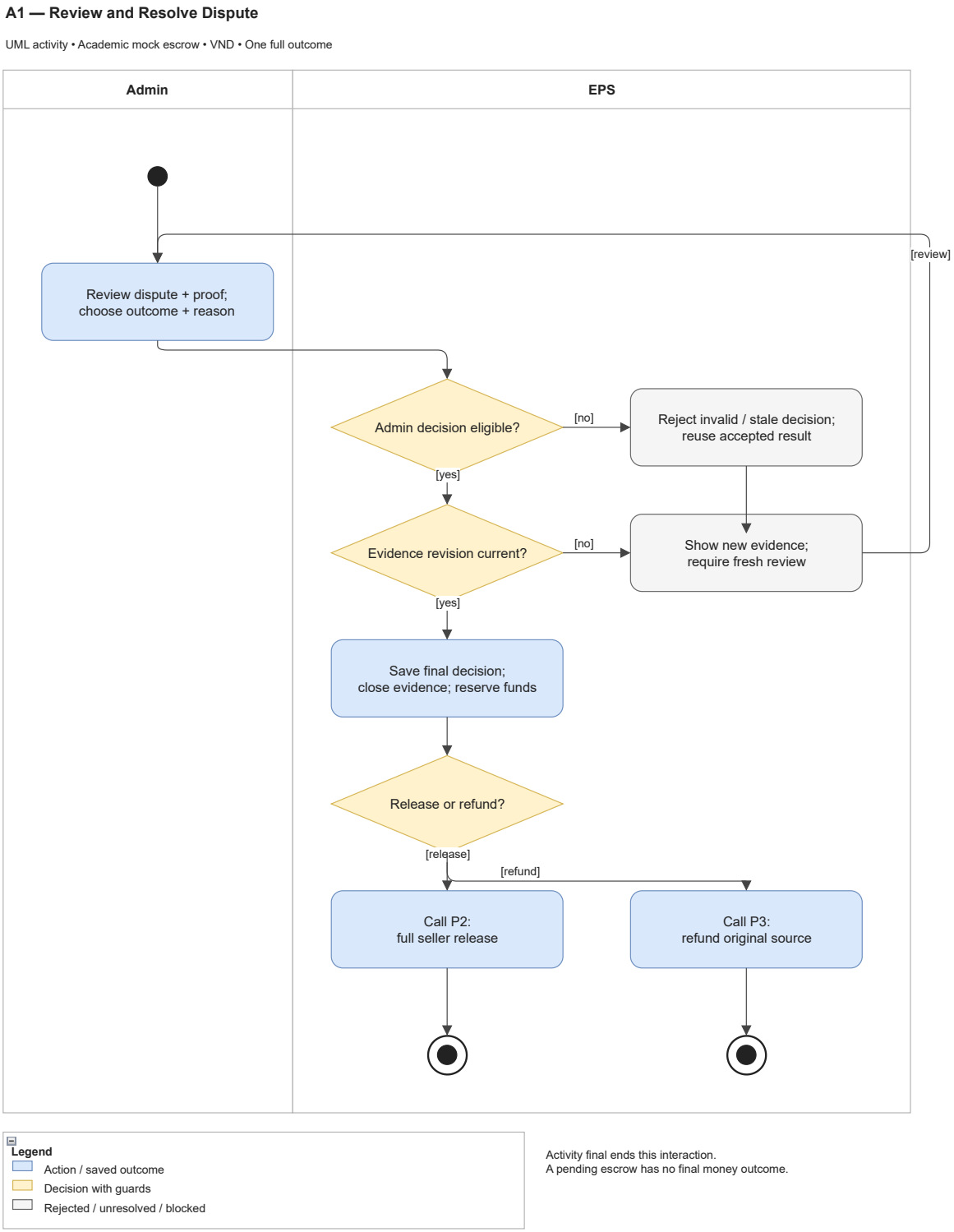


Figure 3.21: A1: Review and Resolve Dispute

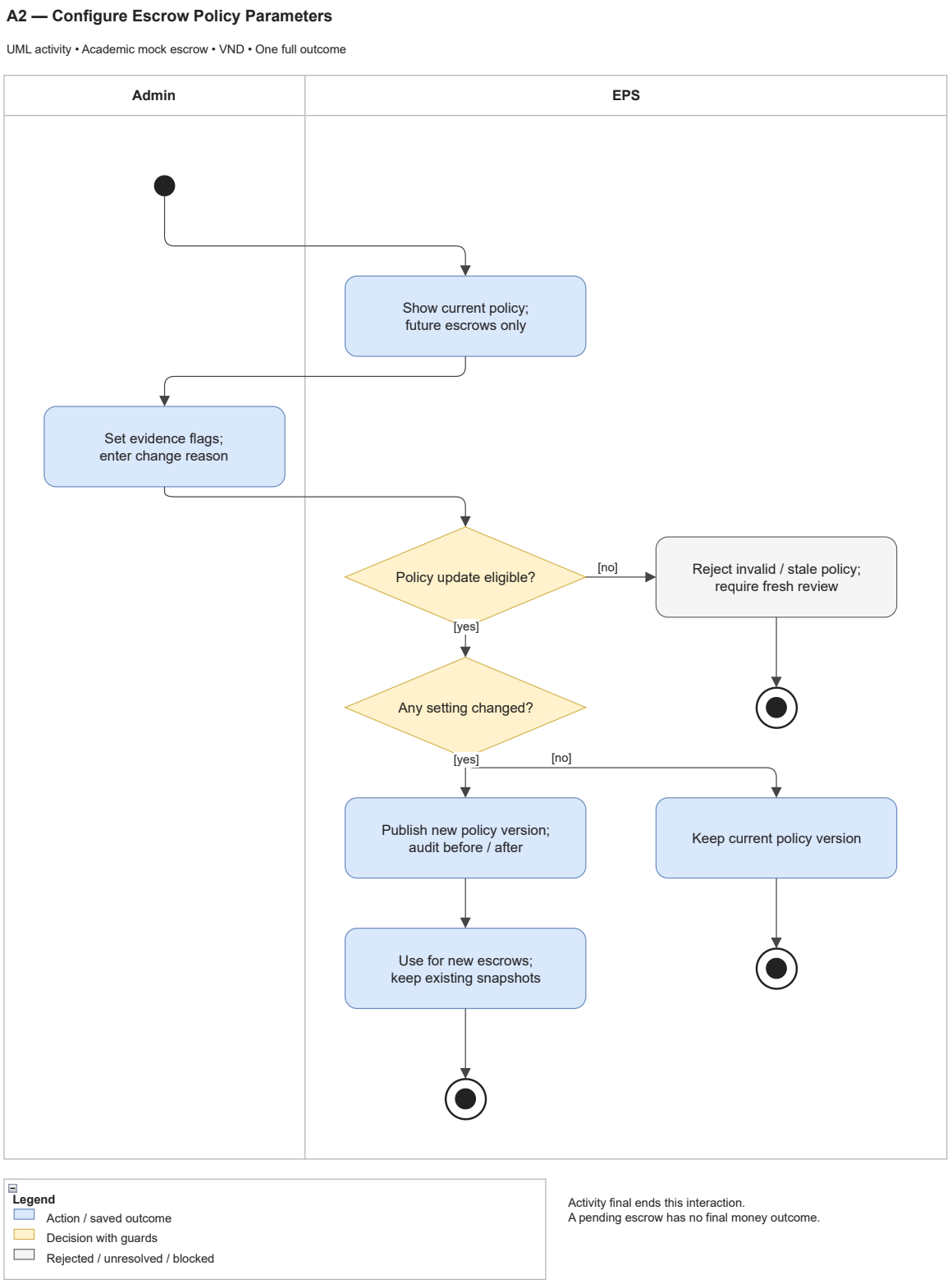


Figure 3.22: A2: Configure Escrow Policy Parameters

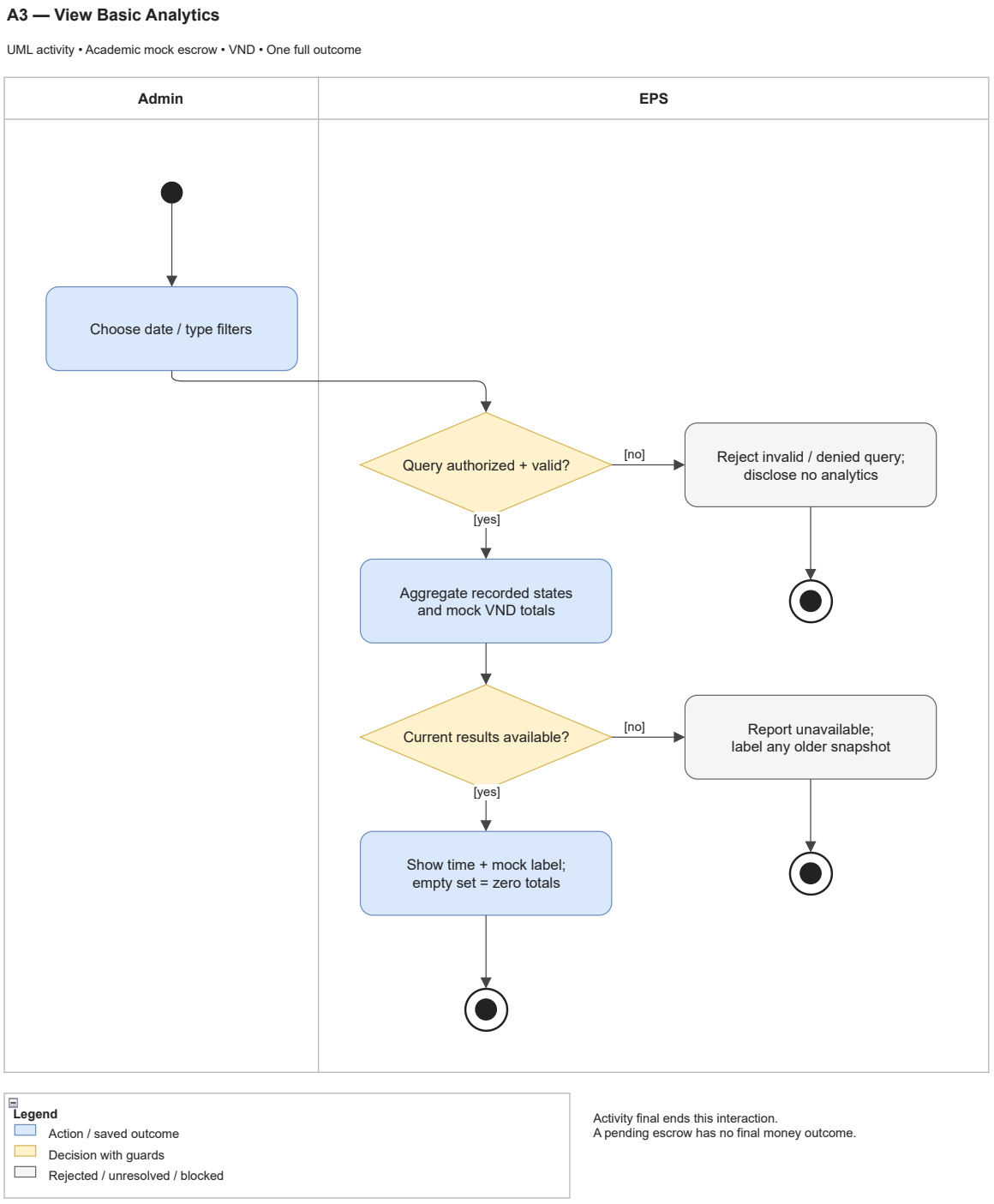


Figure 3.23: A3: View Basic Analytics

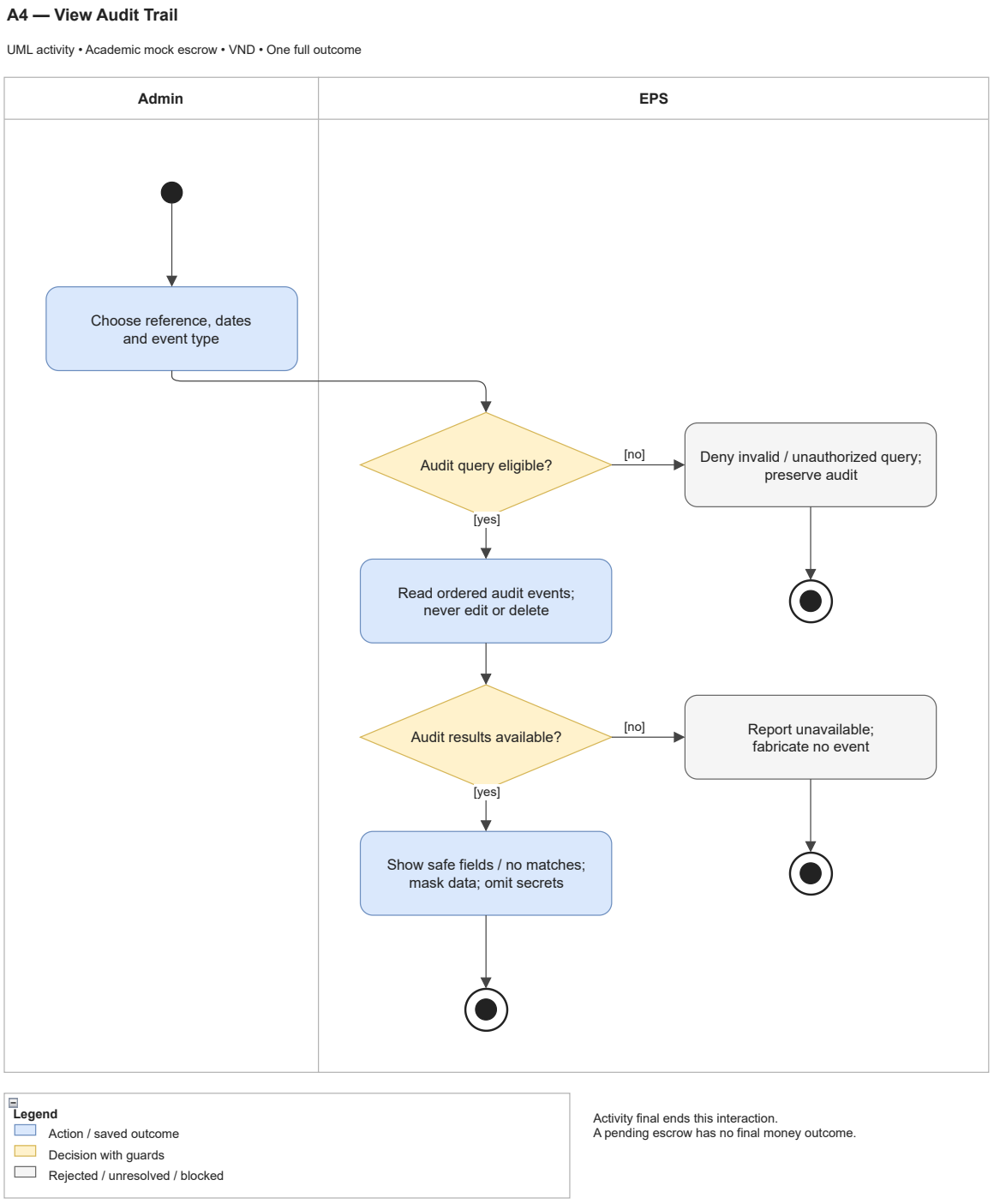


Figure 3.24: A4: View Audit Trail

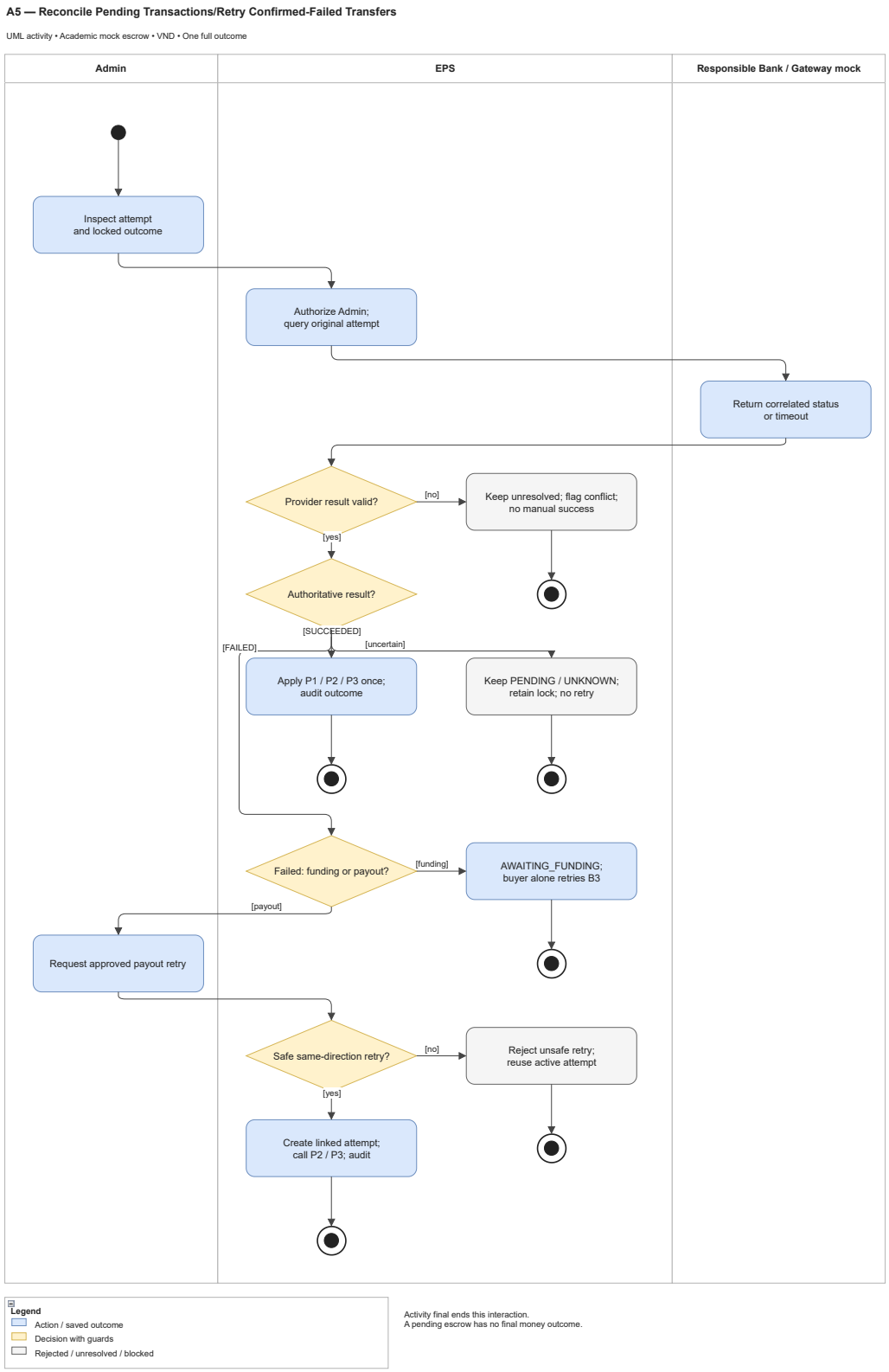


Figure 3.25: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers

3.3.5 Payment Integration's Use cases

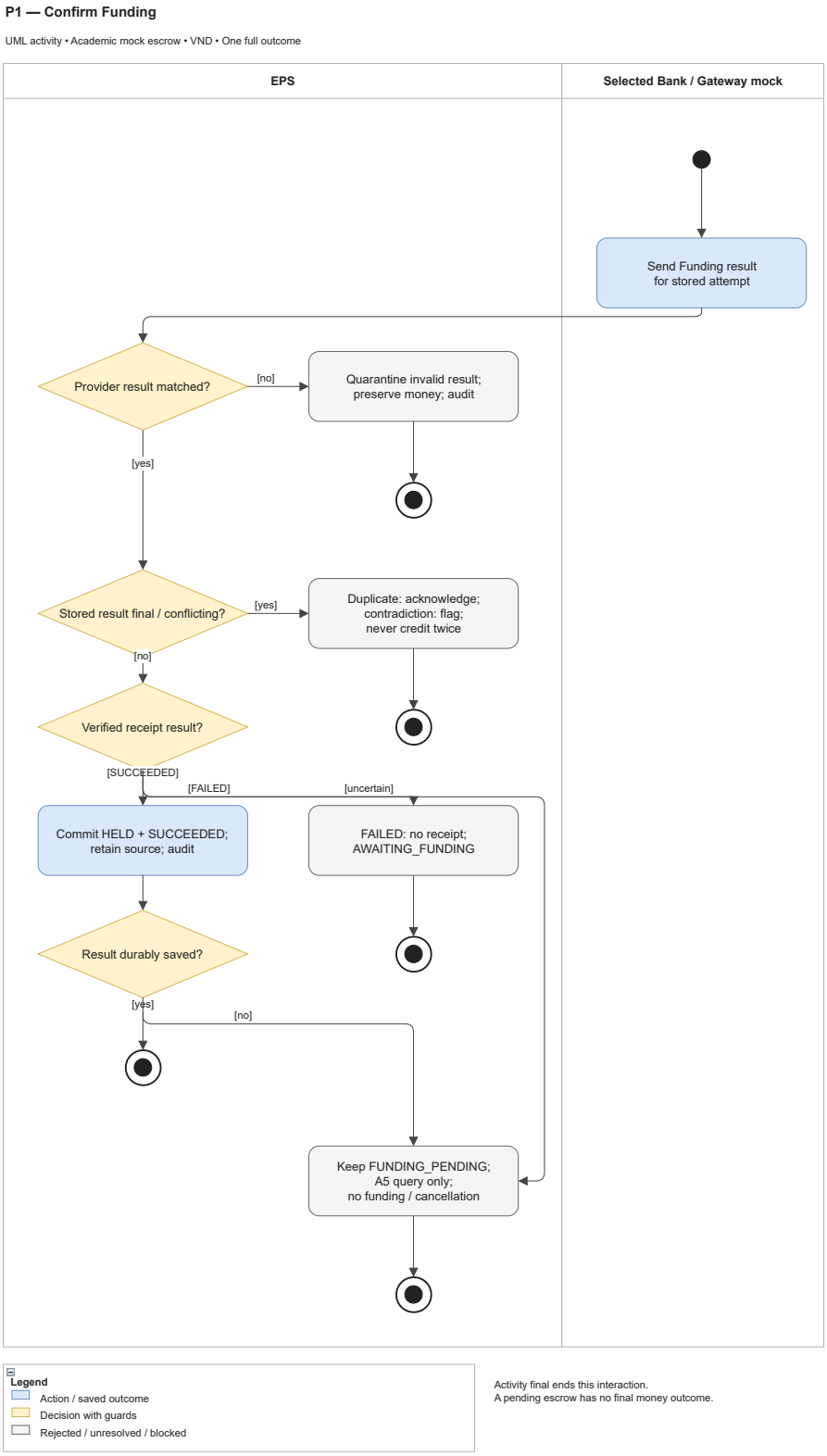


Figure 3.26: P1: Confirm Funding

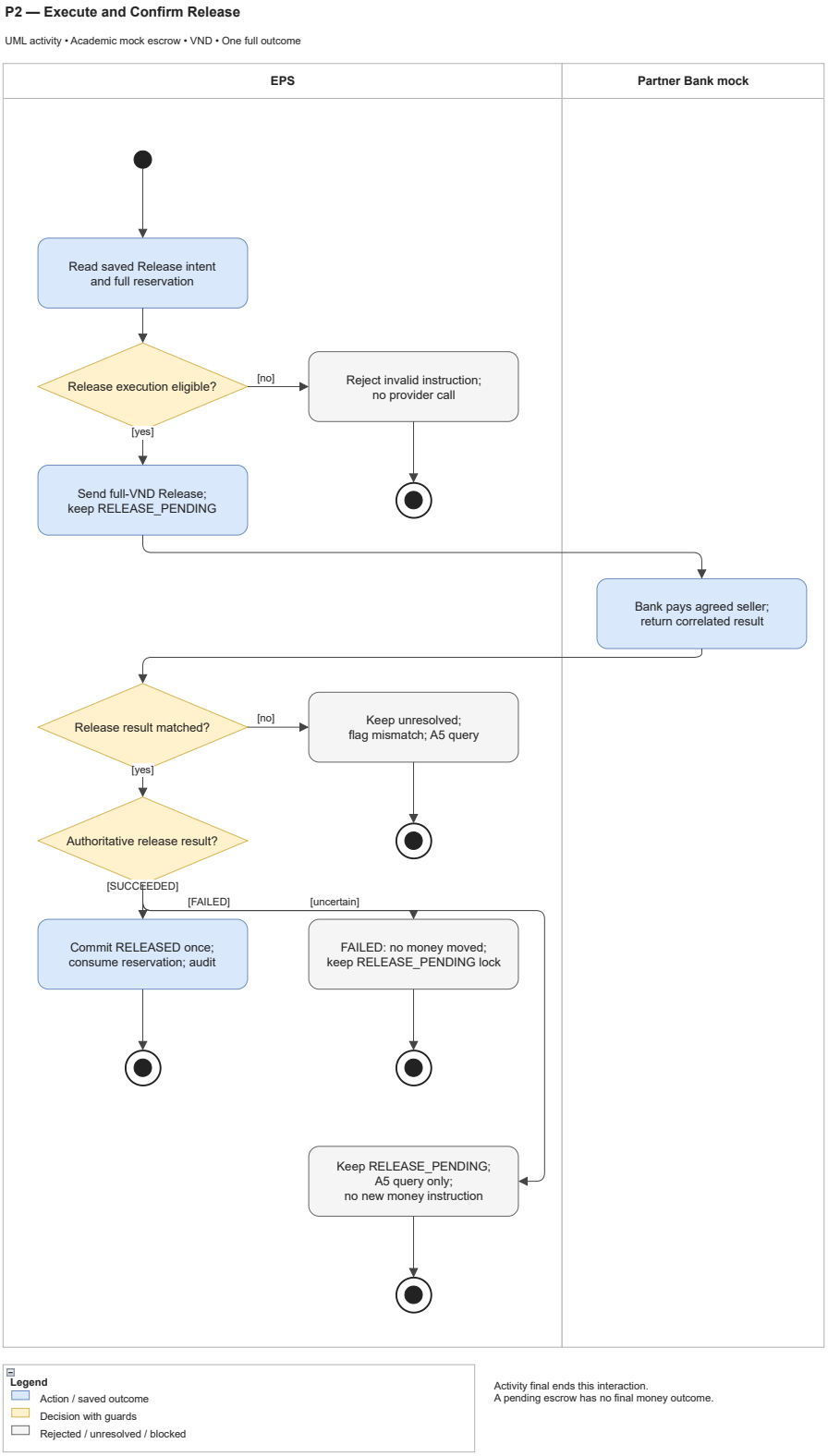


Figure 3.27: P2: Execute and Confirm Release

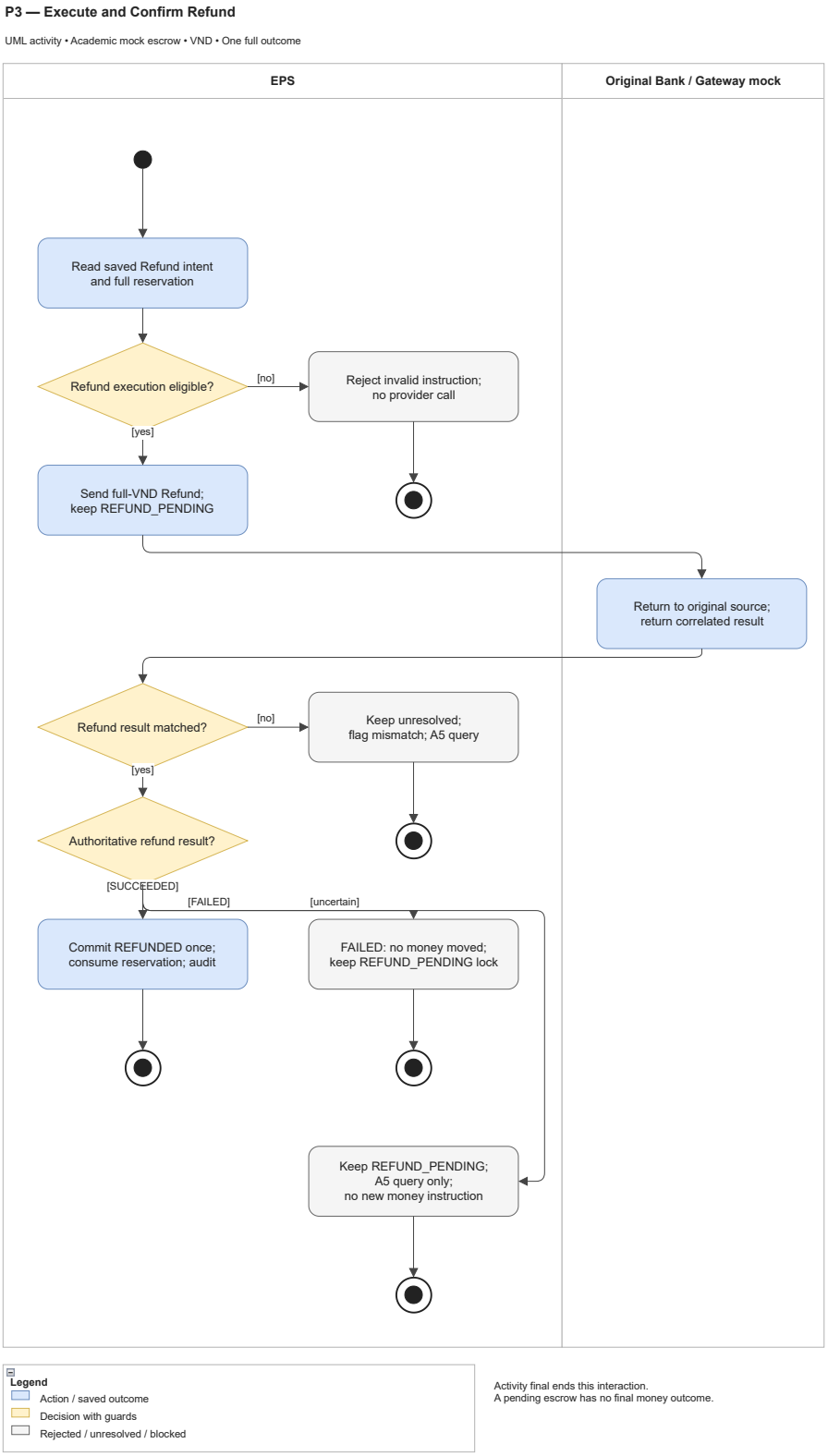


Figure 3.28: P3: Execute and Confirm Refund

3.4 Sequence Diagram

Sequence diagrams show actor, boundary, control and entity interactions. Large flows continue in the referenced modules; B1.API and U4.API use the same business rules as their user-interface variants.

3.4.1 User's Use cases

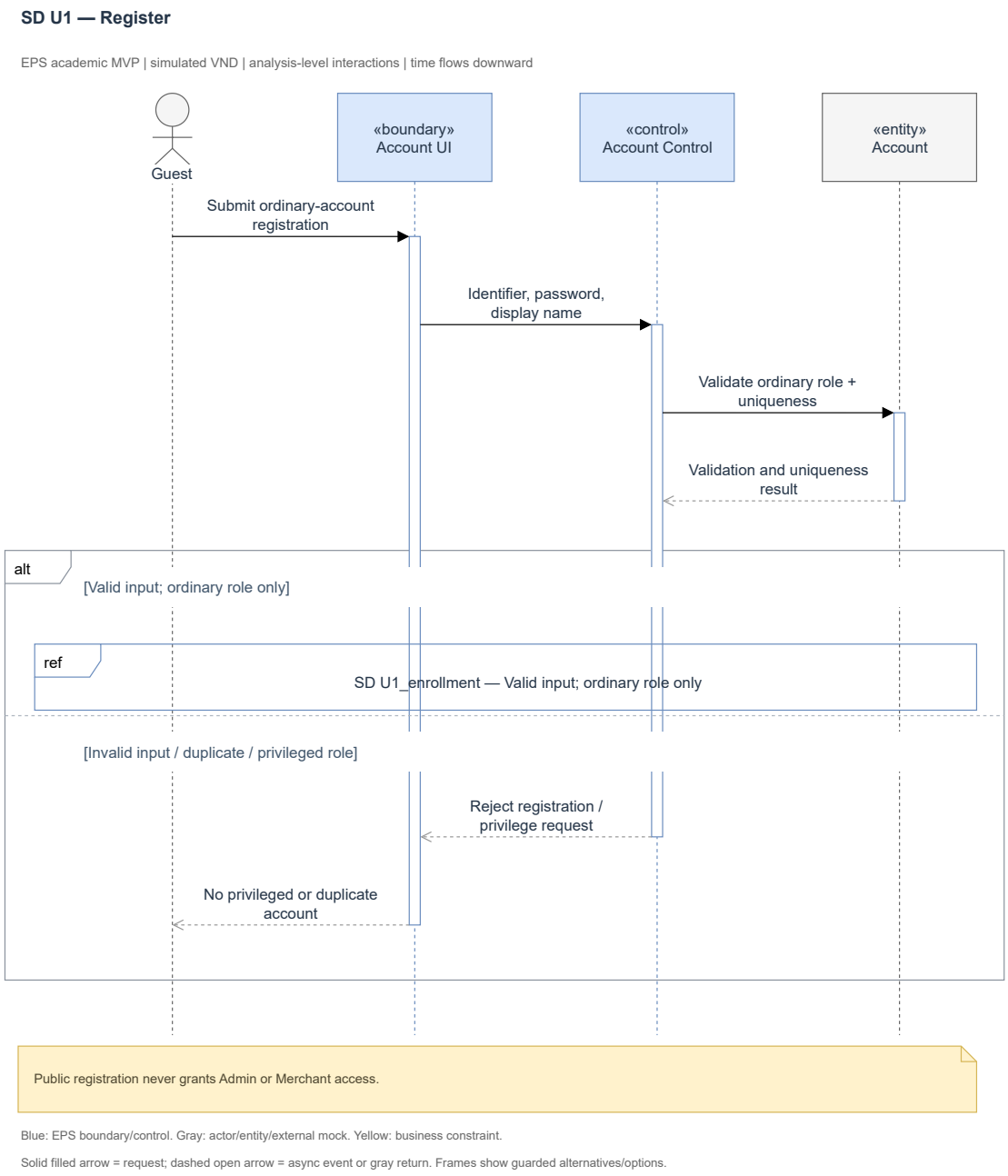


Figure 3.29: U1: Register

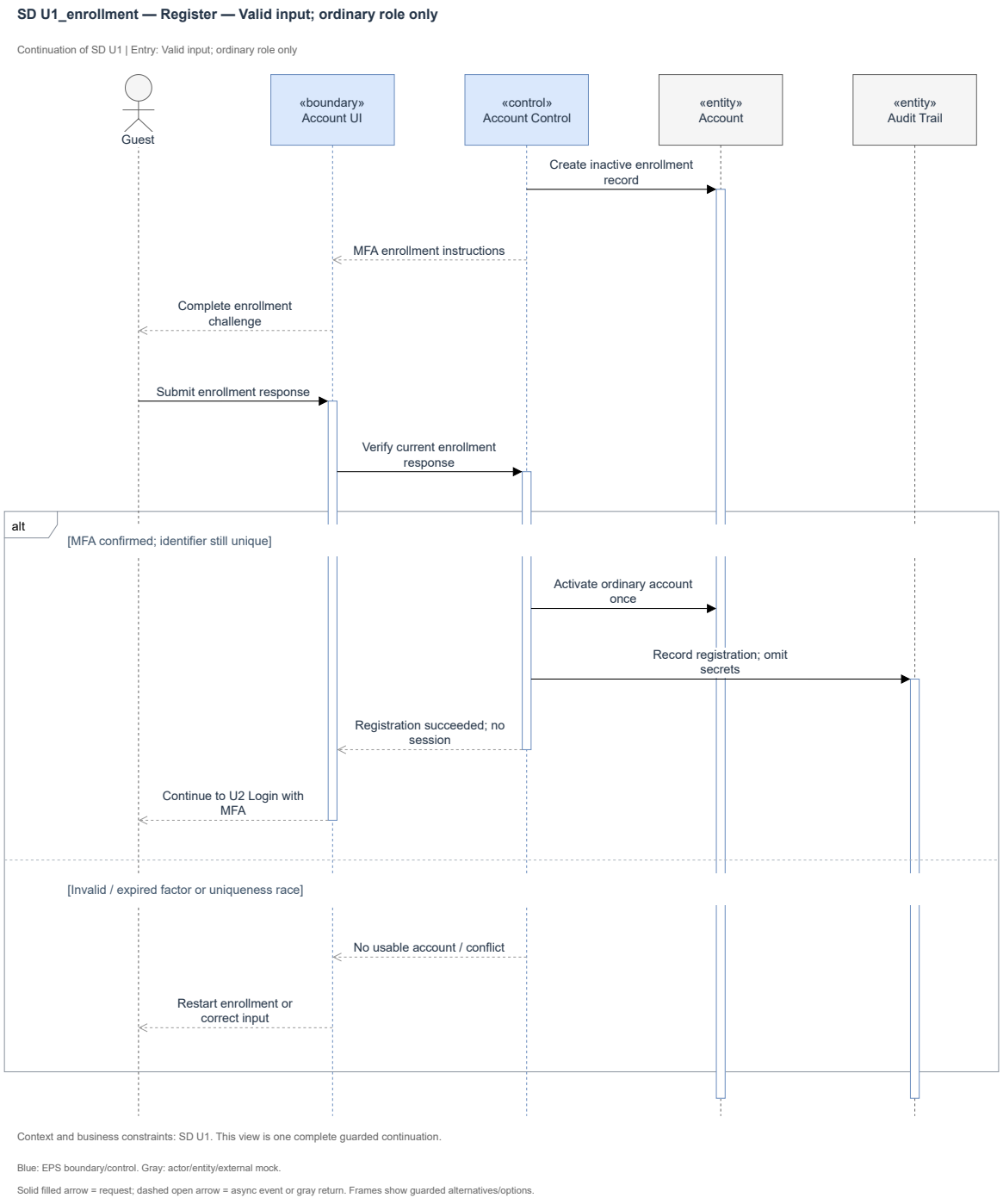


Figure 3.30: U1: Register (continued: U1_enrollment)

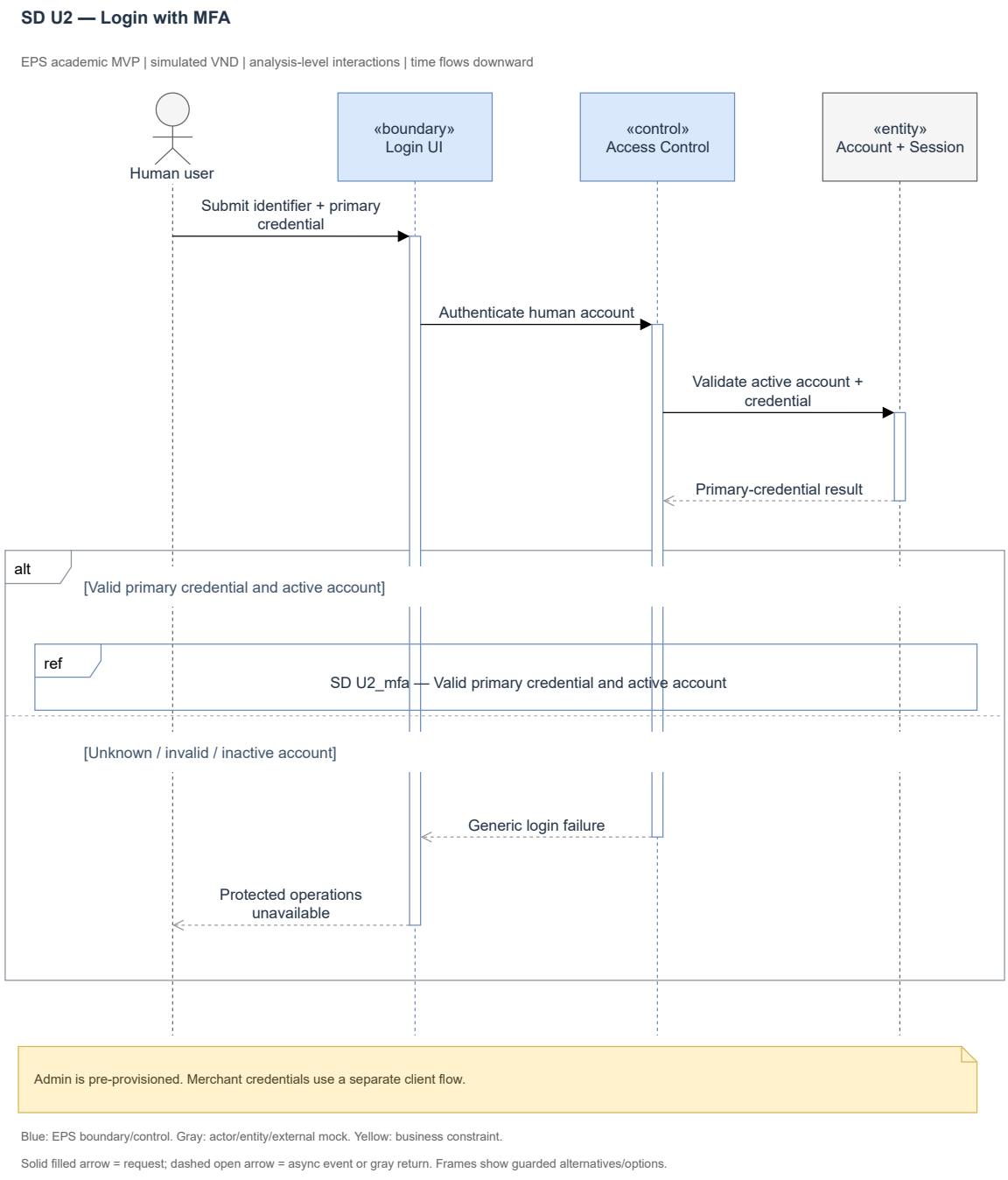


Figure 3.31: U2: Login with MFA

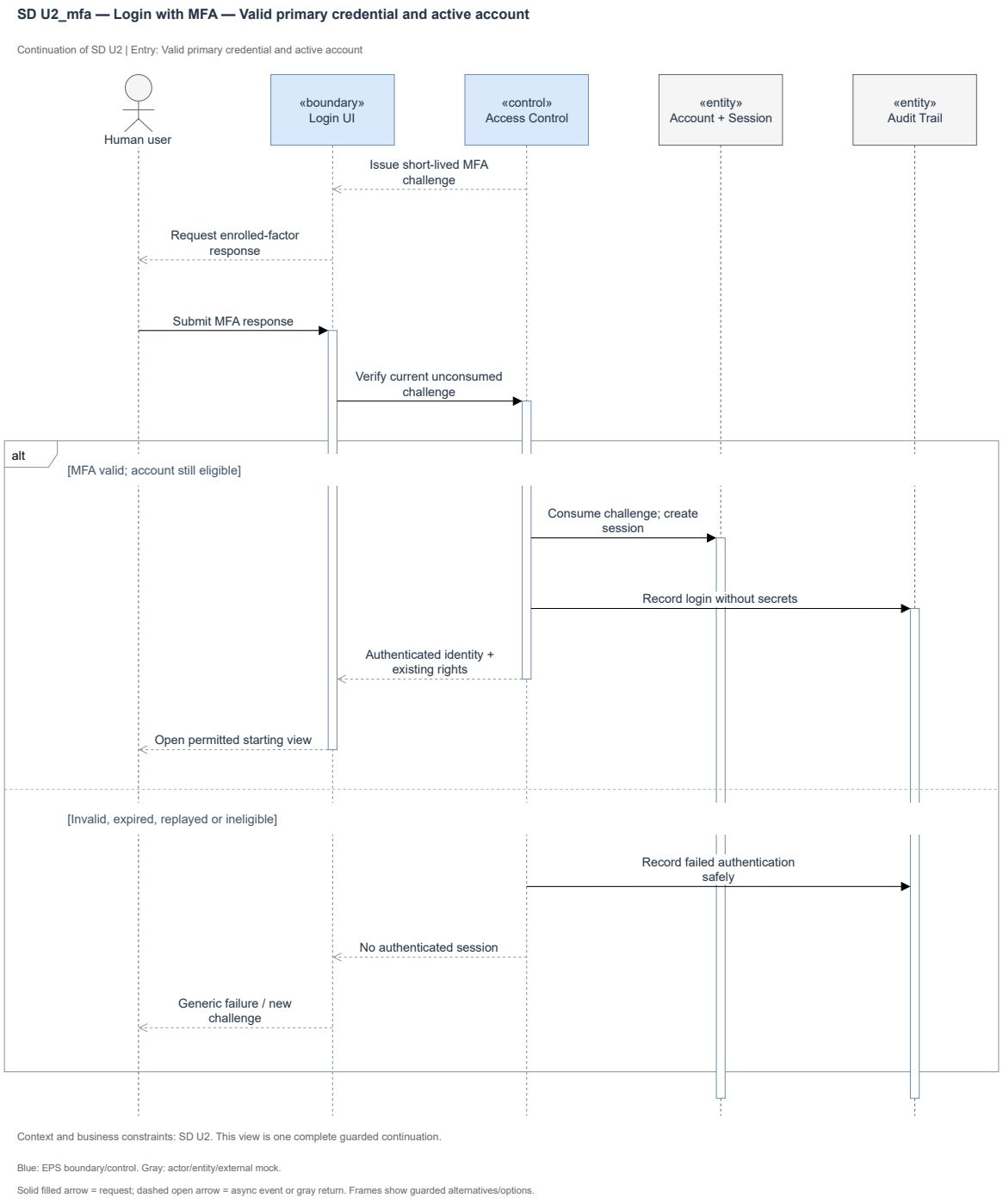


Figure 3.32: U2: Login with MFA (continued: U2_mfa)

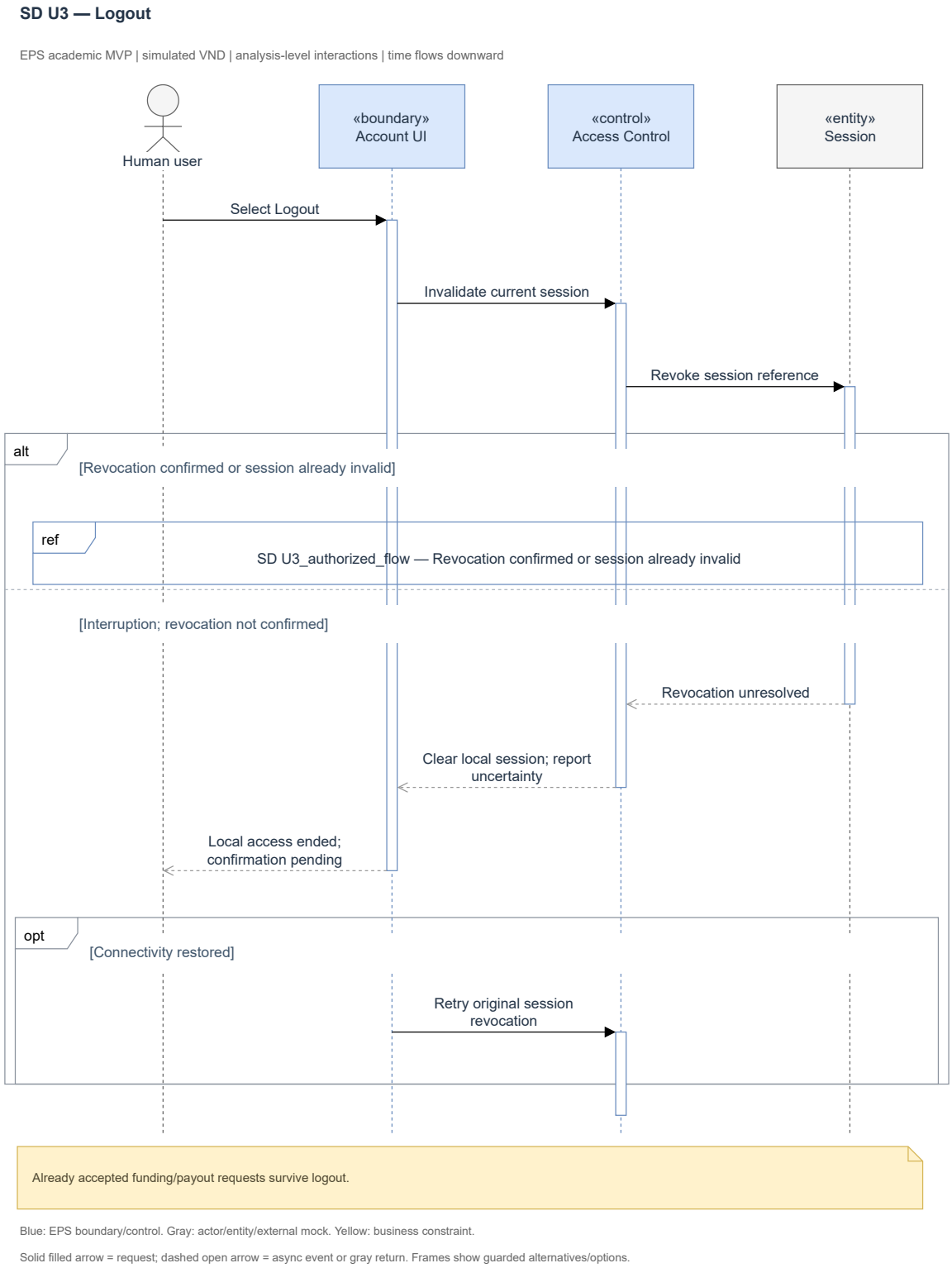


Figure 3.33: U3: Logout

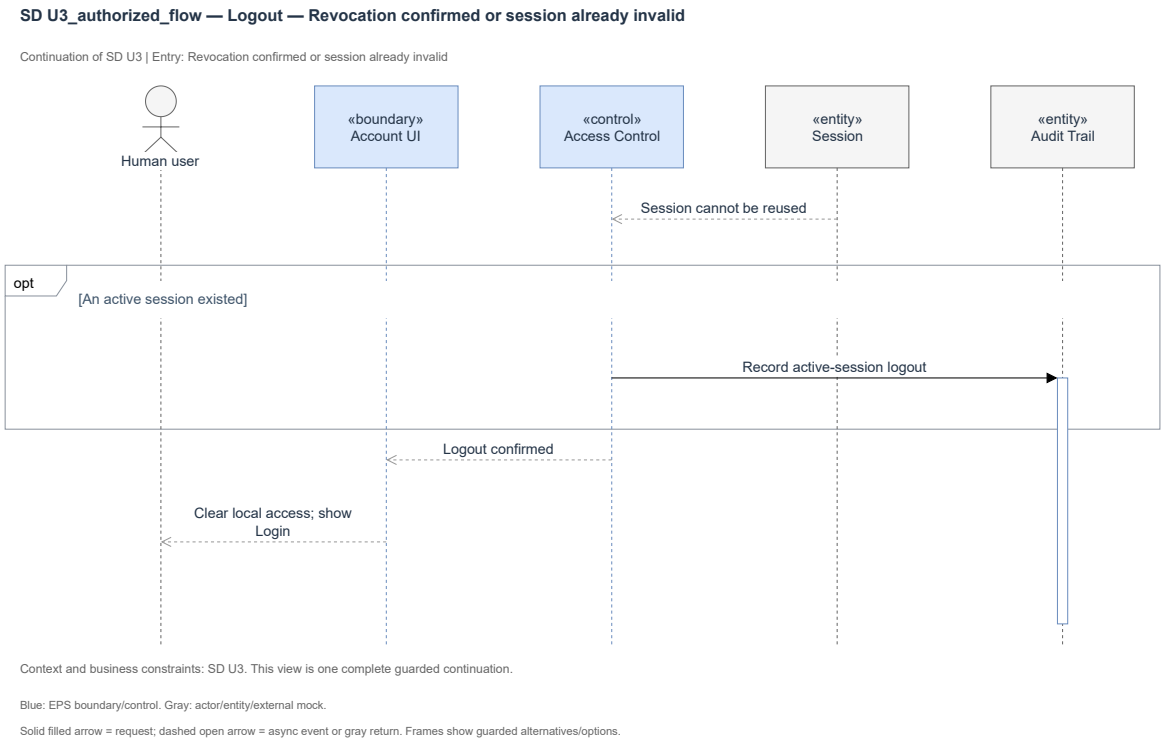


Figure 3.34: U3: Logout (continued: U3_authorized_flow)

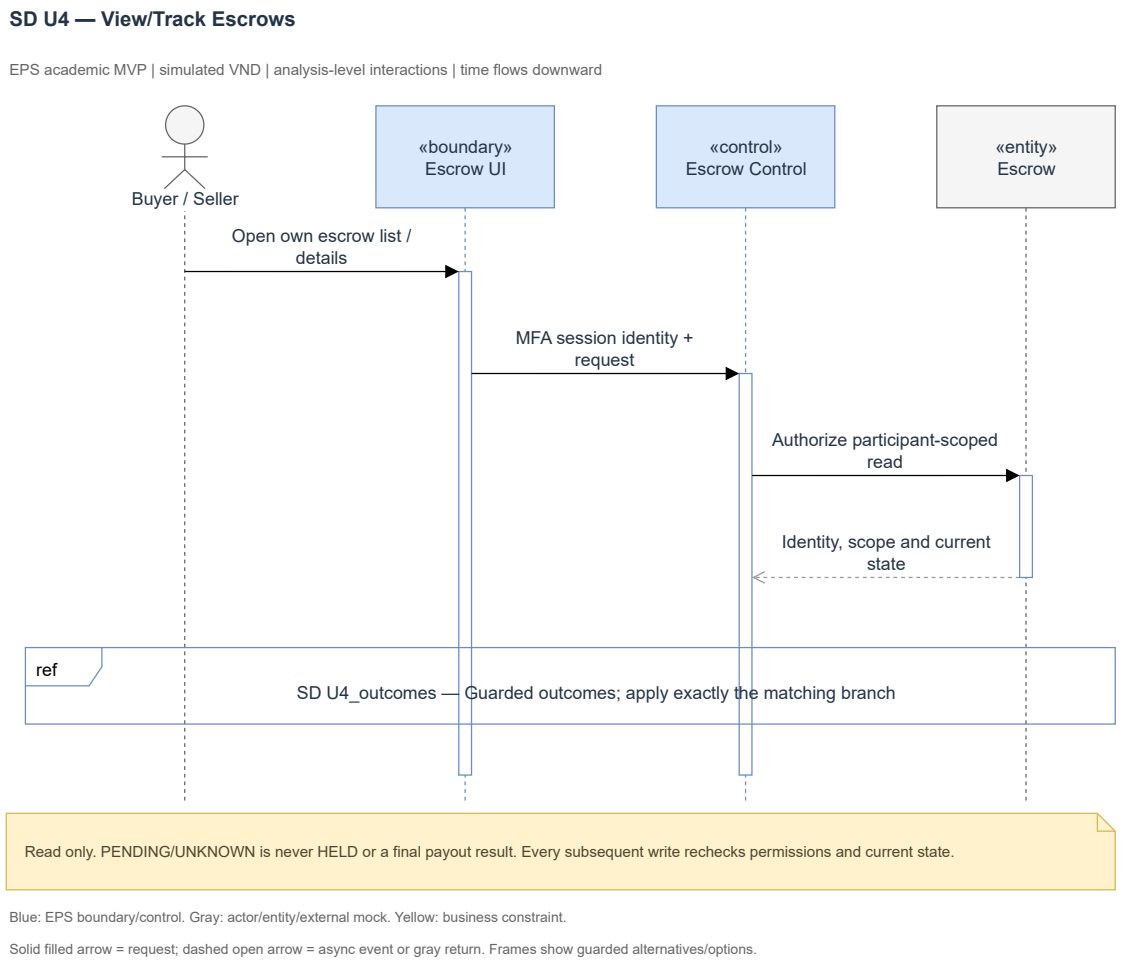


Figure 3.35: U4: View/Track Escrows

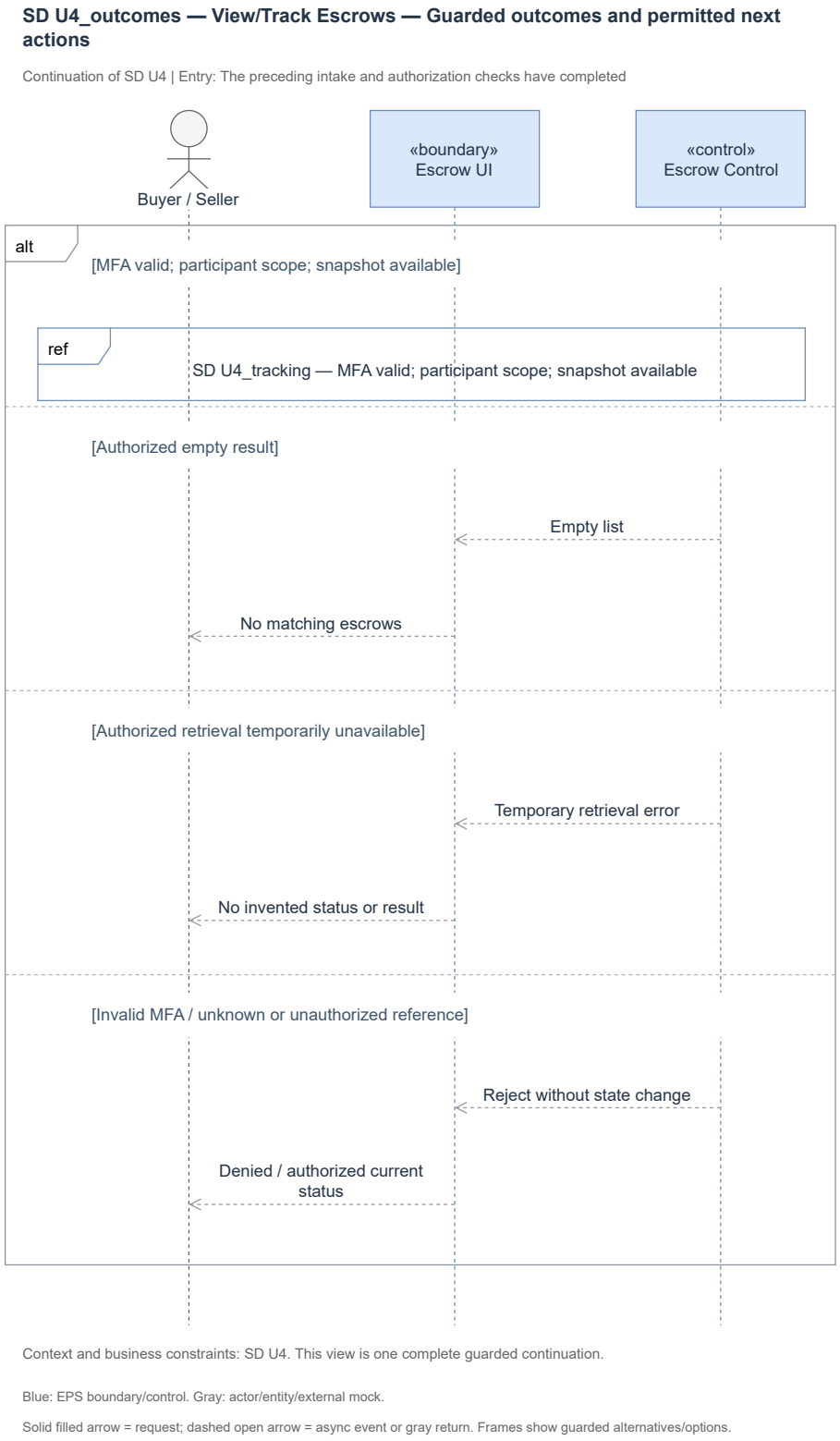


Figure 3.36: U4: View/Track Escrows (continued: U4_outcomes)

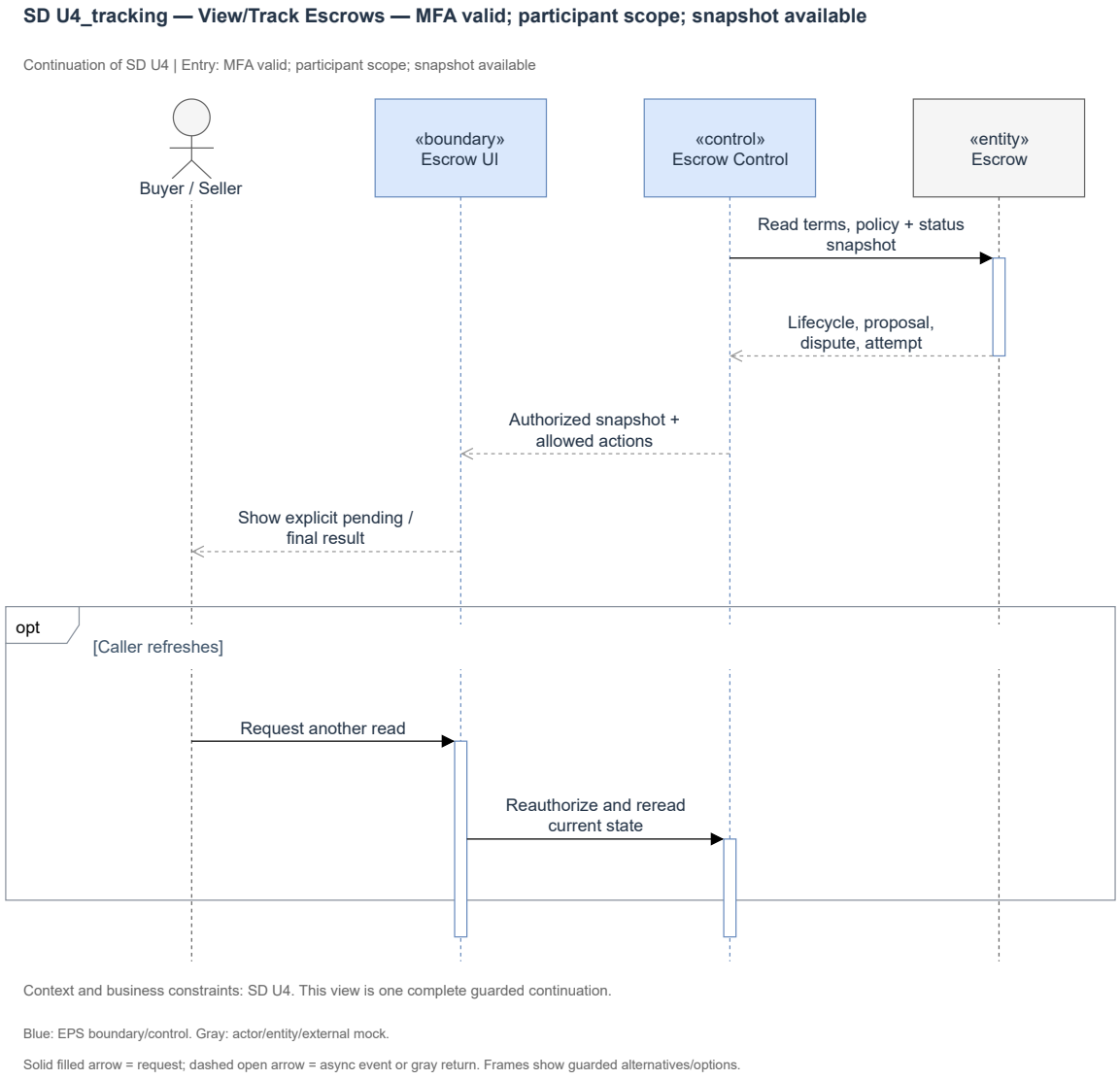


Figure 3.37: U4: View/Track Escrows (continued: U4_tracking)

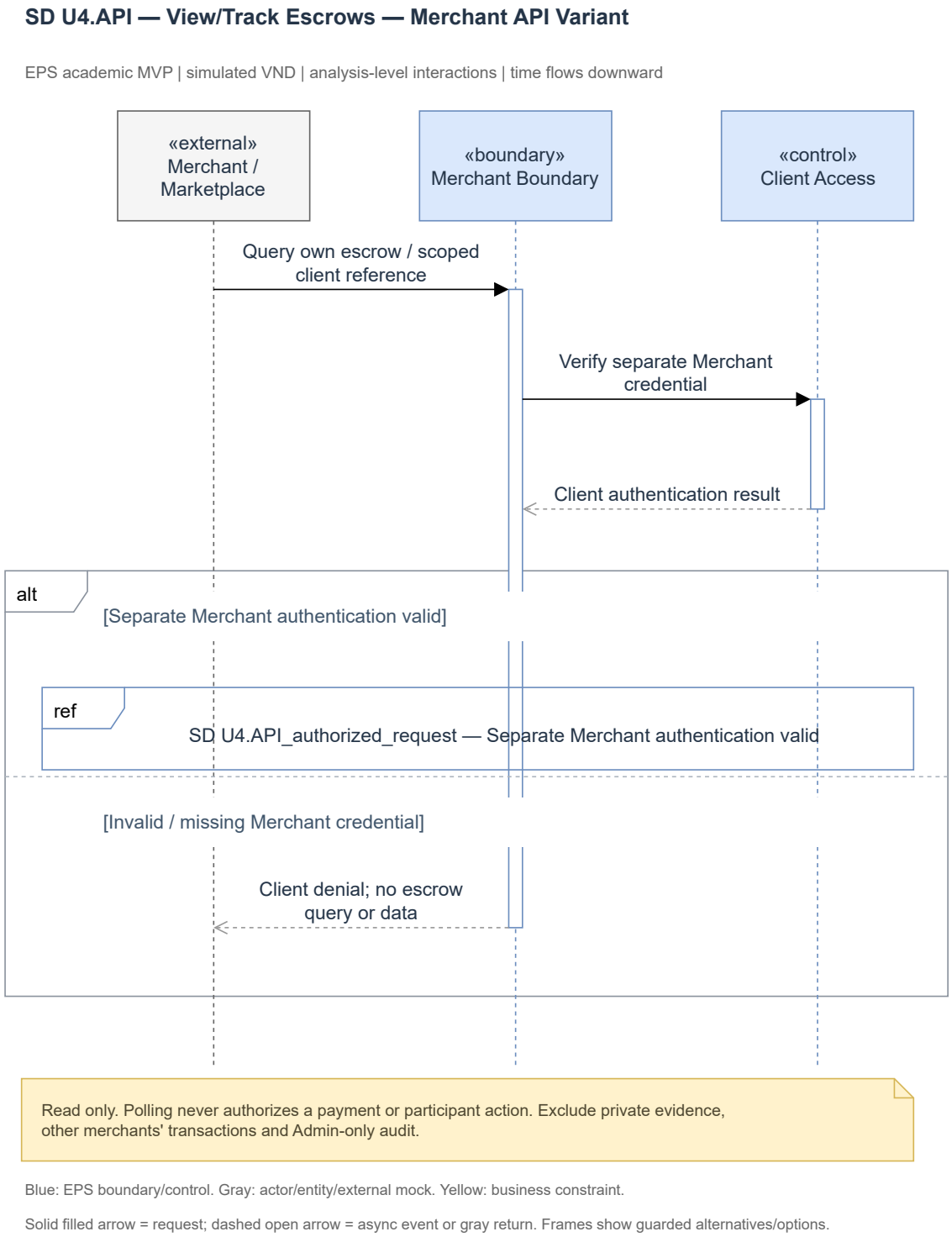


Figure 3.38: U4.API: View/Track Escrows

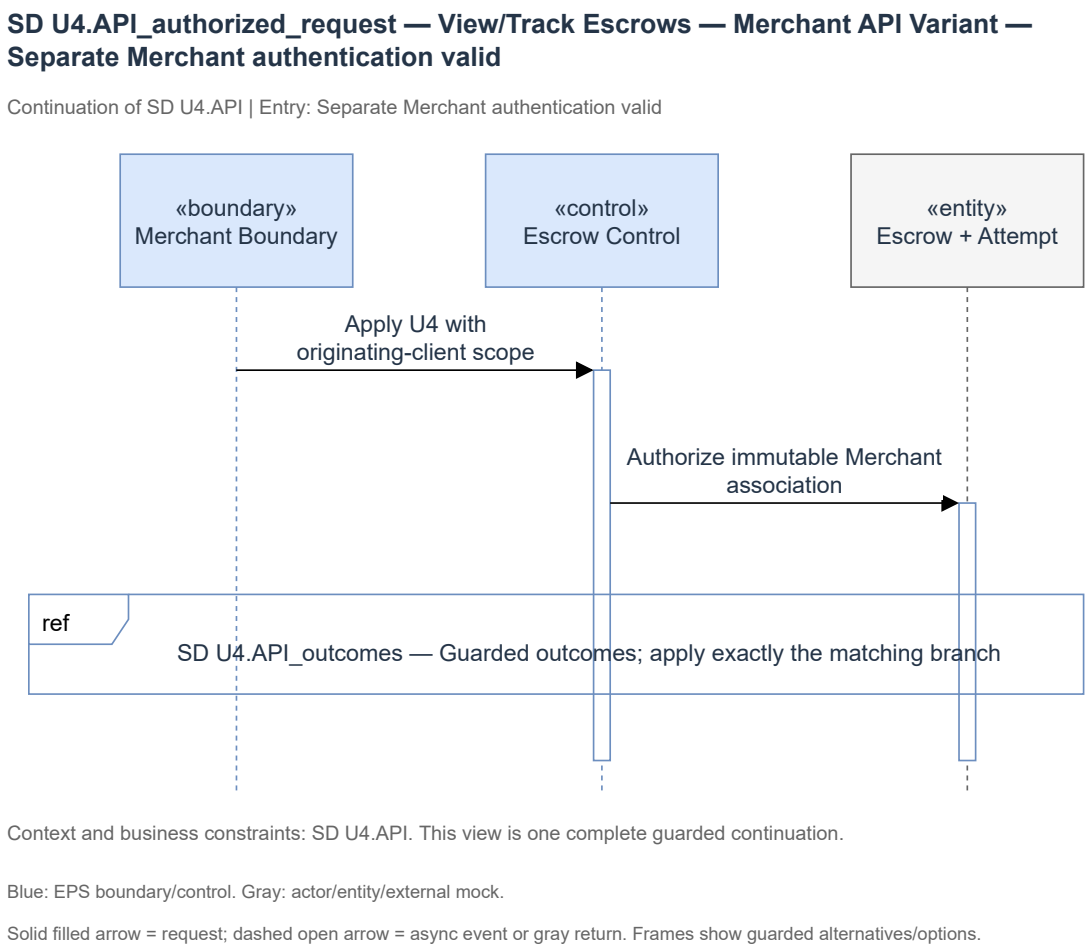


Figure 3.39: U4.API: View/Track Escrows (continued: U4.API_authorized_request)

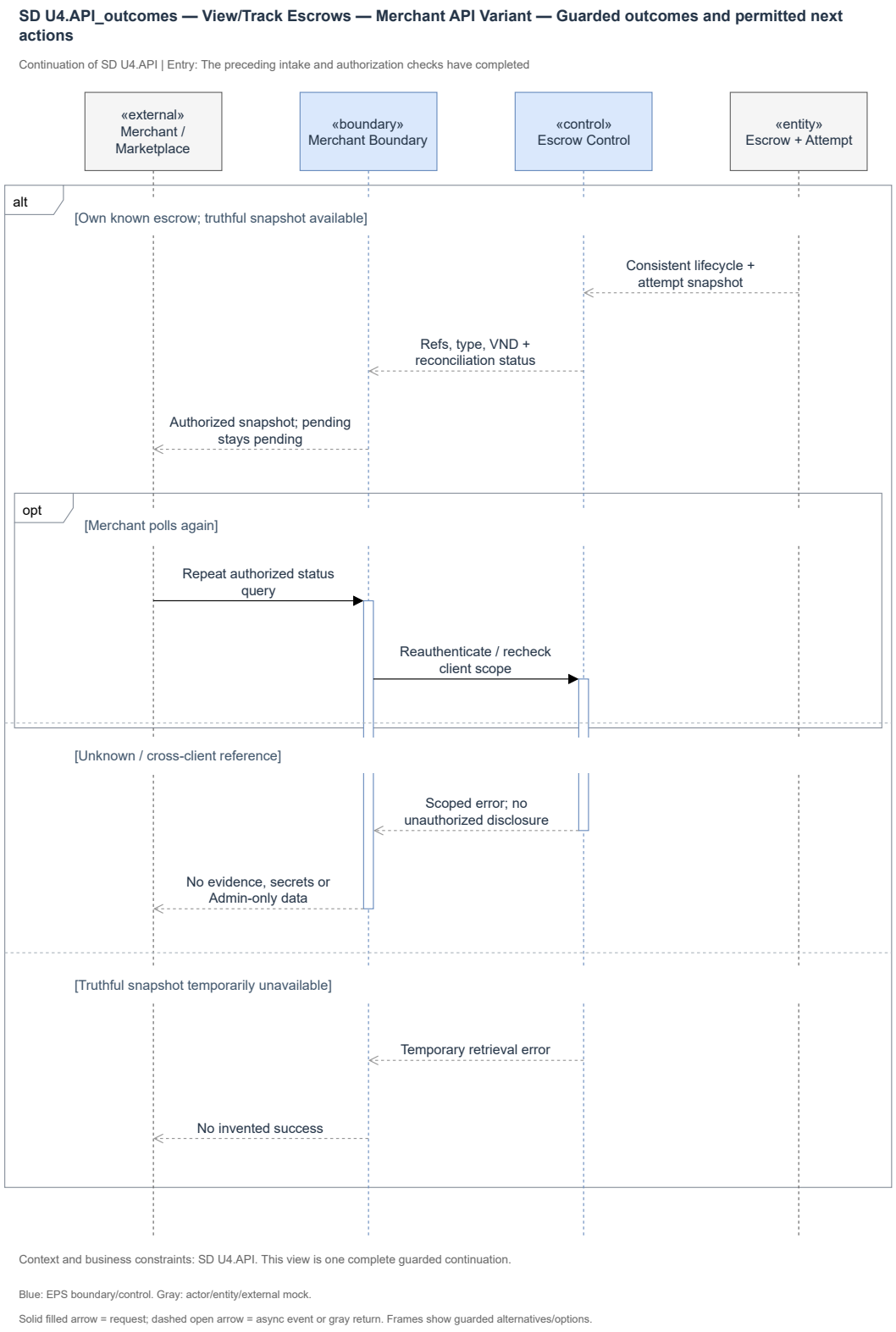


Figure 3.40: U4.API: View/Track Escrows (continued: U4.API_outcomes)

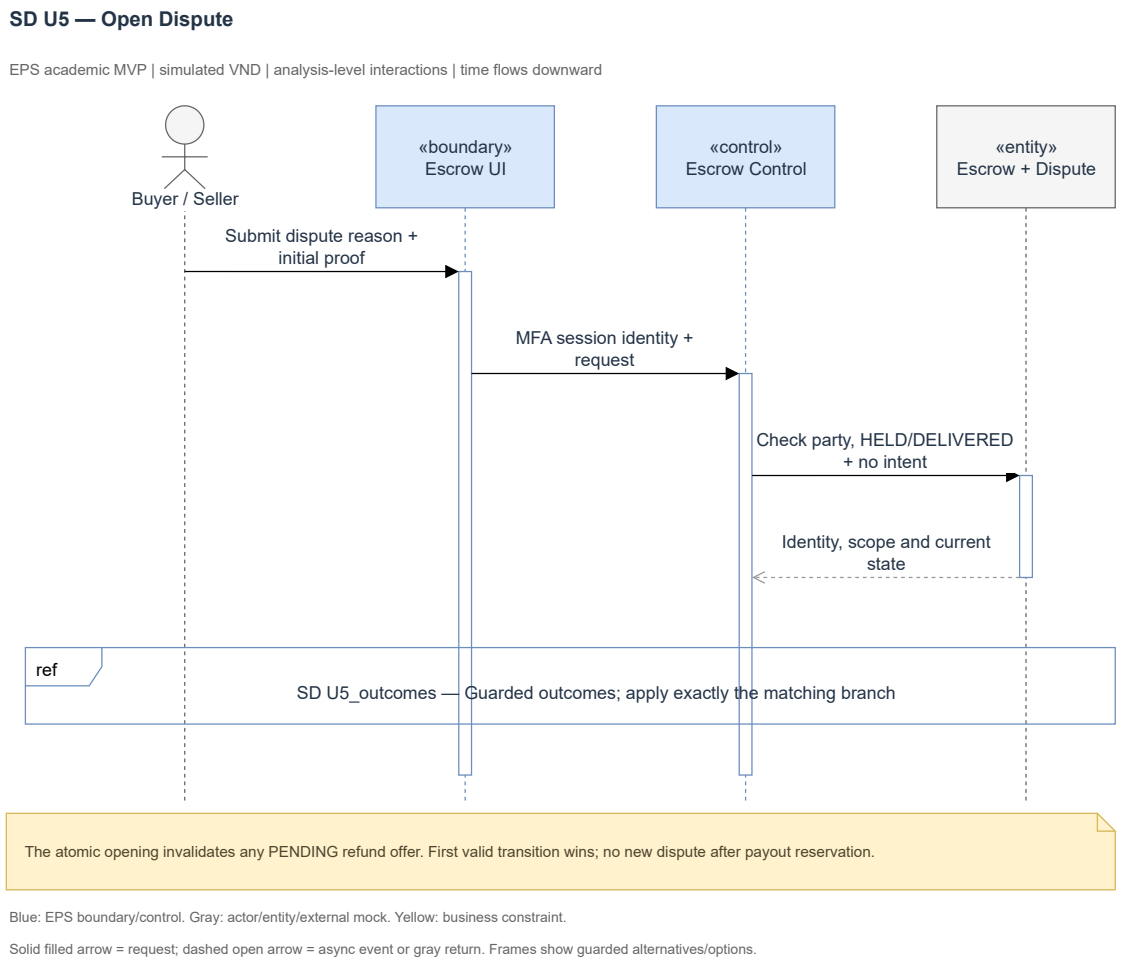


Figure 3.41: U5: Open Dispute

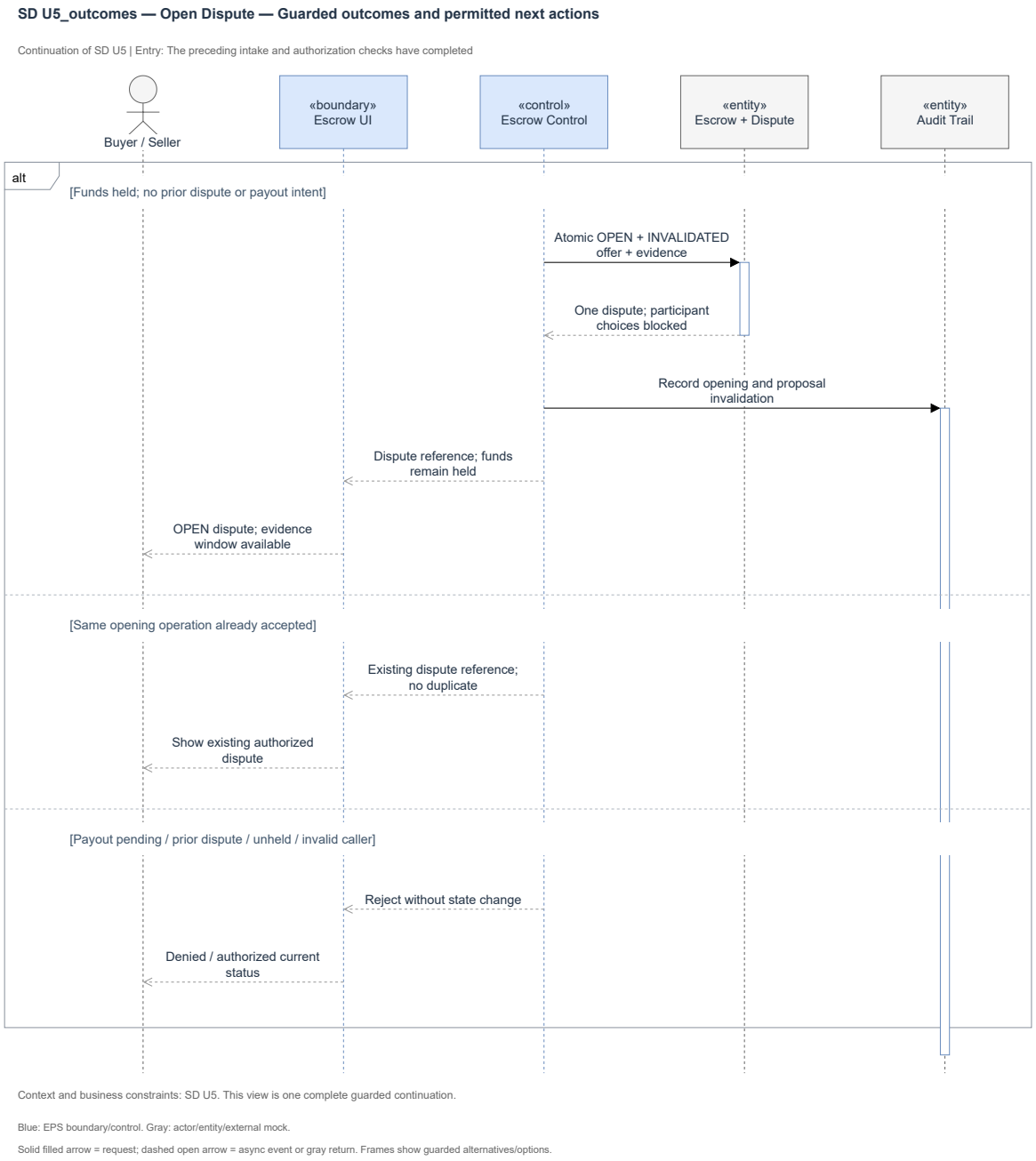


Figure 3.42: U5: Open Dispute (continued: U5_outcomes)

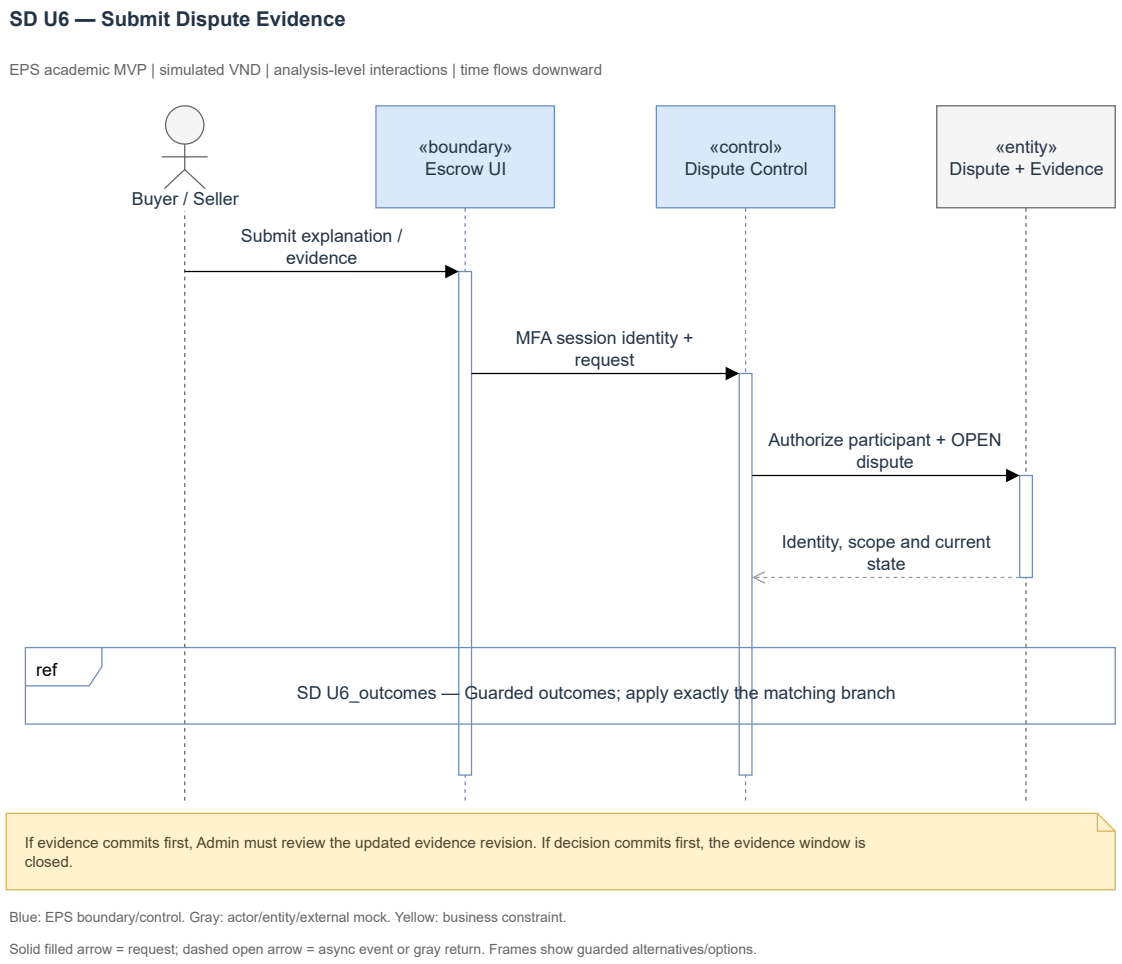


Figure 3.43: U6: Submit Dispute Evidence

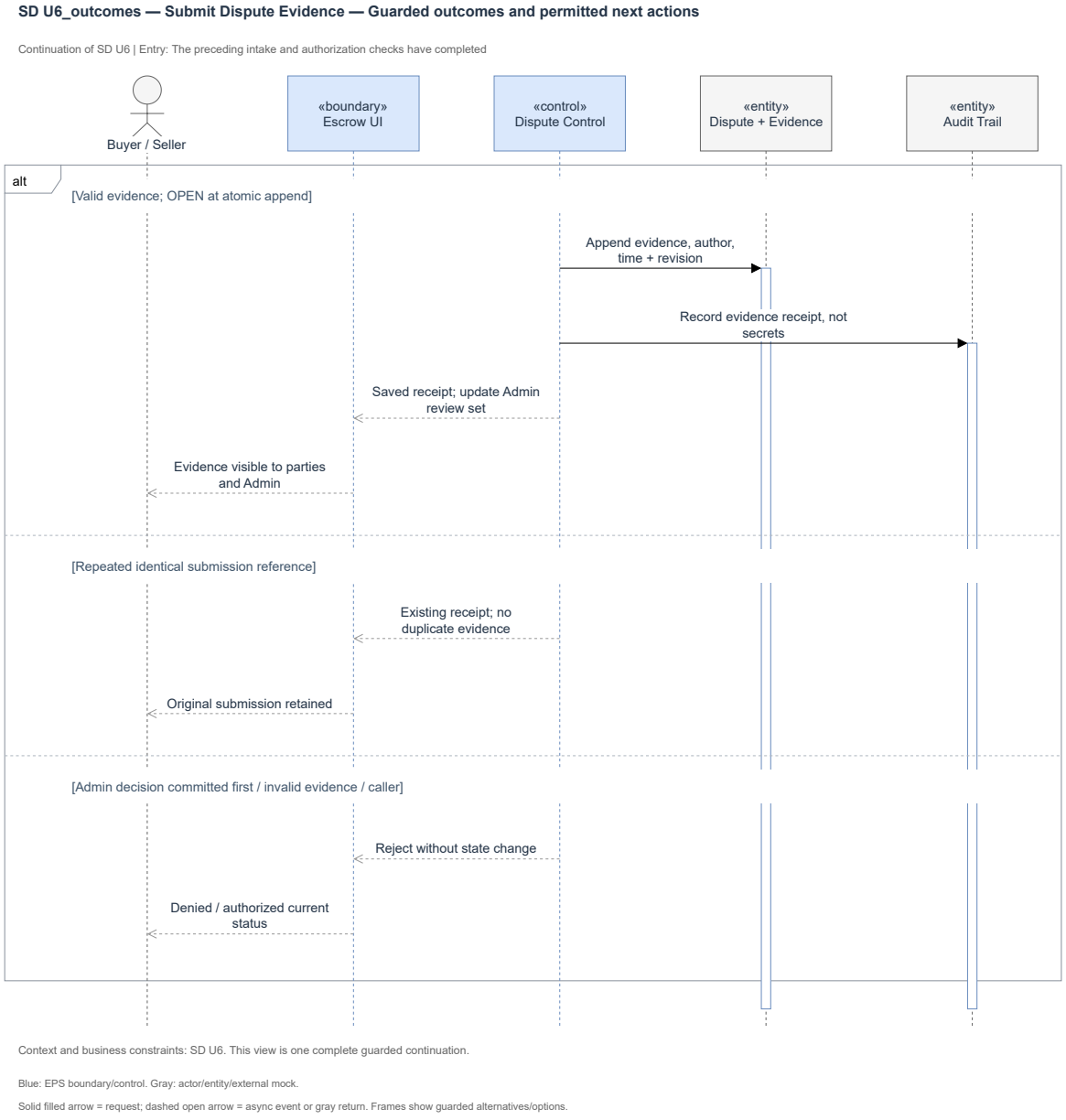


Figure 3.44: U6: Submit Dispute Evidence (continued: U6_outcomes)

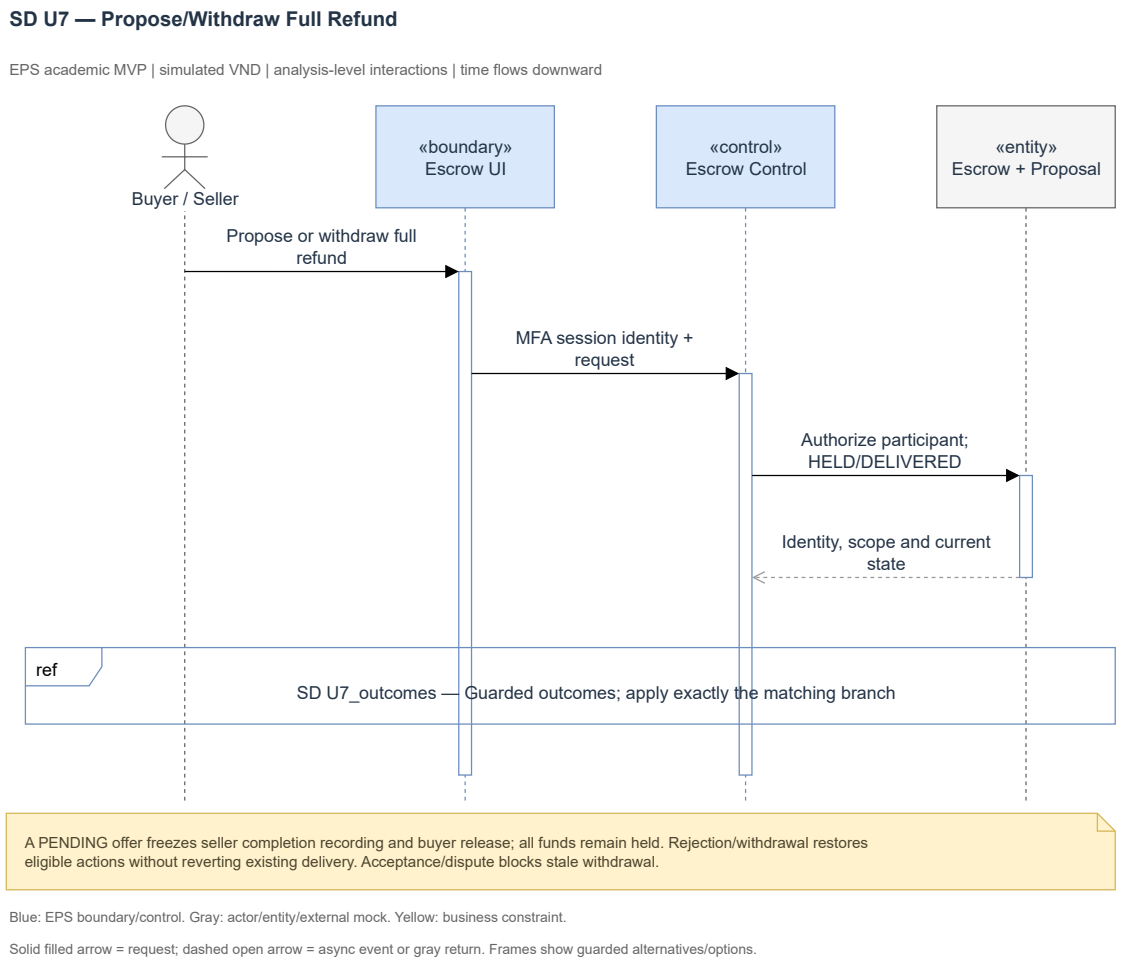


Figure 3.45: U7: Propose/Withdraw Full Refund

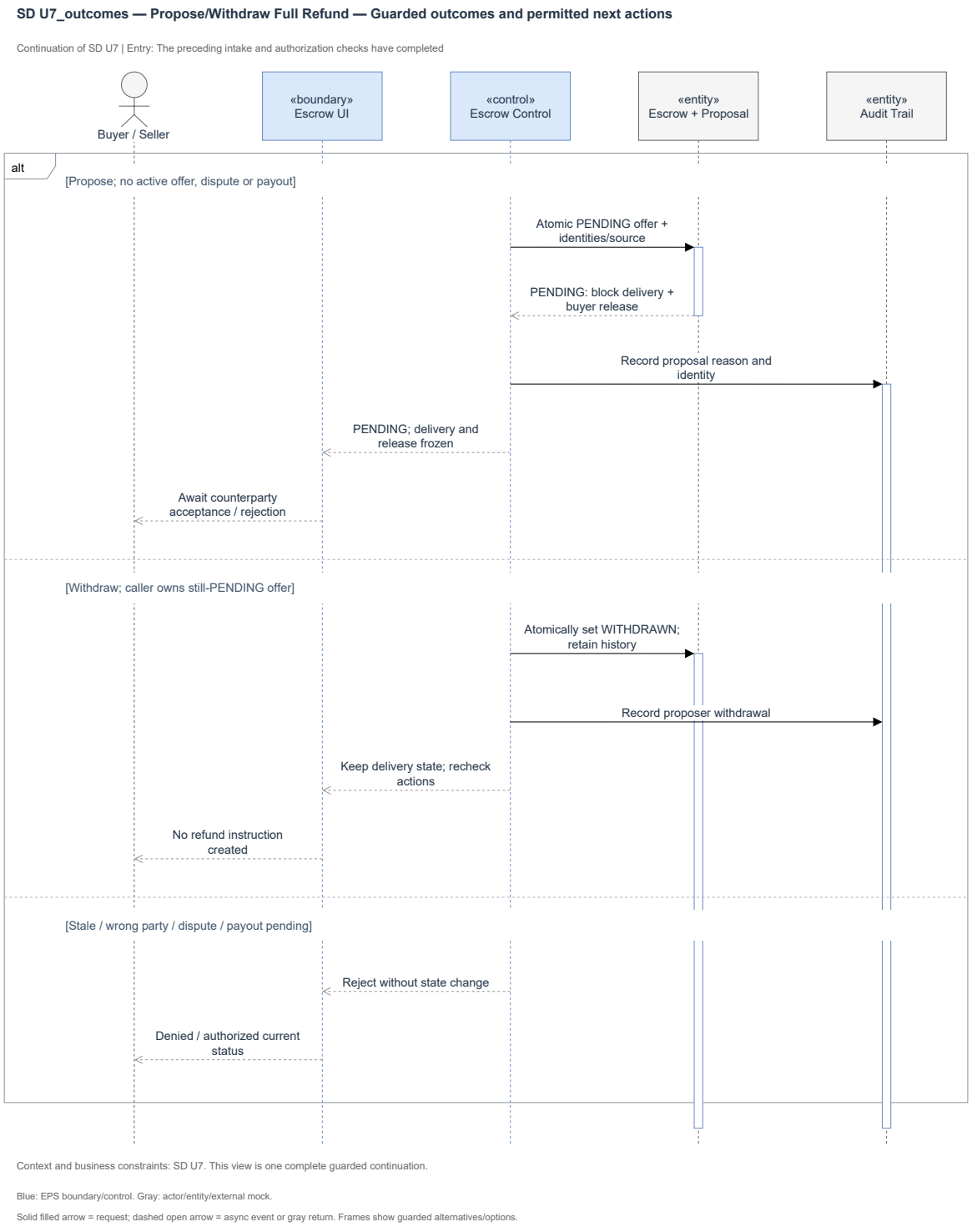


Figure 3.46: U7: Propose/Withdraw Full Refund (continued: U7_outcomes)

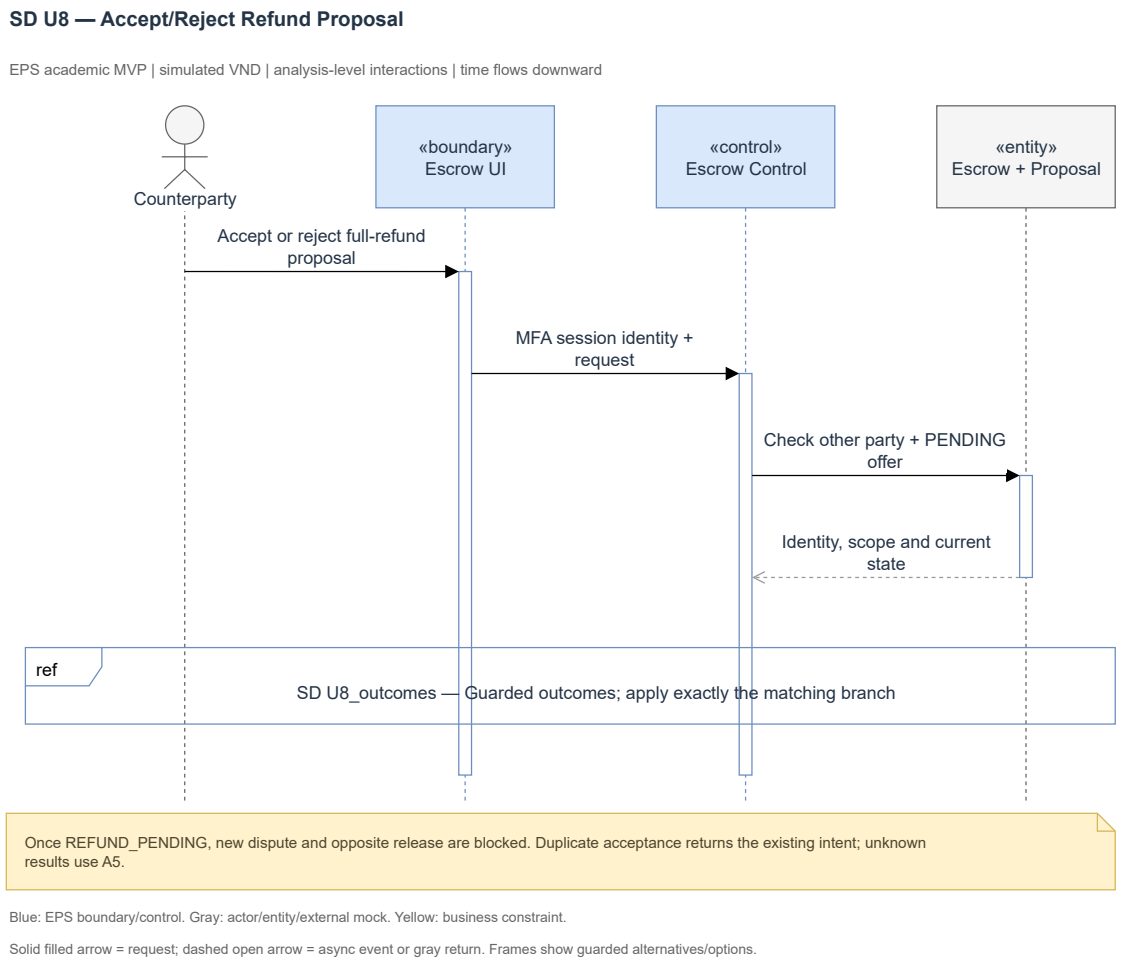


Figure 3.47: U8: Accept/Reject Refund Proposal

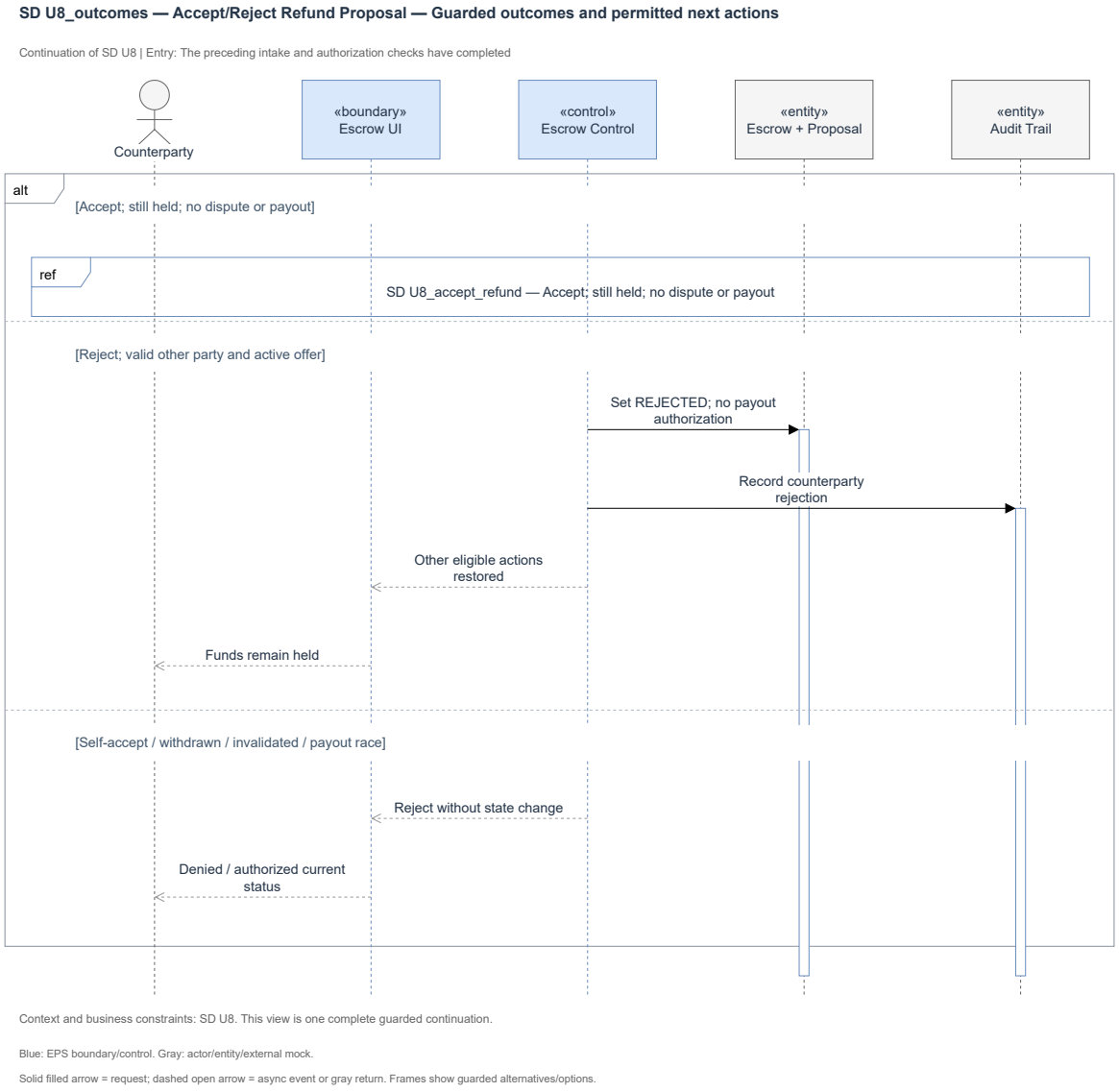


Figure 3.48: U8: Accept/Reject Refund Proposal (continued: U8_outcomes)

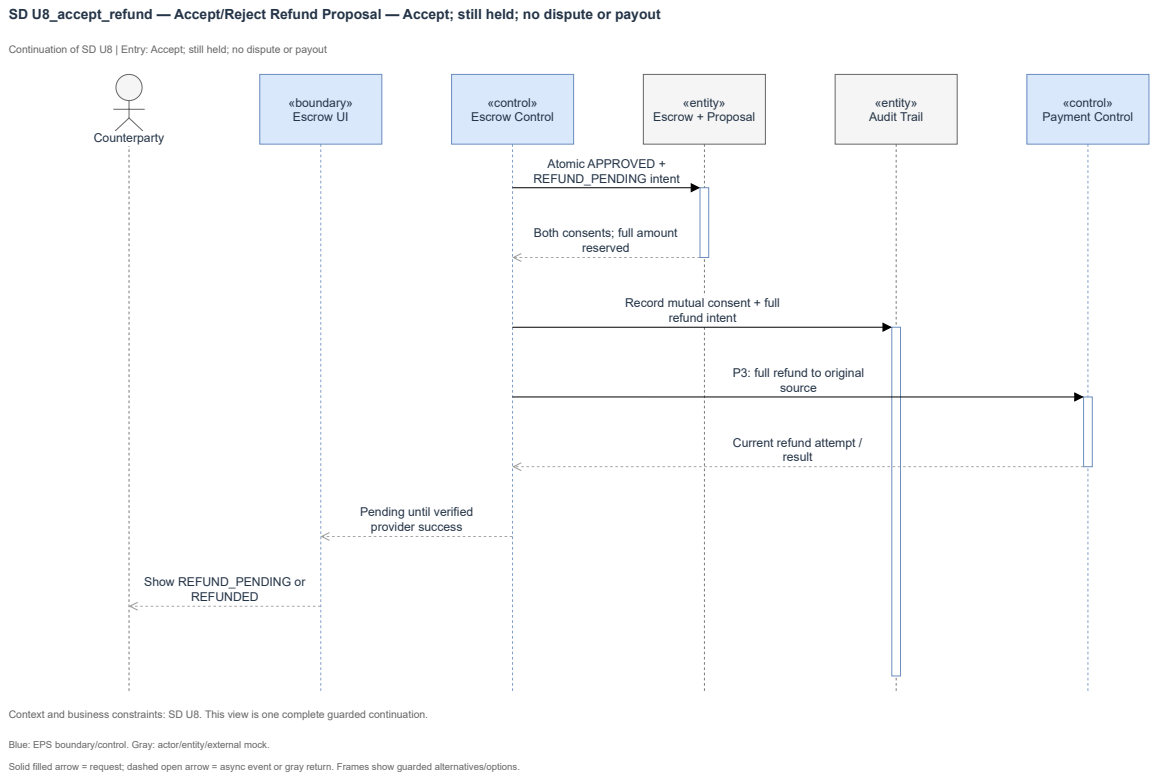


Figure 3.49: U8: Accept/Reject Refund Proposal (continued: U8_accept_refund)

3.4.2 Buyer's Use cases

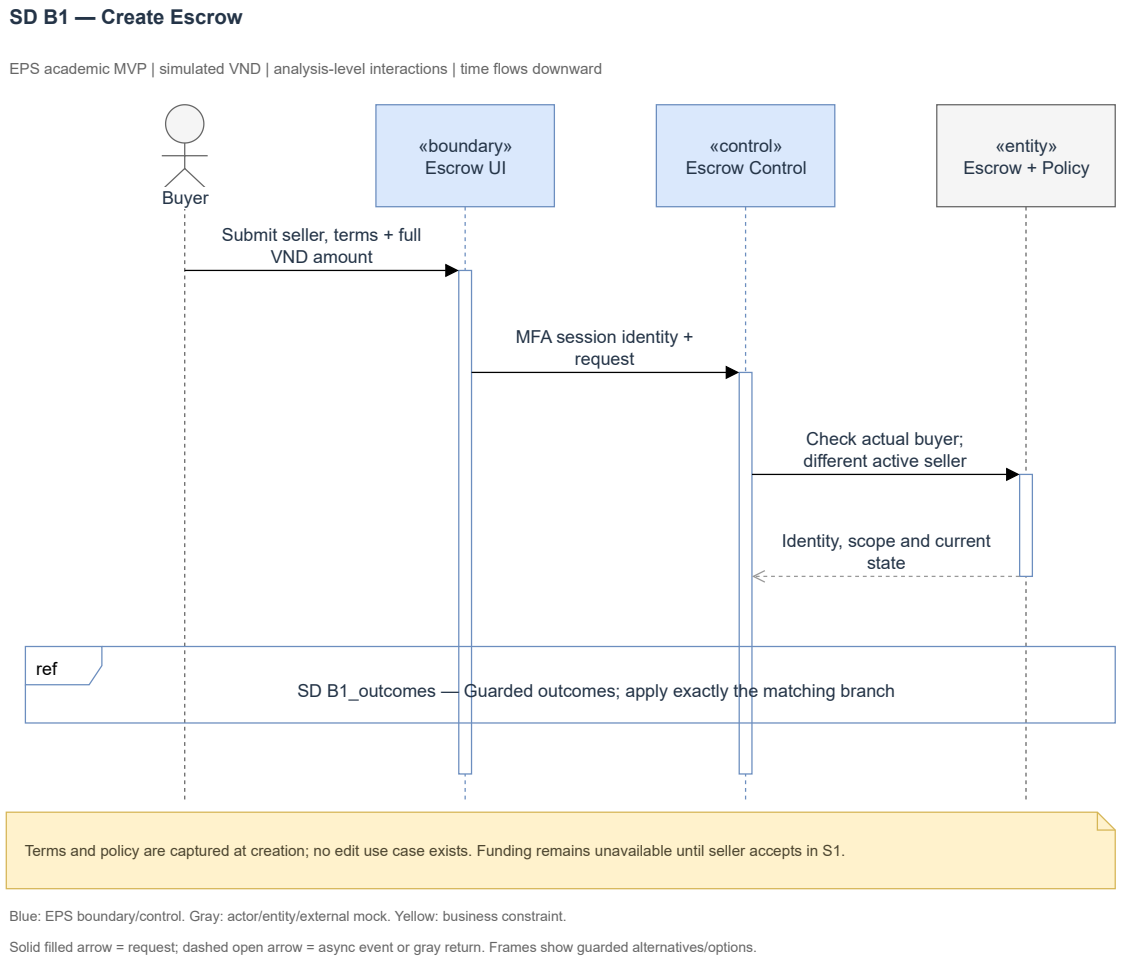


Figure 3.50: B1: Create Escrow

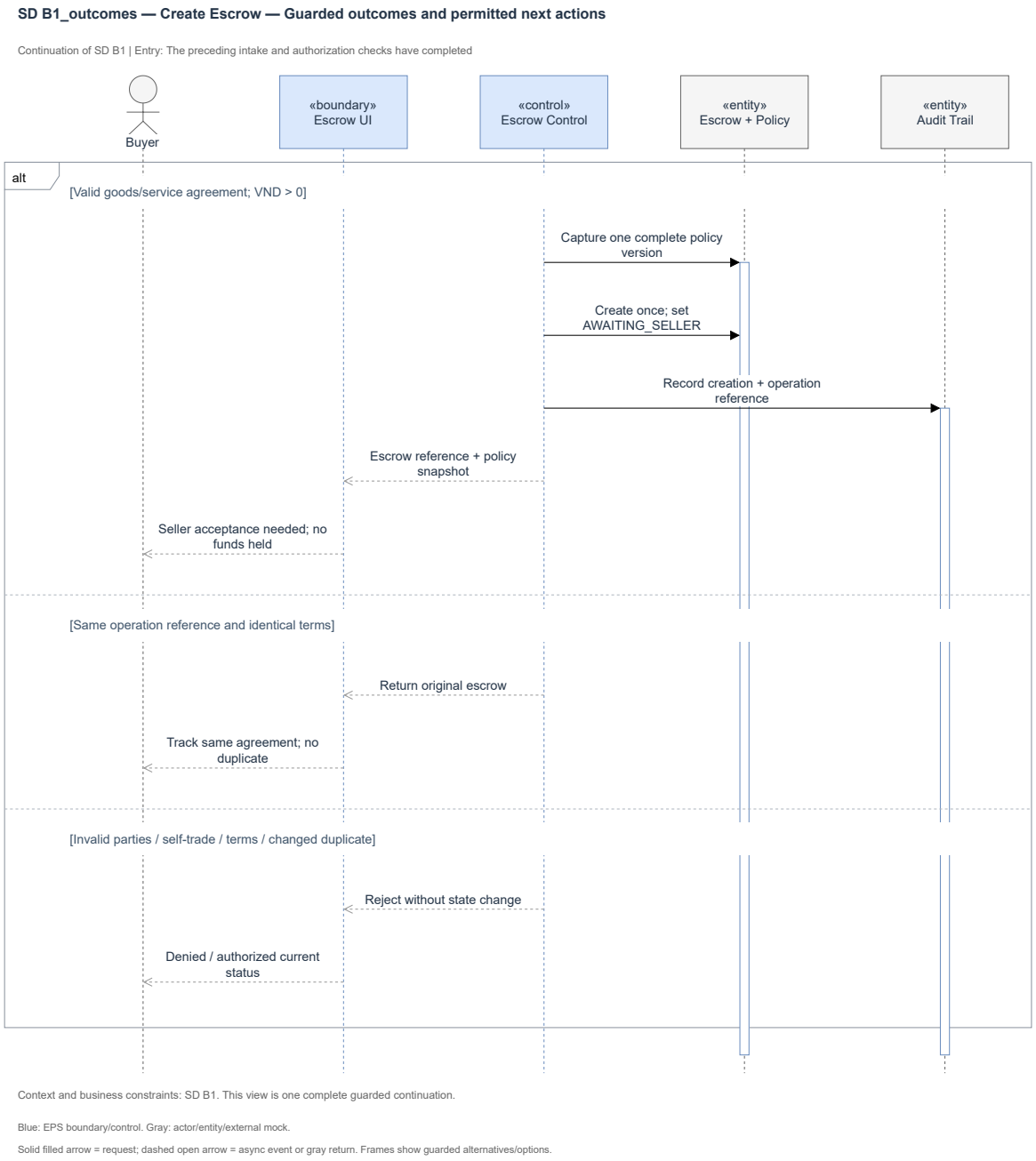


Figure 3.51: B1: Create Escrow (continued: B1_outcomes)

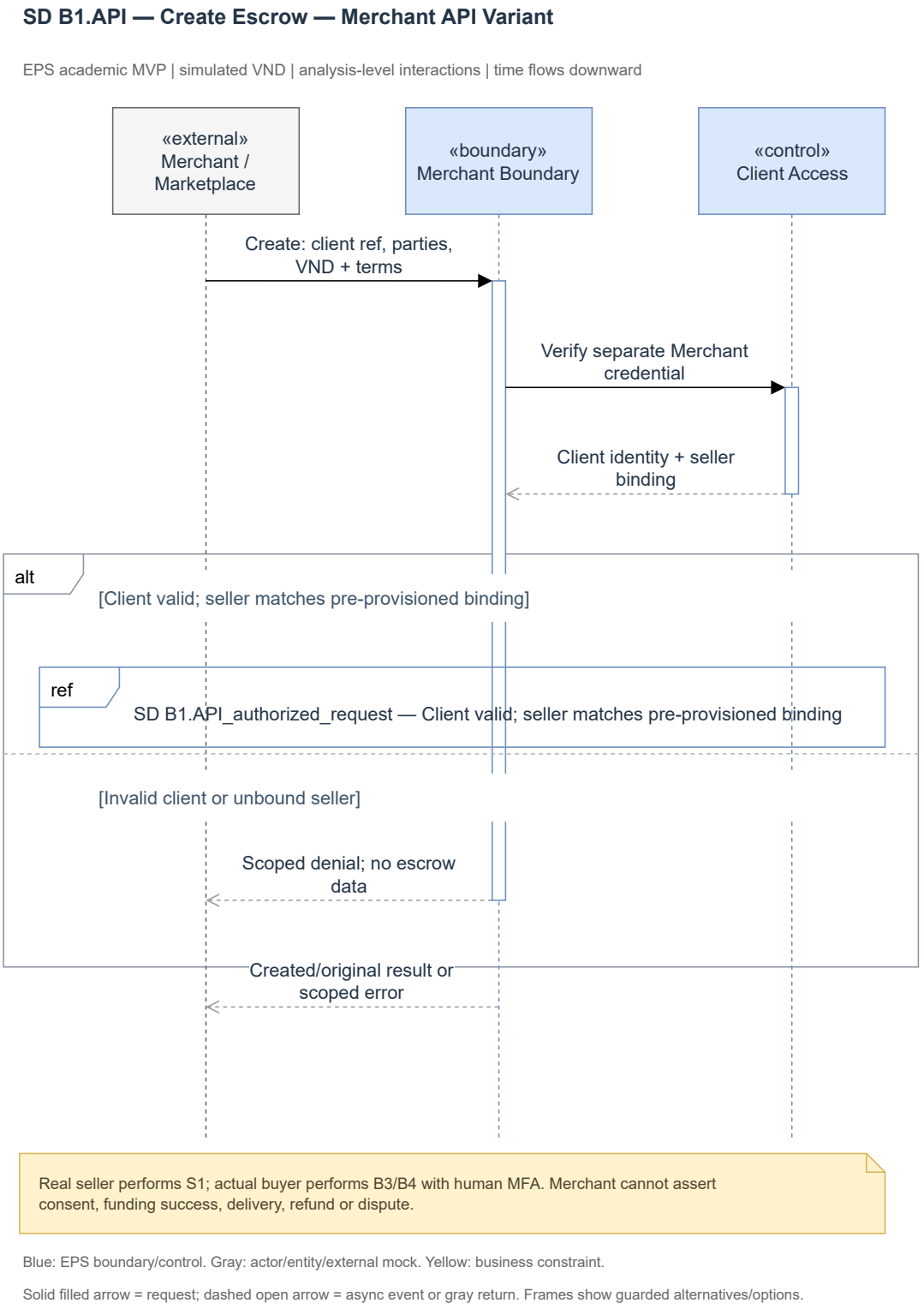


Figure 3.52: B1.API: Create Escrow

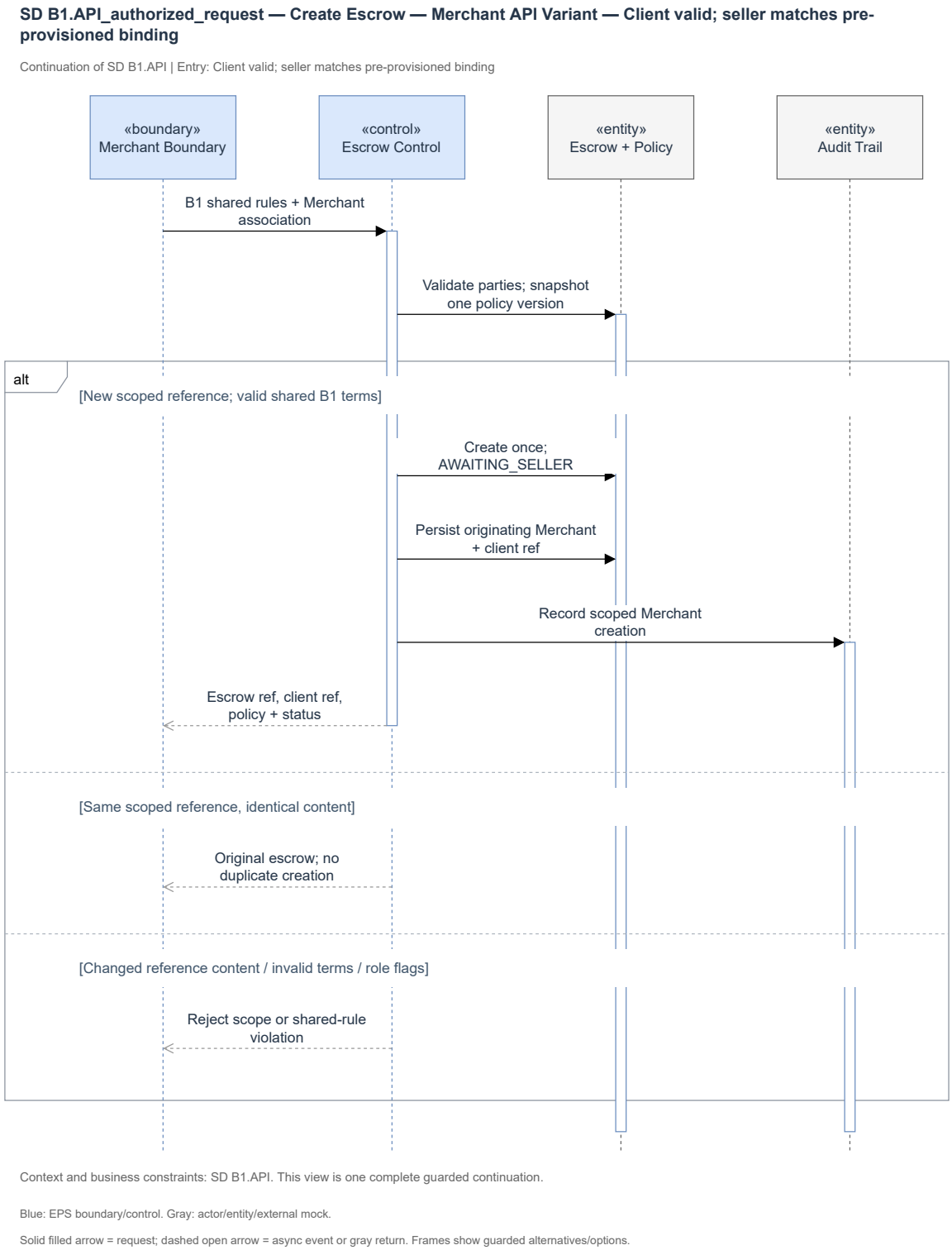


Figure 3.53: B1.API: Create Escrow (continued: B1.API_authorized_request)

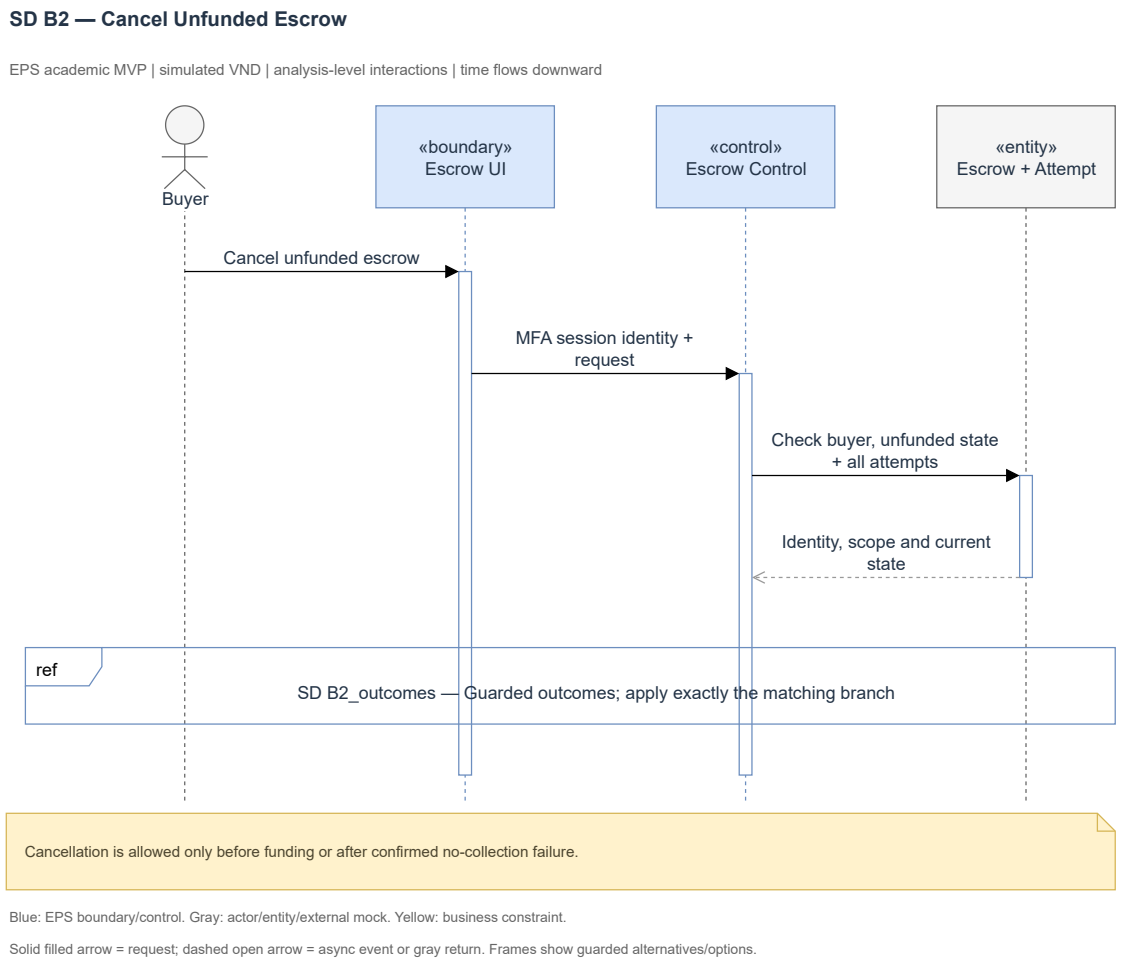


Figure 3.54: B2: Cancel Unfunded Escrow

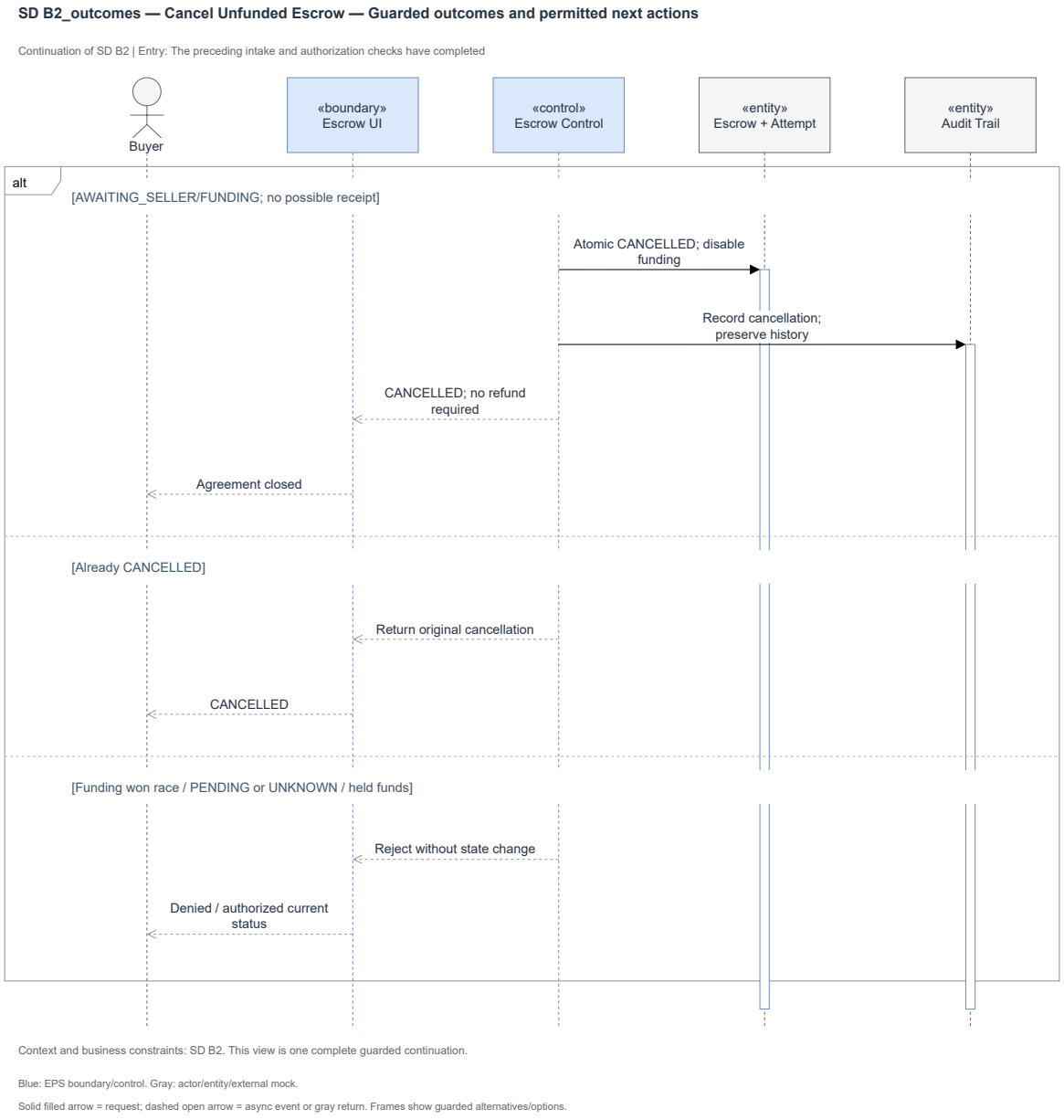


Figure 3.55: B2: Cancel Unfunded Escrow (continued: B2_outcomes)

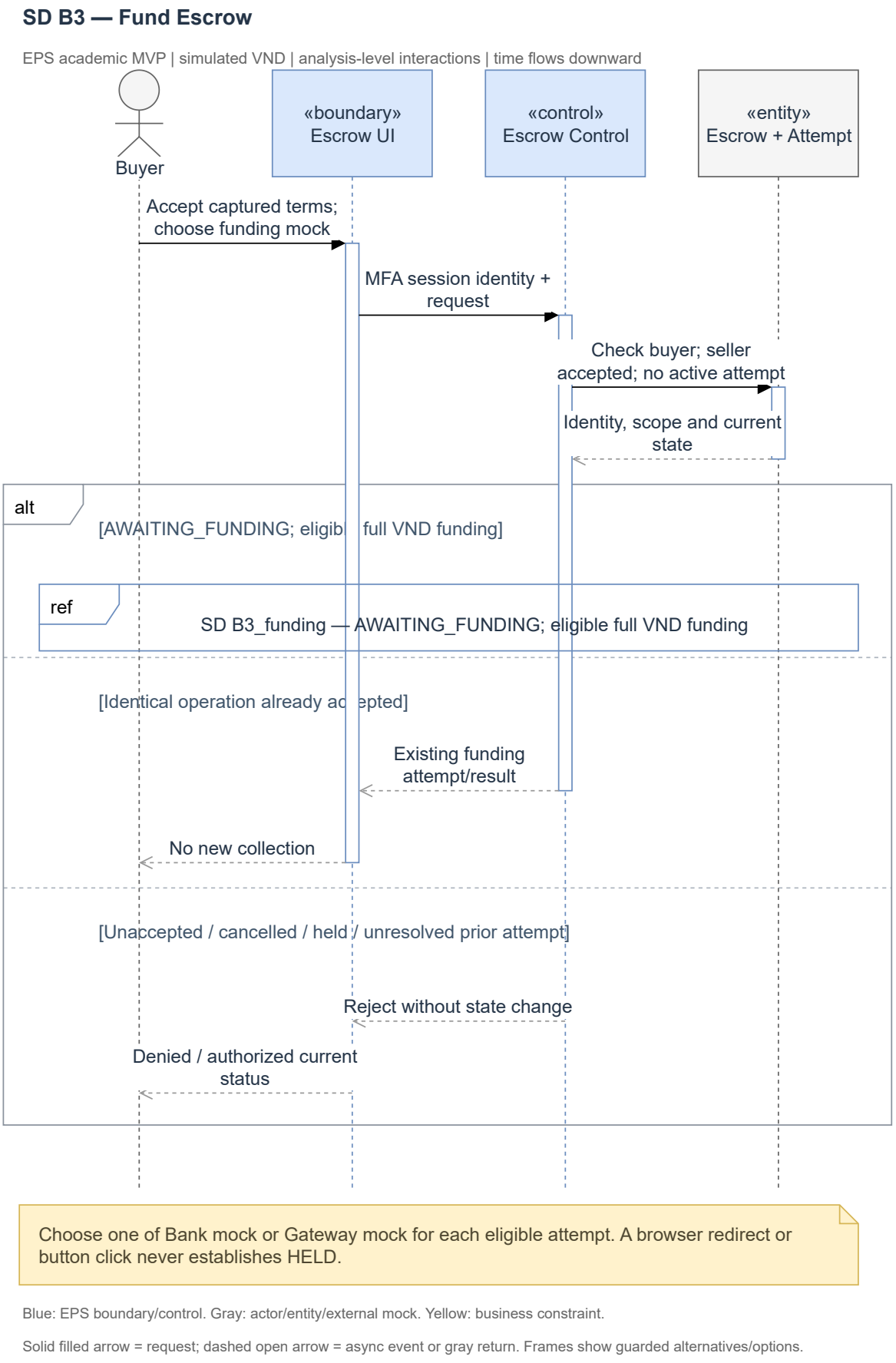


Figure 3.56: B3: Fund Escrow

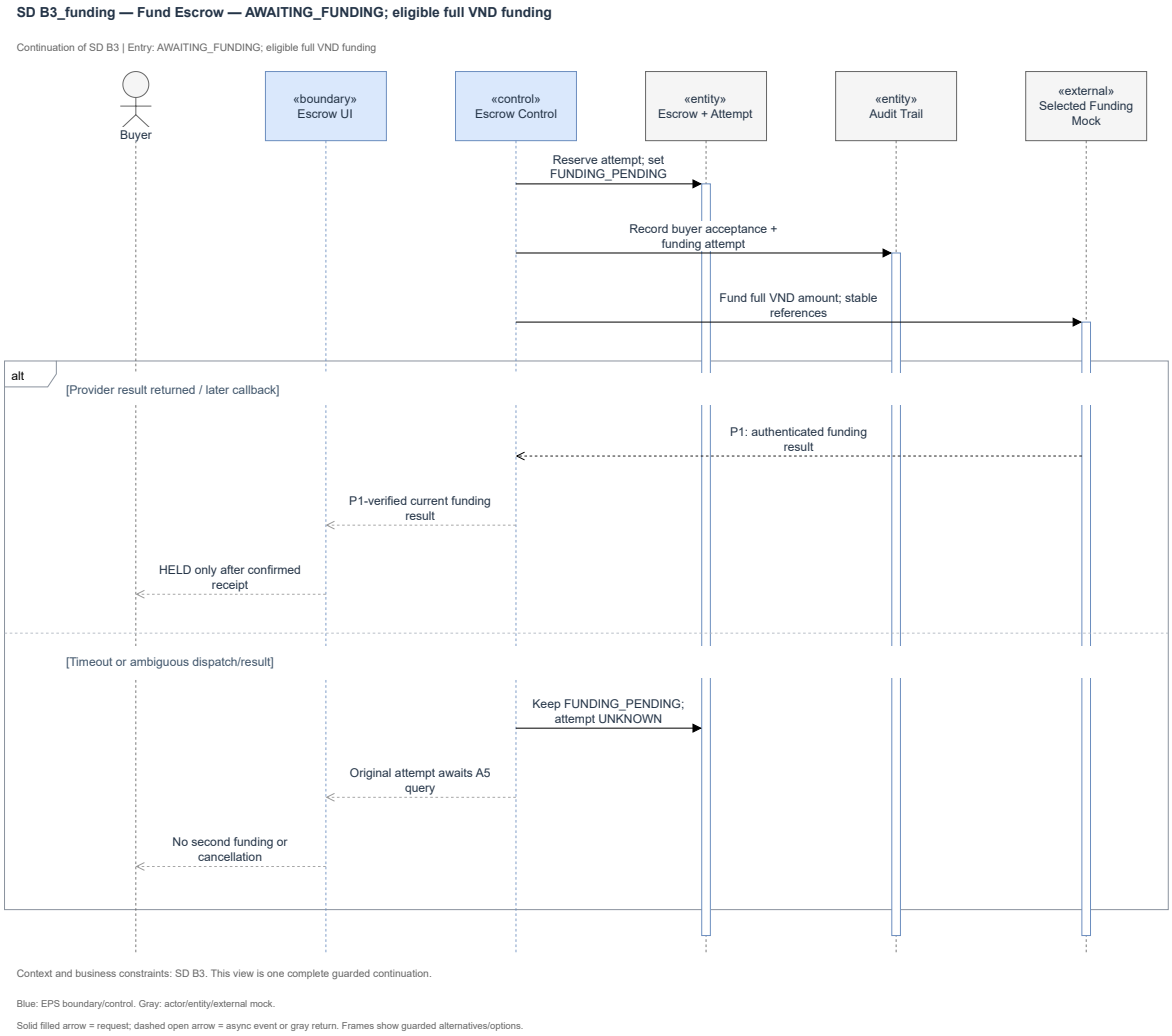


Figure 3.57: B3: Fund Escrow (continued: B3_funding)

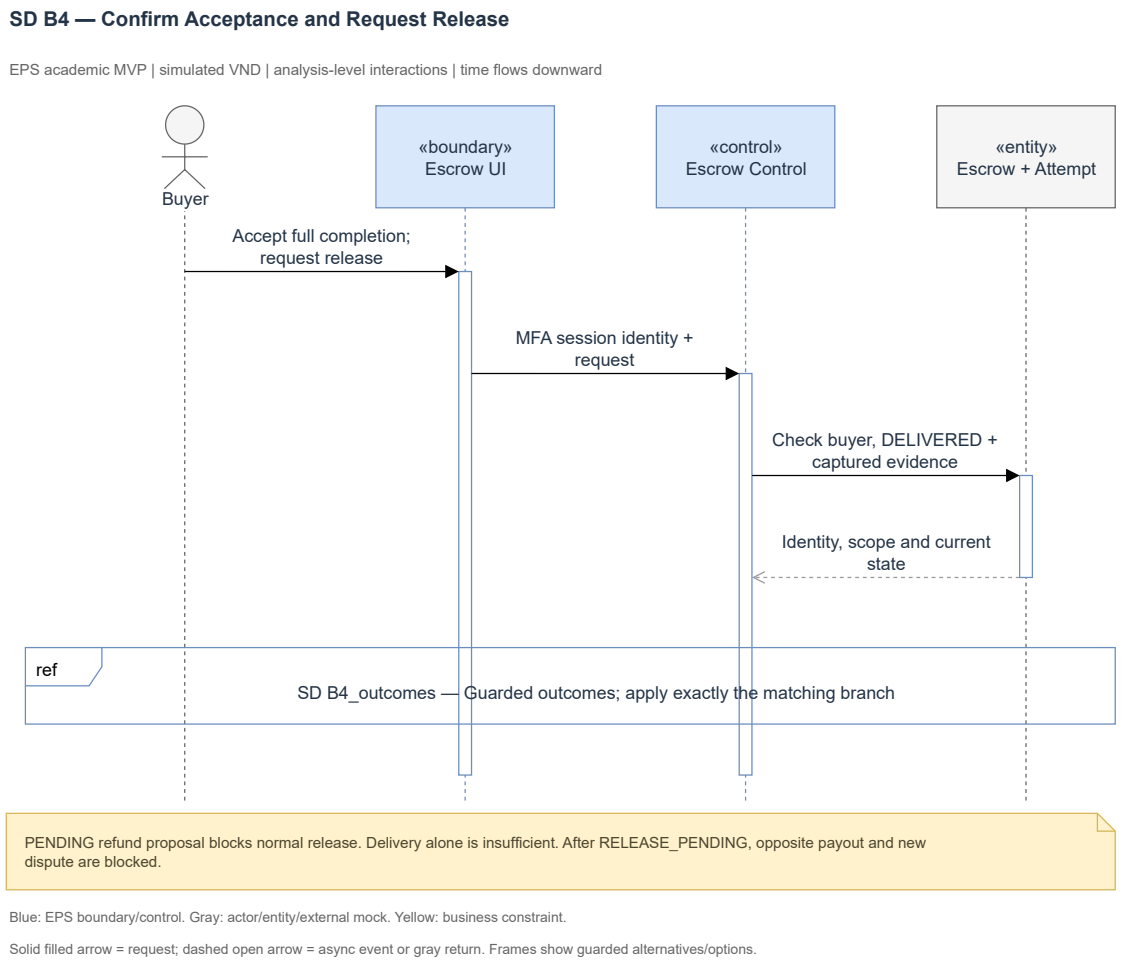


Figure 3.58: B4: Confirm Acceptance and Request Release

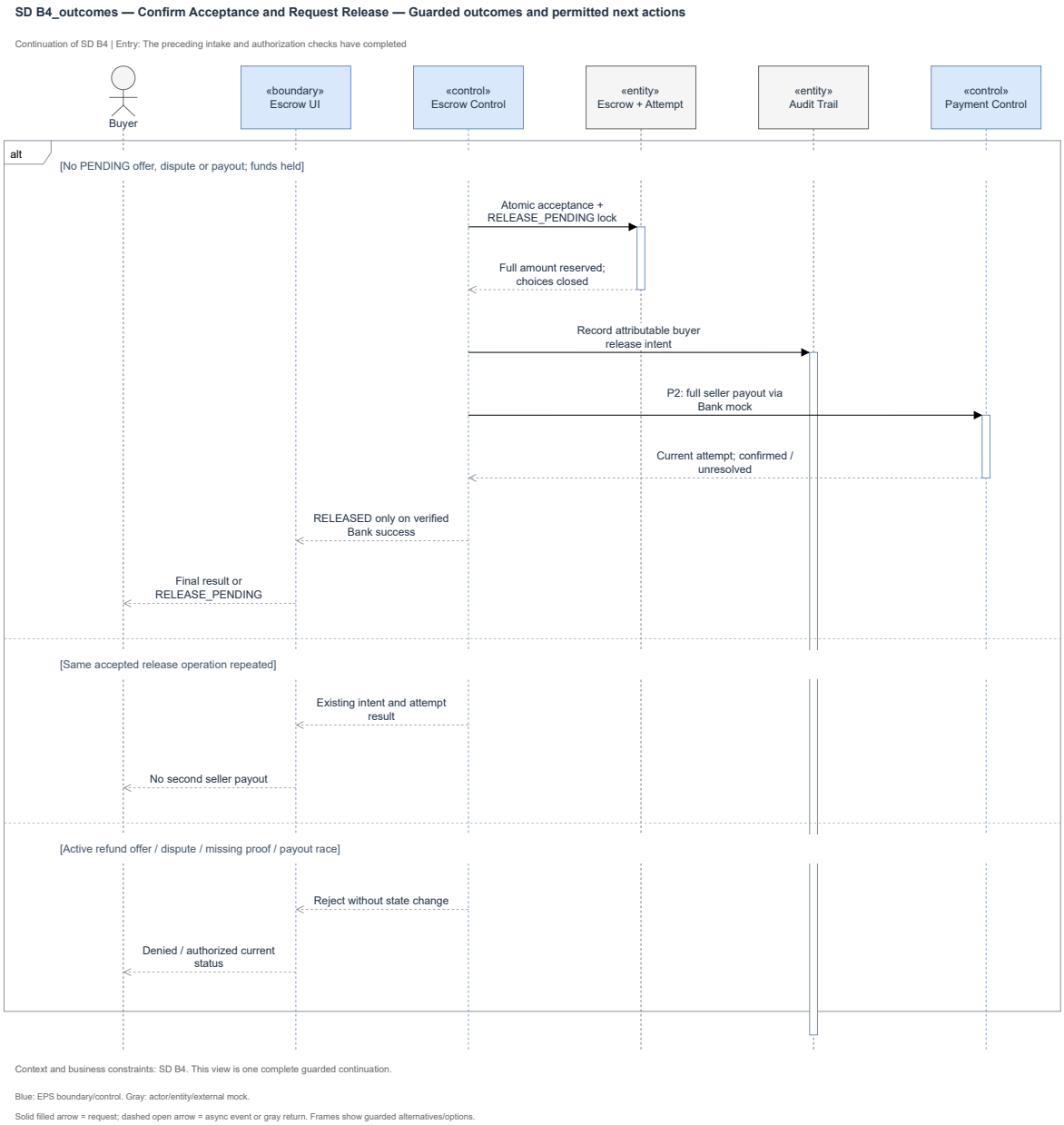


Figure 3.59: B4: Confirm Acceptance and Request Release (continued: B4_outcomes)

3.4.3 Seller's Use cases

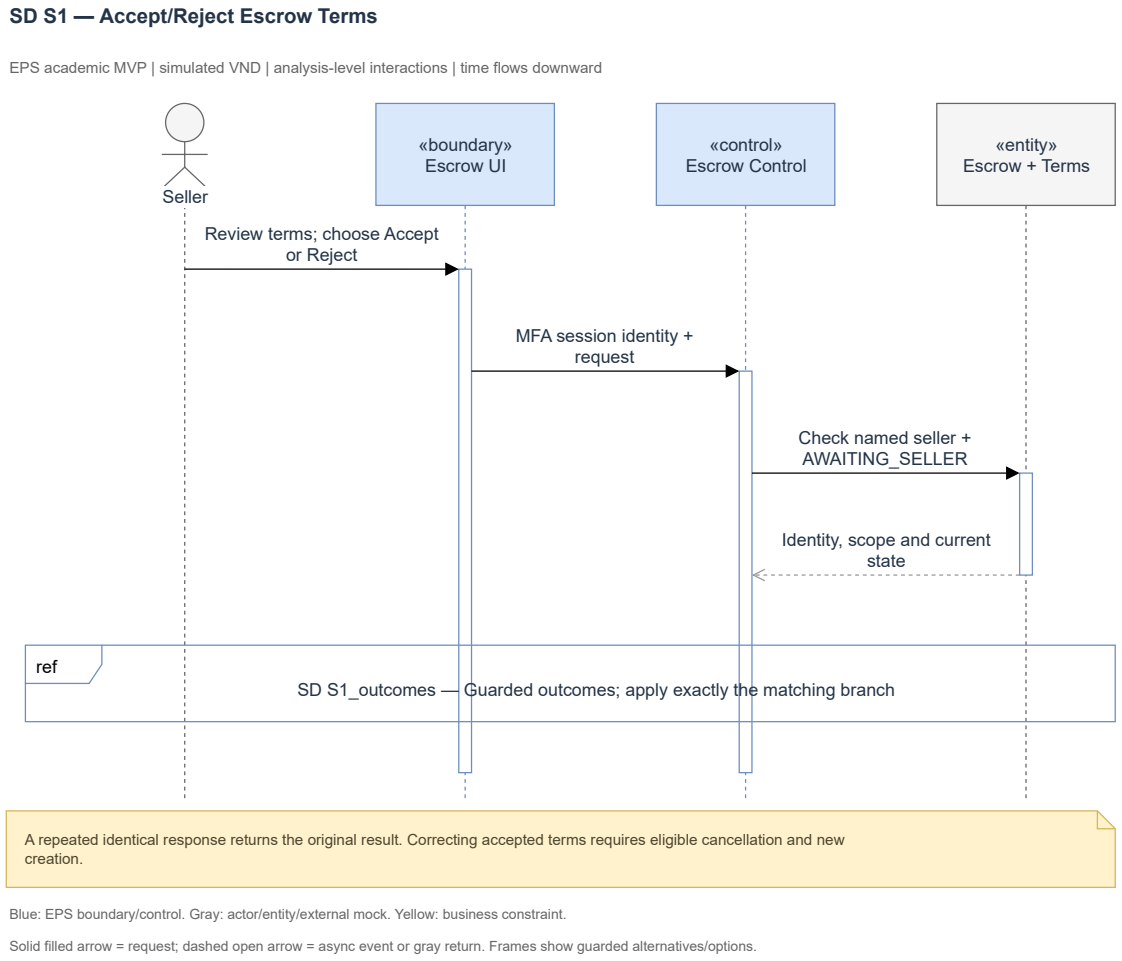


Figure 3.60: S1: Accept/Reject Escrow Terms

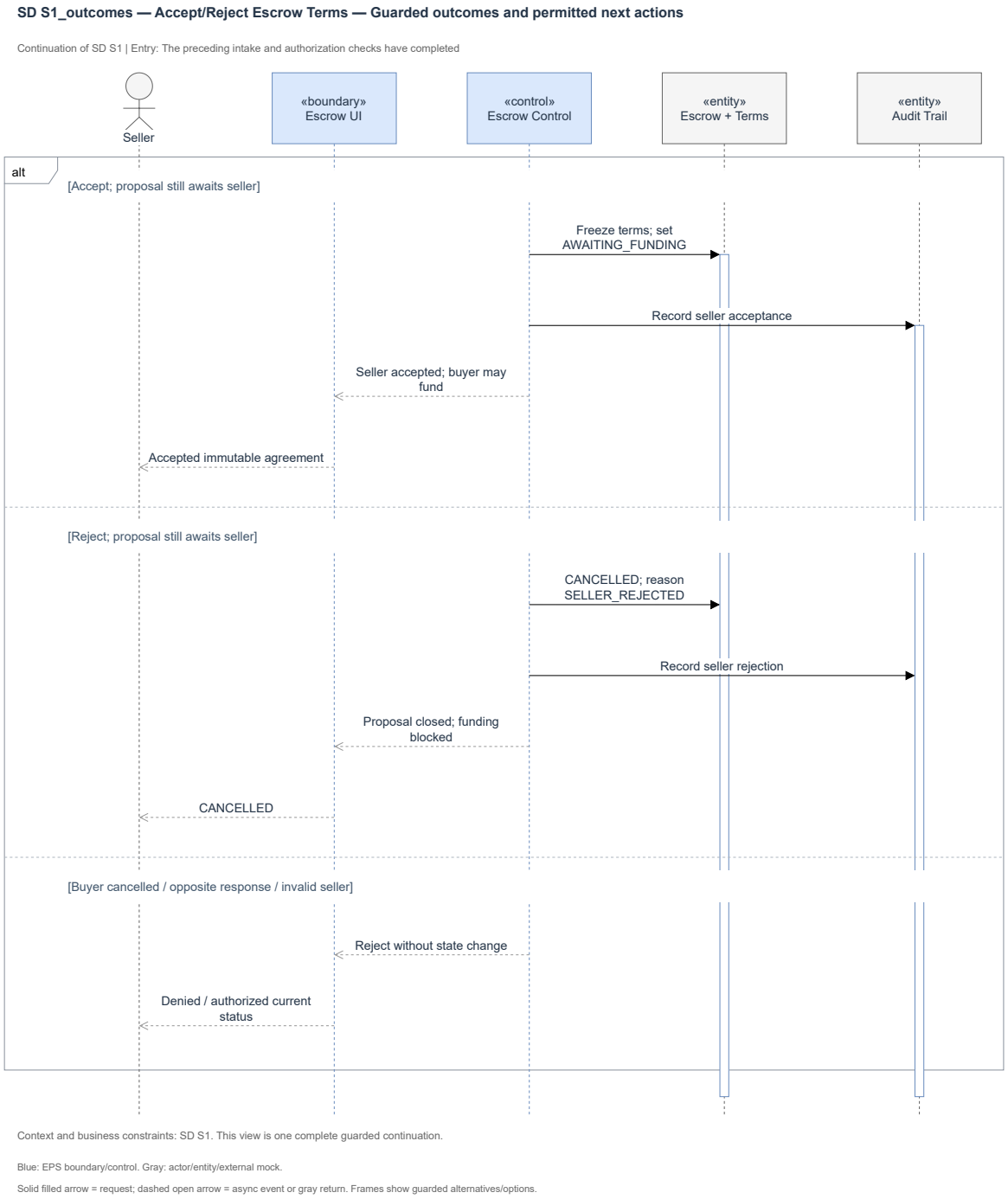


Figure 3.61: S1: Accept/Reject Escrow Terms (continued: S1_outcomes)

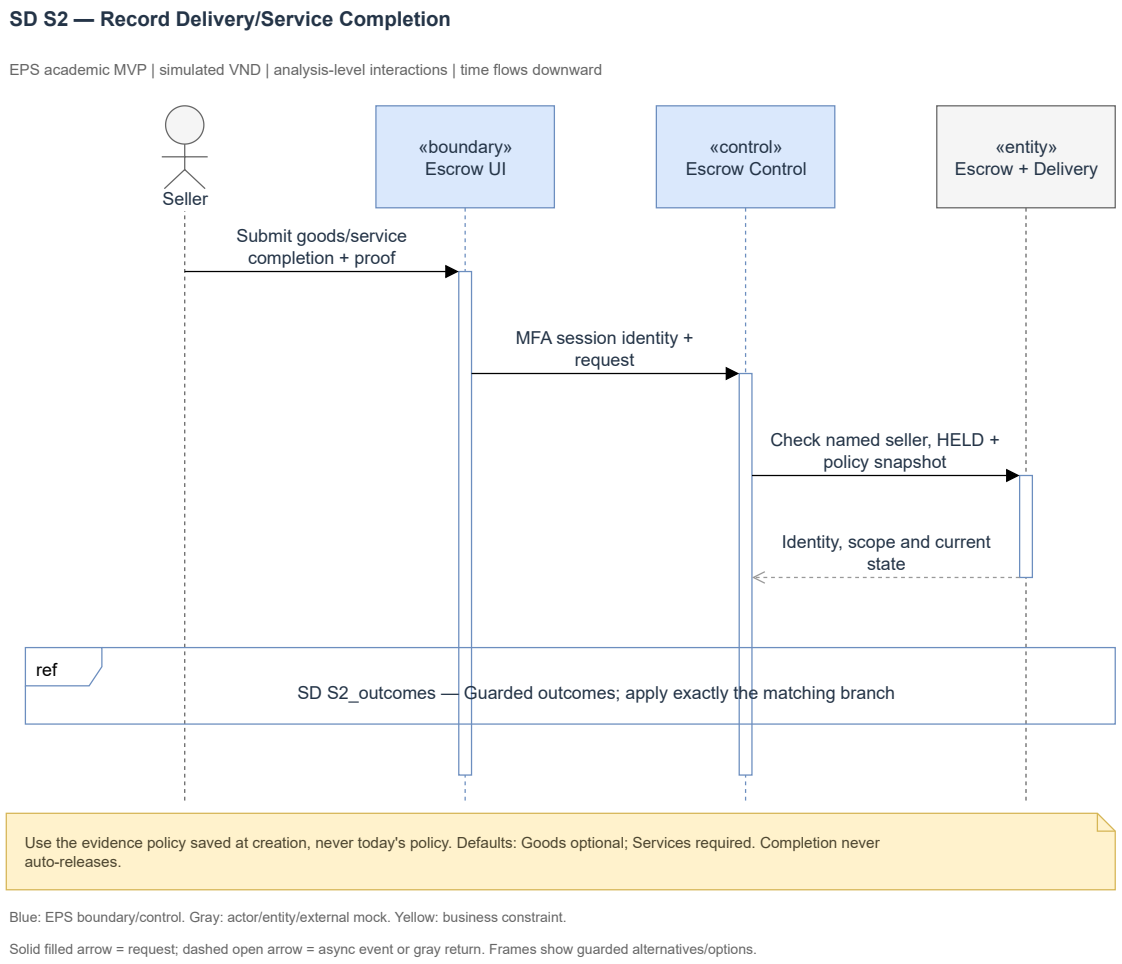


Figure 3.62: S2: Record Delivery/Service Completion

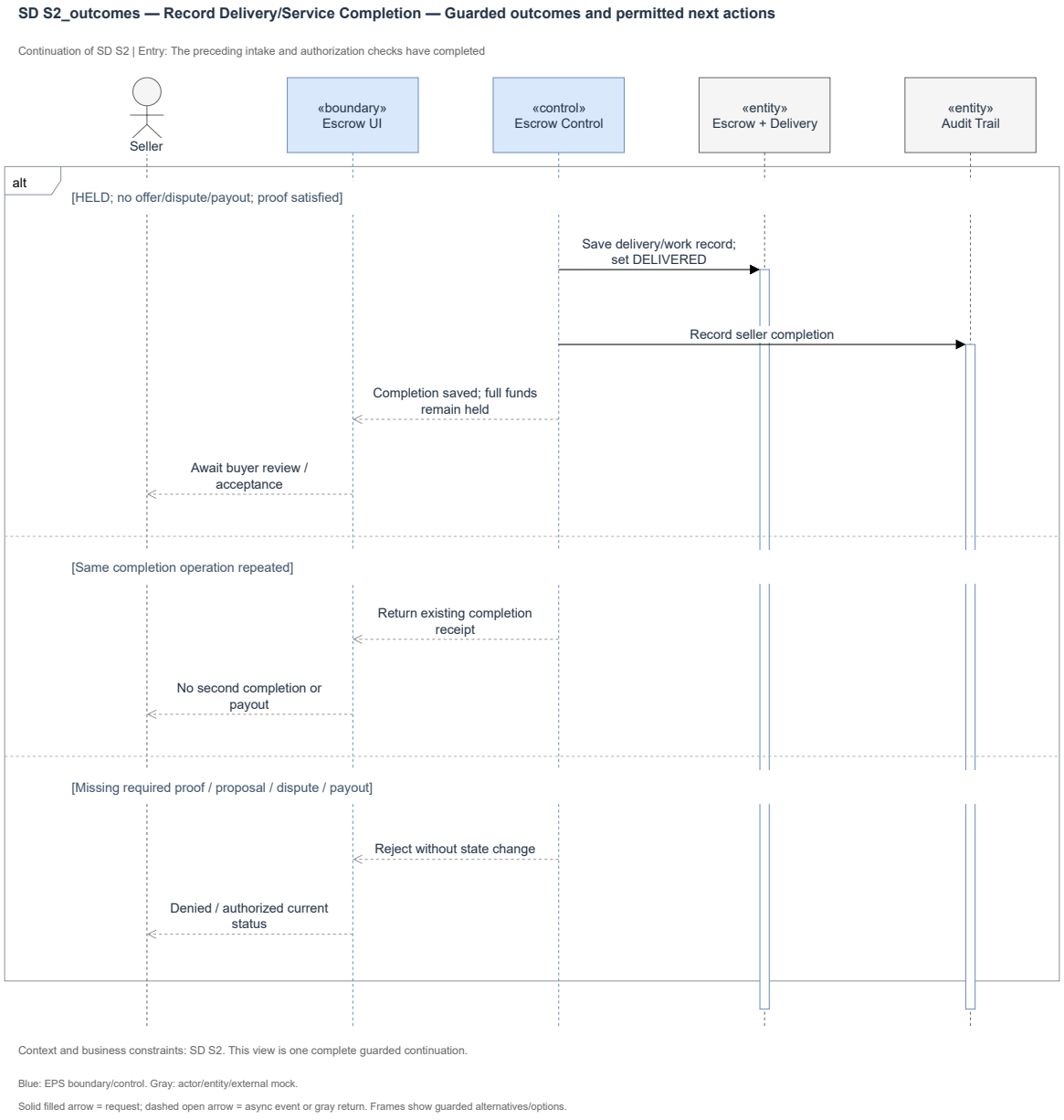


Figure 3.63: S2: Record Delivery/Service Completion (continued: S2_outcomes)

3.4.4 Admin's Use cases

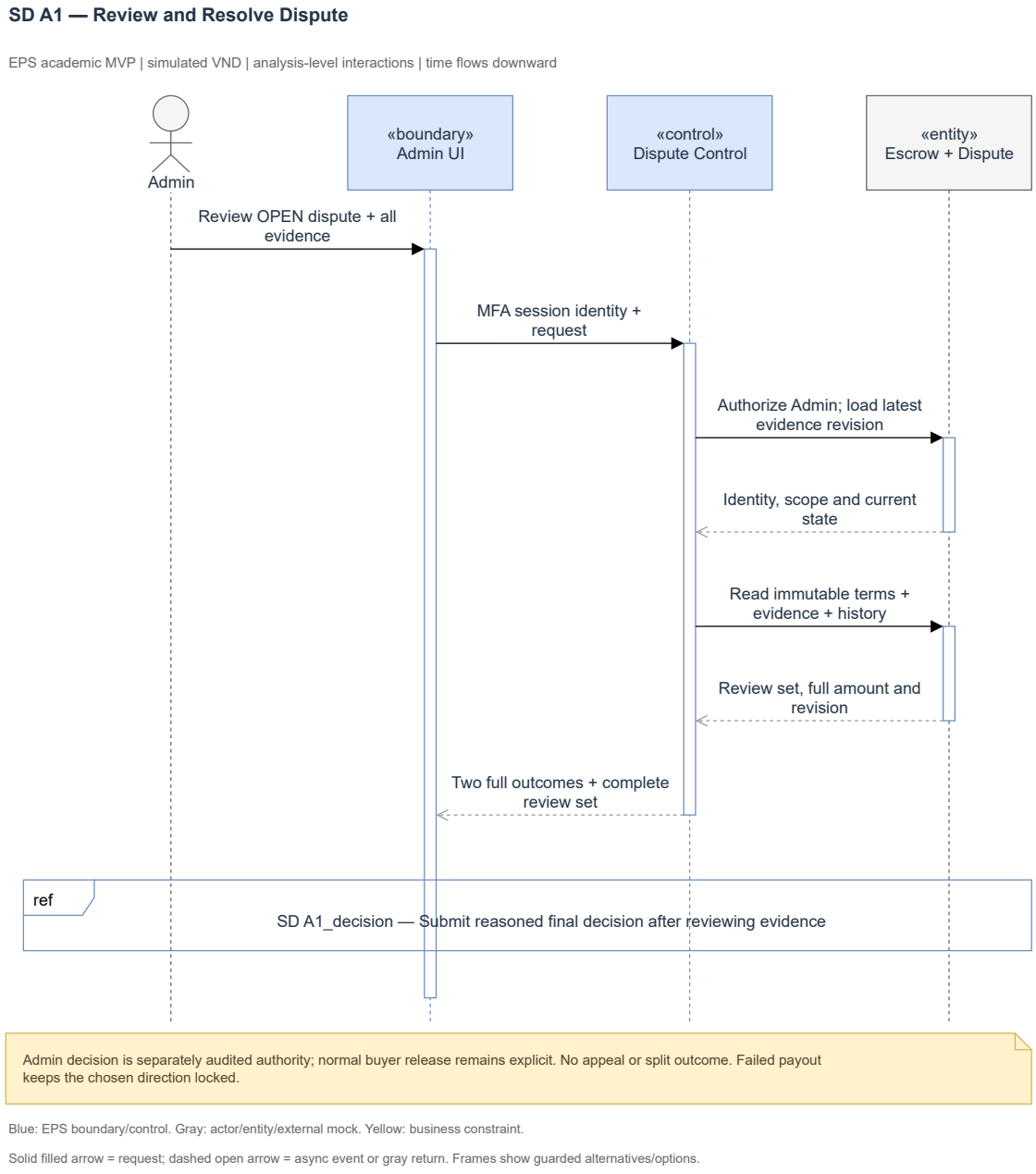


Figure 3.64: A1: Review and Resolve Dispute

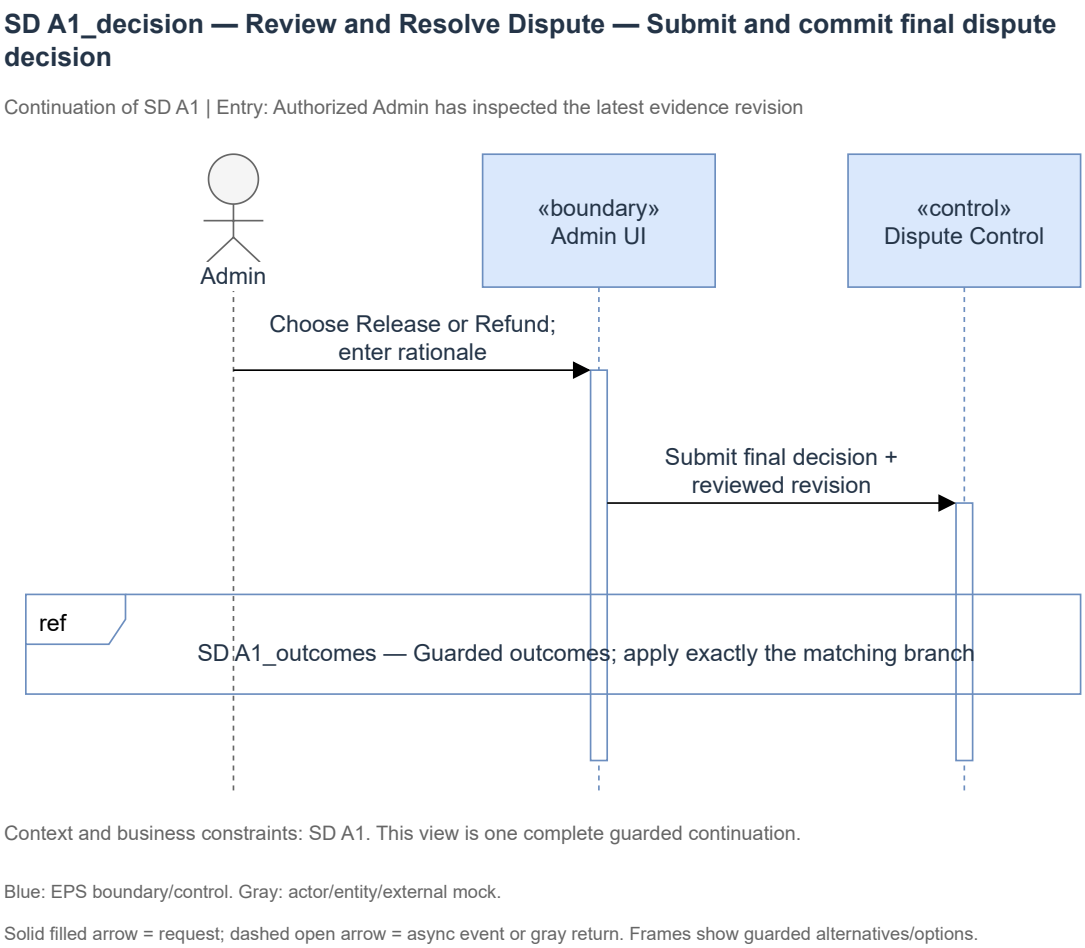


Figure 3.65: A1: Review and Resolve Dispute (continued: A1_decision)

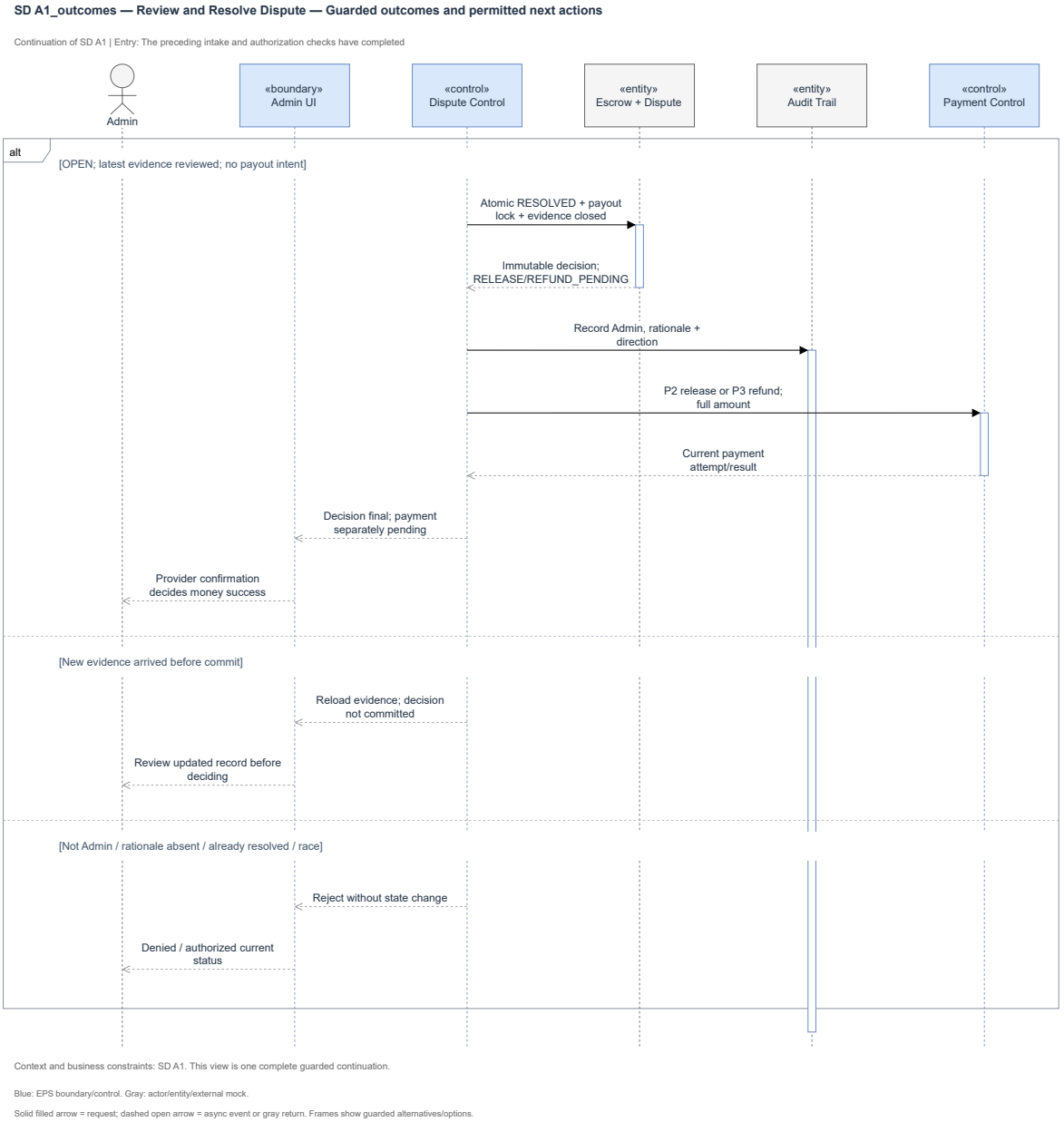


Figure 3.66: A1: Review and Resolve Dispute (continued: A1_outcomes)

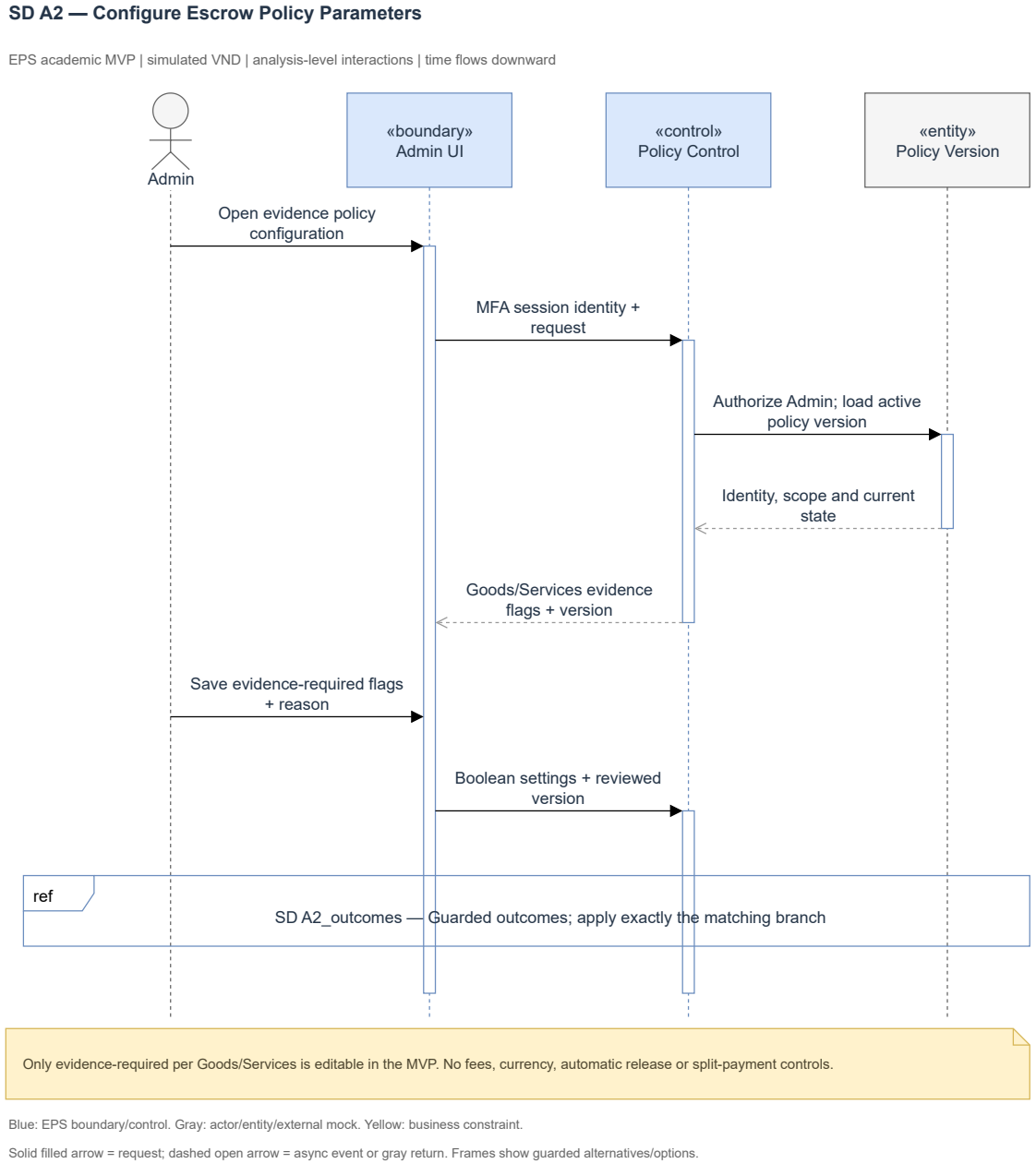


Figure 3.67: A2: Configure Escrow Policy Parameters

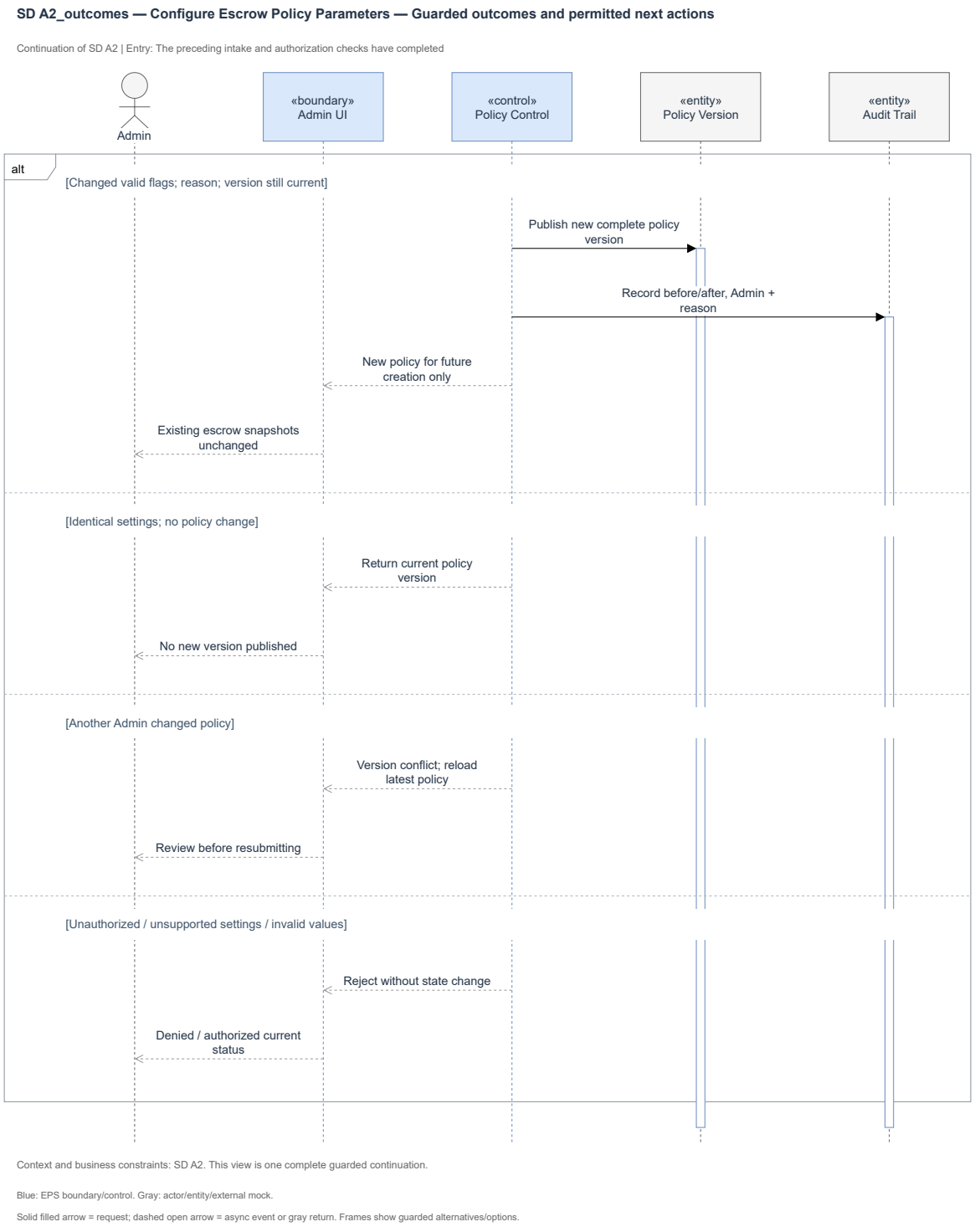


Figure 3.68: A2: Configure Escrow Policy Parameters (continued: A2_outcomes)

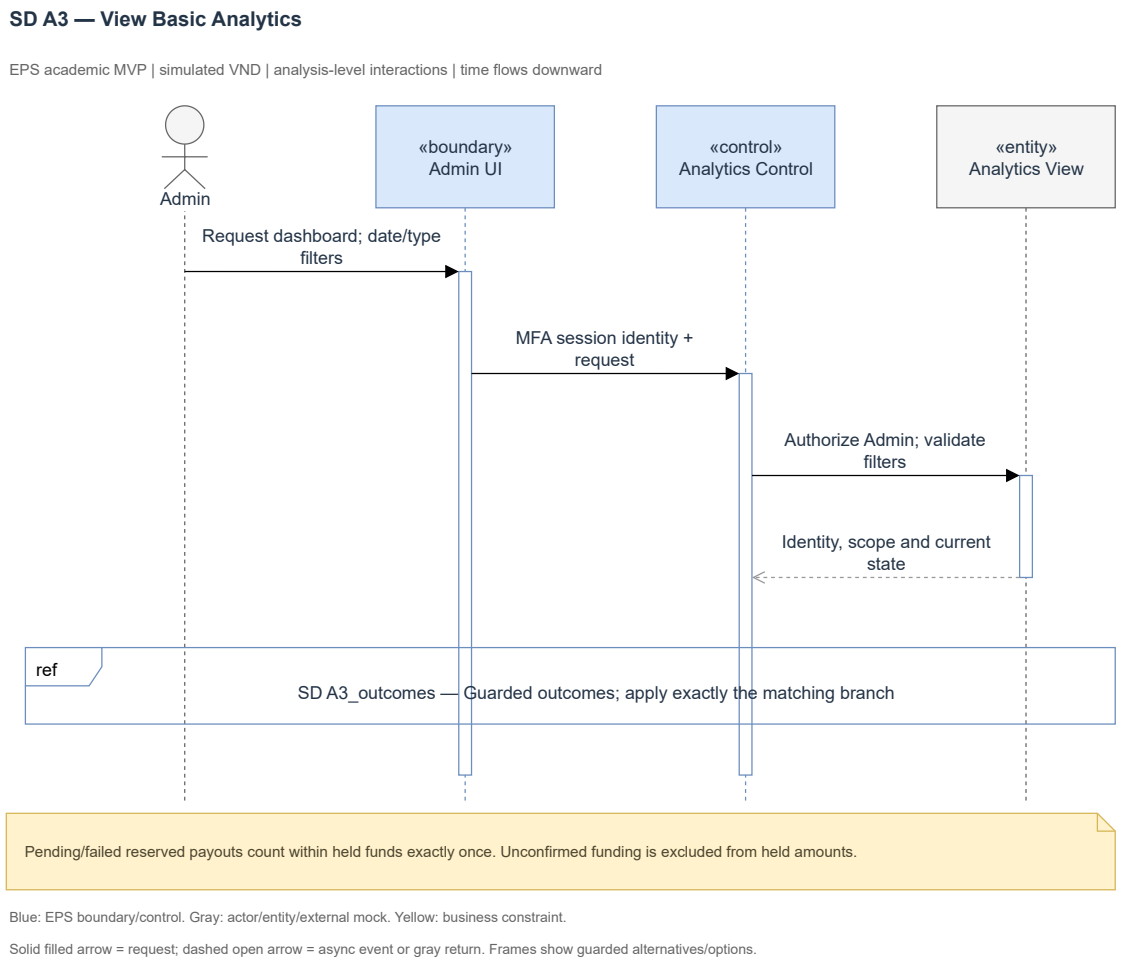


Figure 3.69: A3: View Basic Analytics

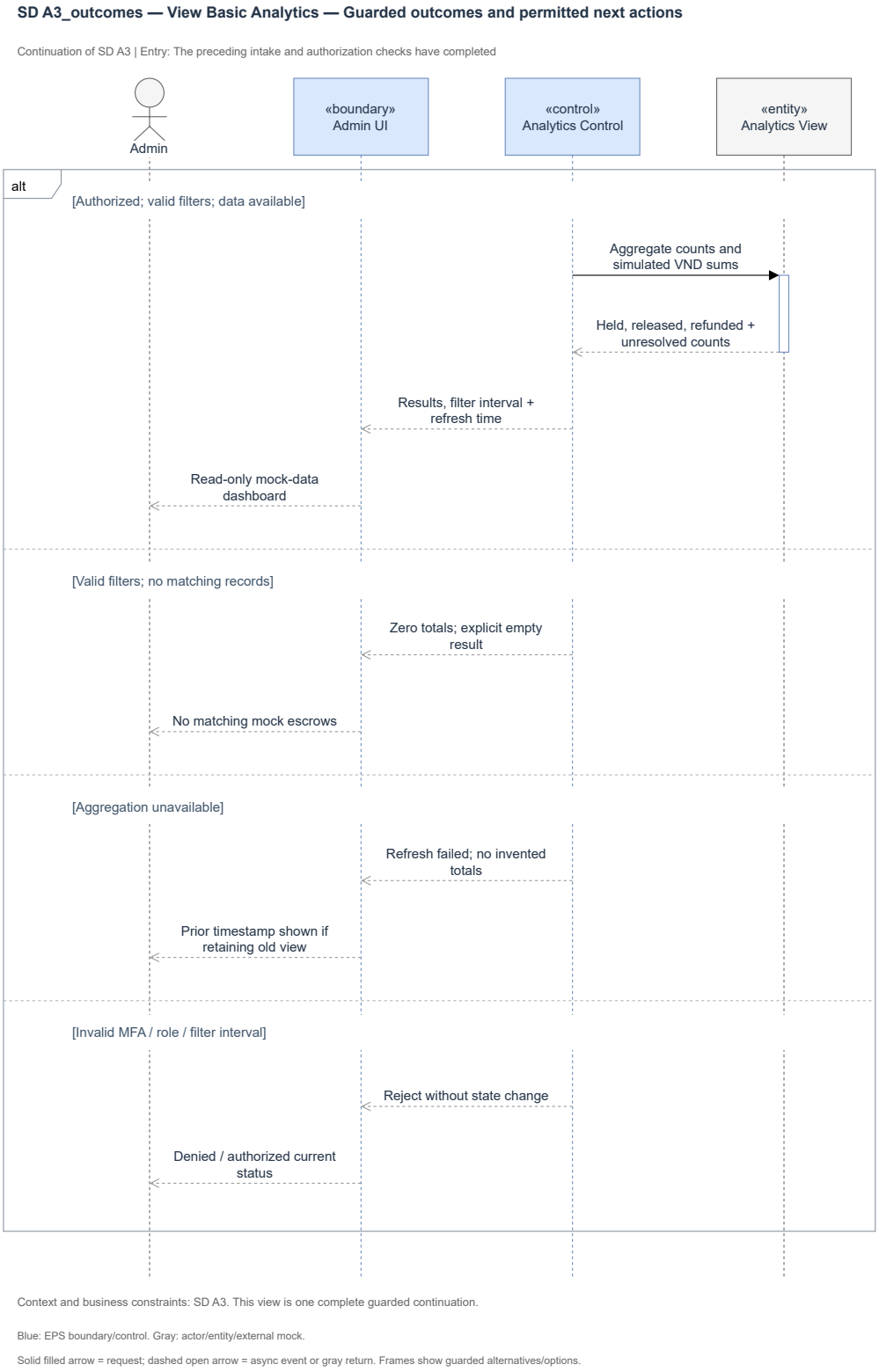


Figure 3.70: A3: View Basic Analytics (continued: A3_outcomes)

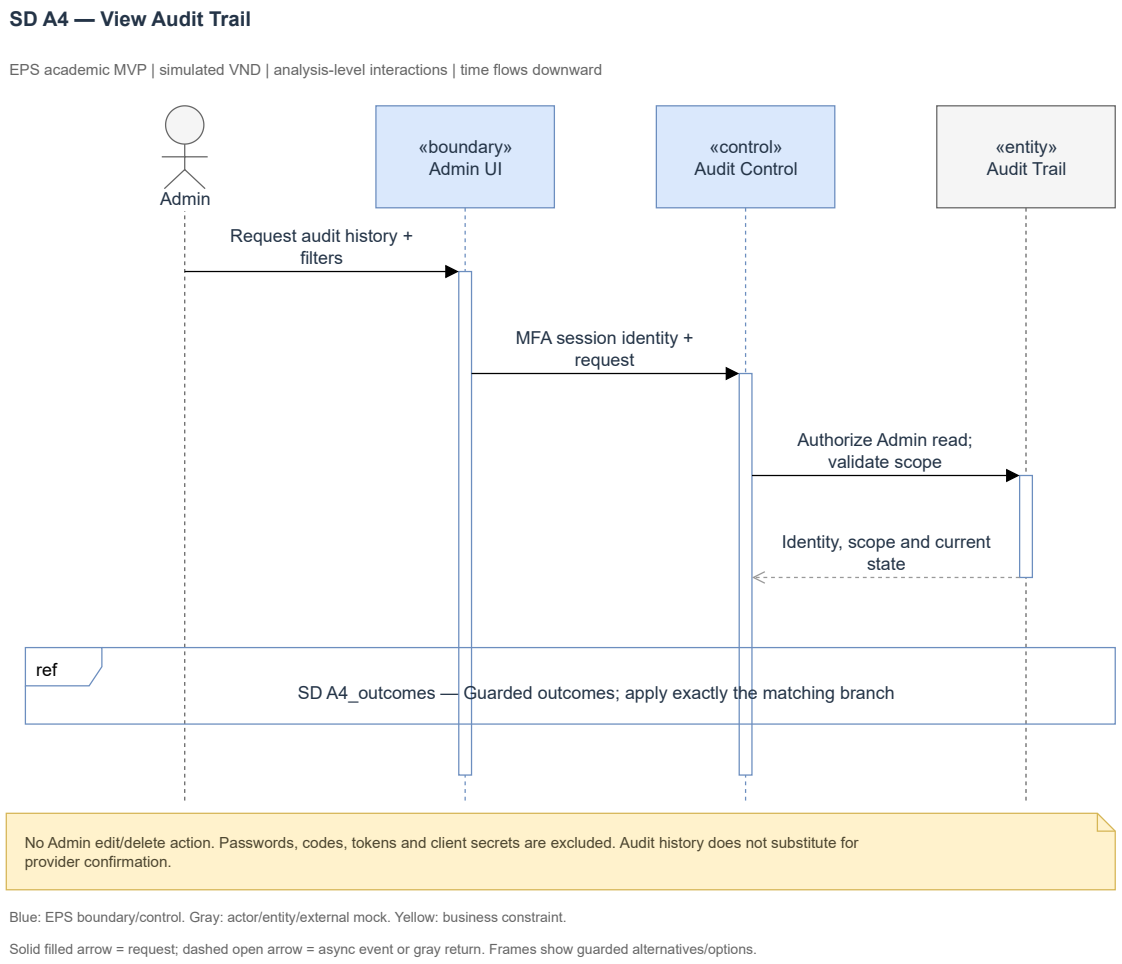


Figure 3.71: A4: View Audit Trail

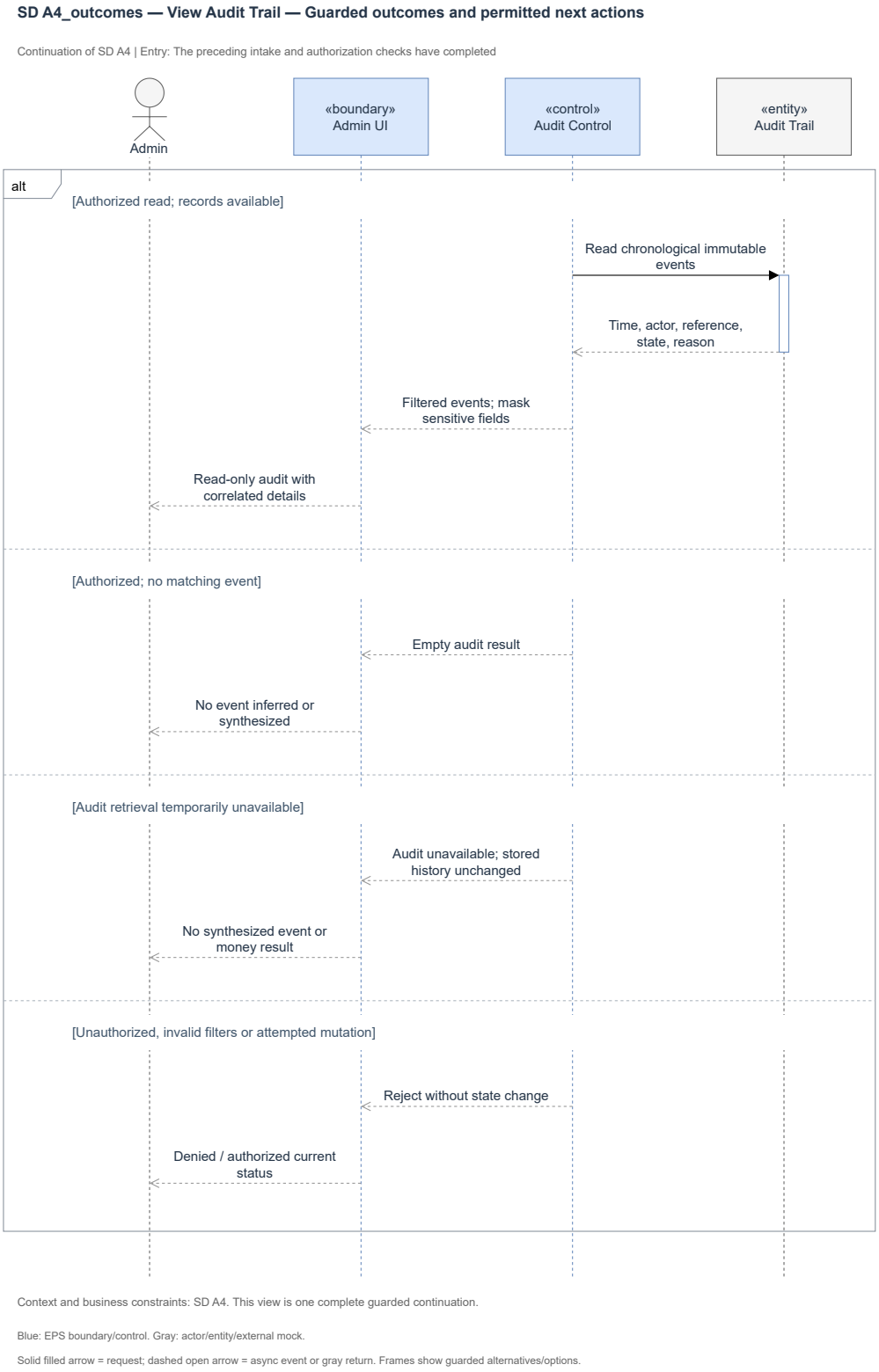


Figure 3.72: A4: View Audit Trail (continued: A4_outcomes)

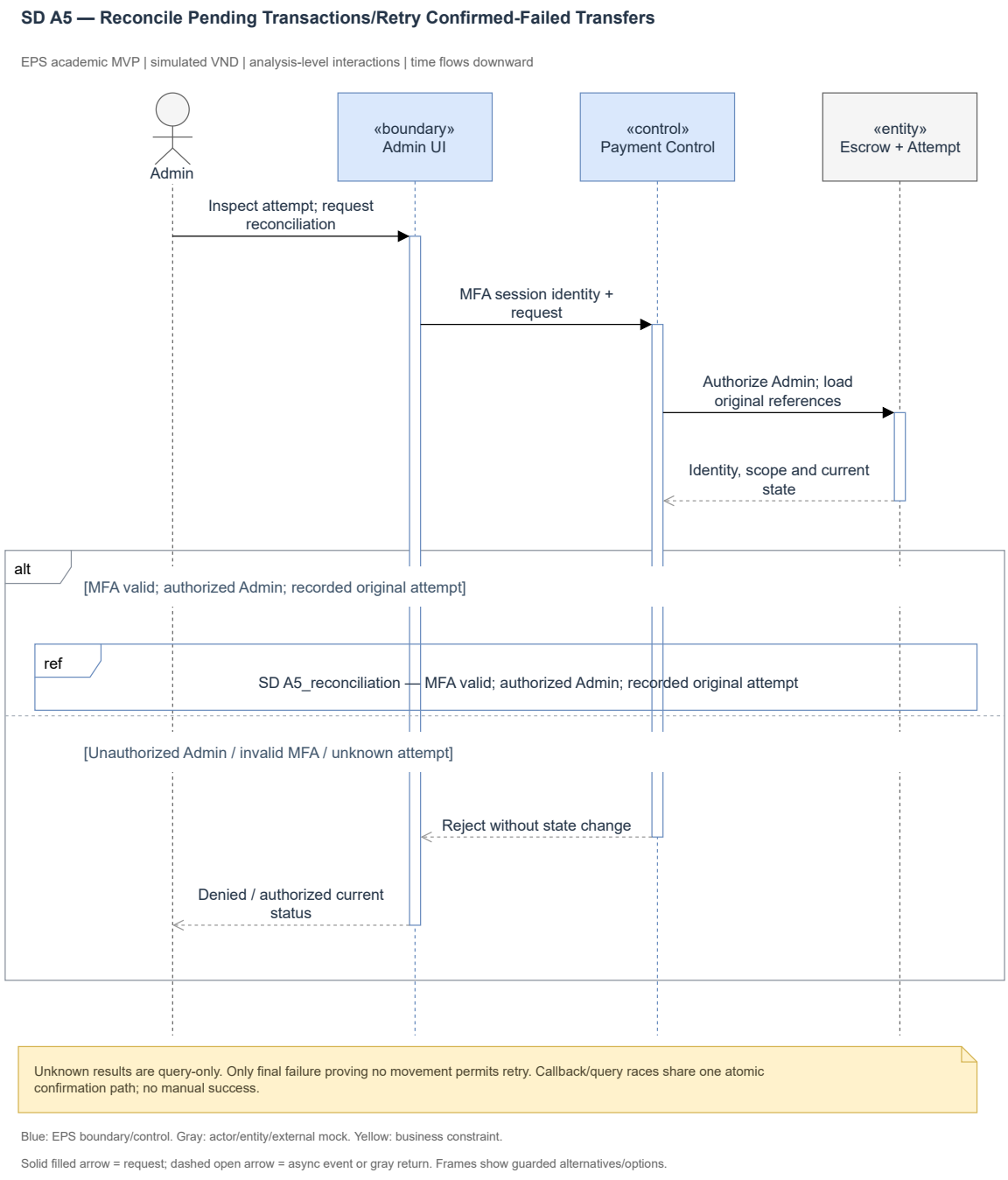


Figure 3.73: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers

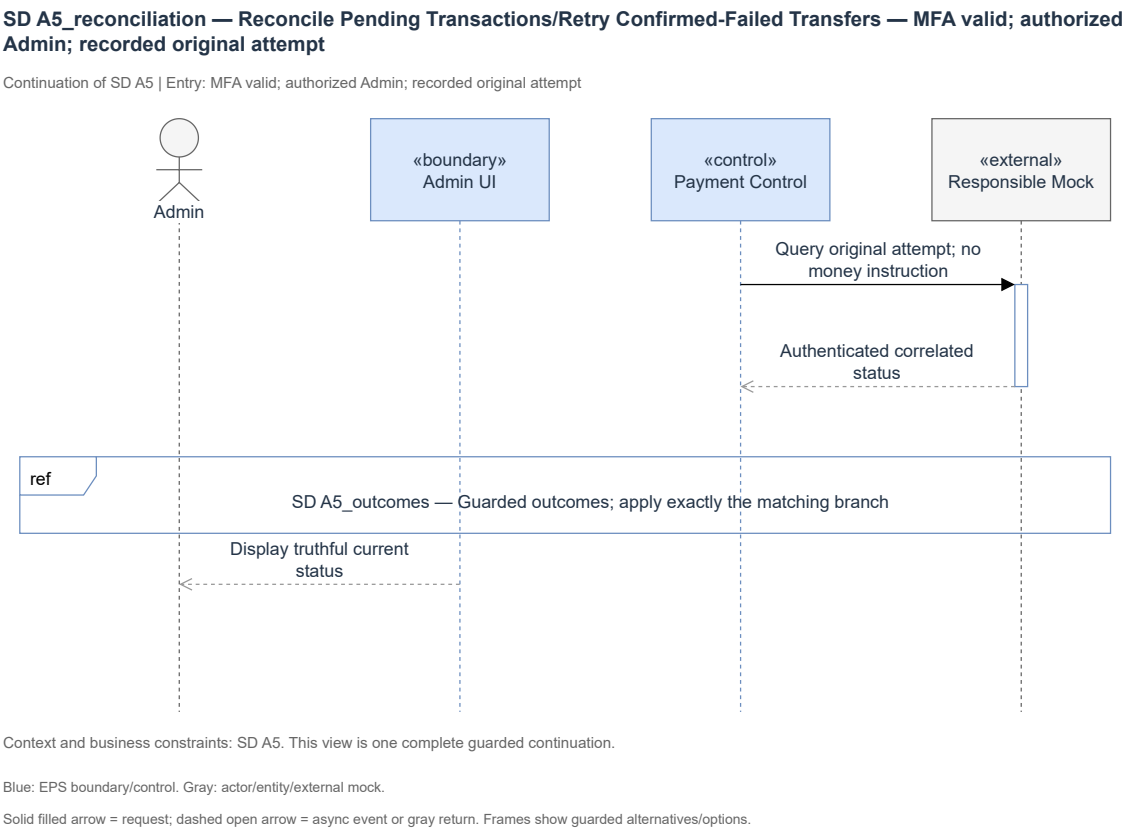


Figure 3.74: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers (continued: A5_reconciliation)

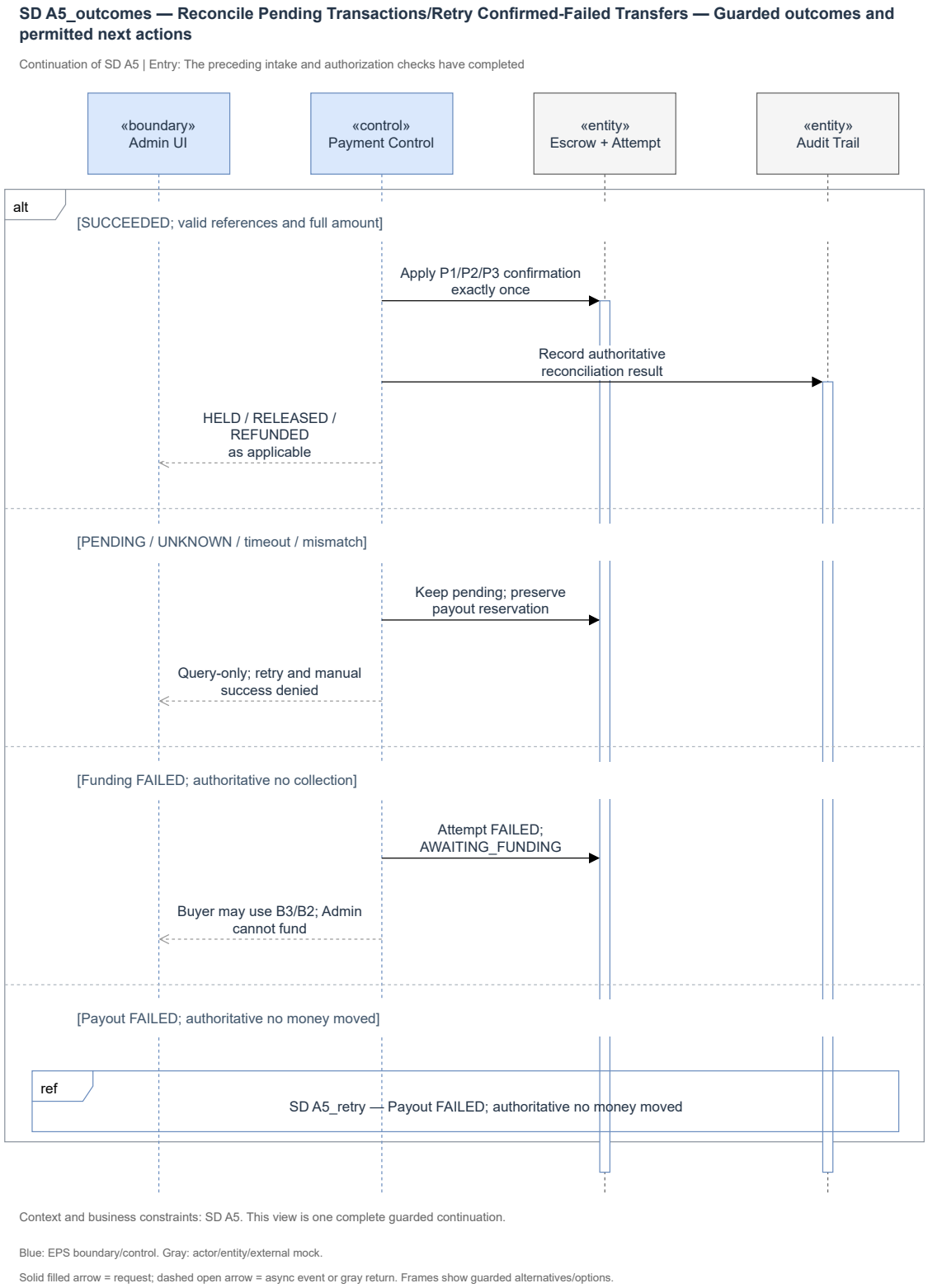


Figure 3.75: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers (continued: A5_outcomes)

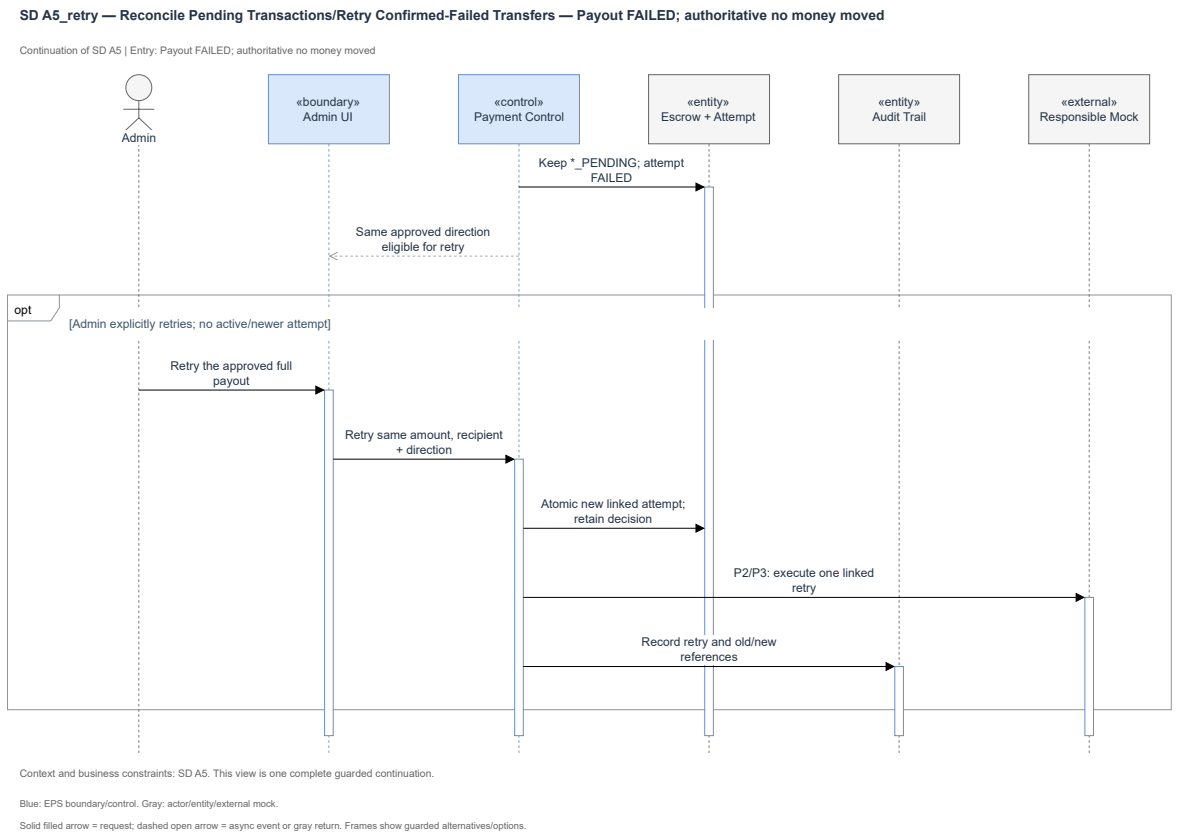


Figure 3.76: A5: Reconcile Pending Transactions/Retry Confirmed-Failed Transfers (continued: A5_retry)

3.4.5 Payment Integration's Use cases

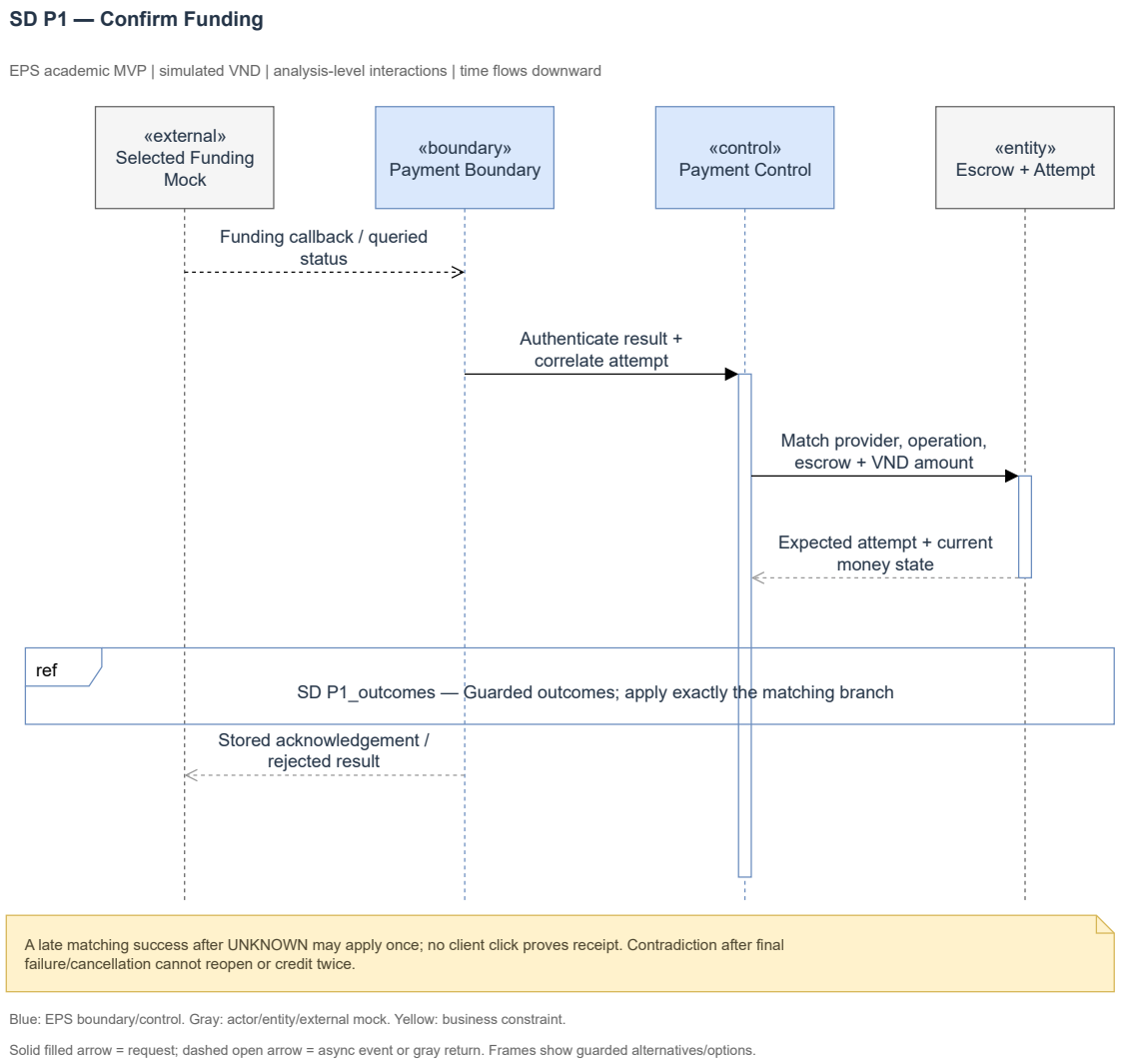


Figure 3.77: P1: Confirm Funding

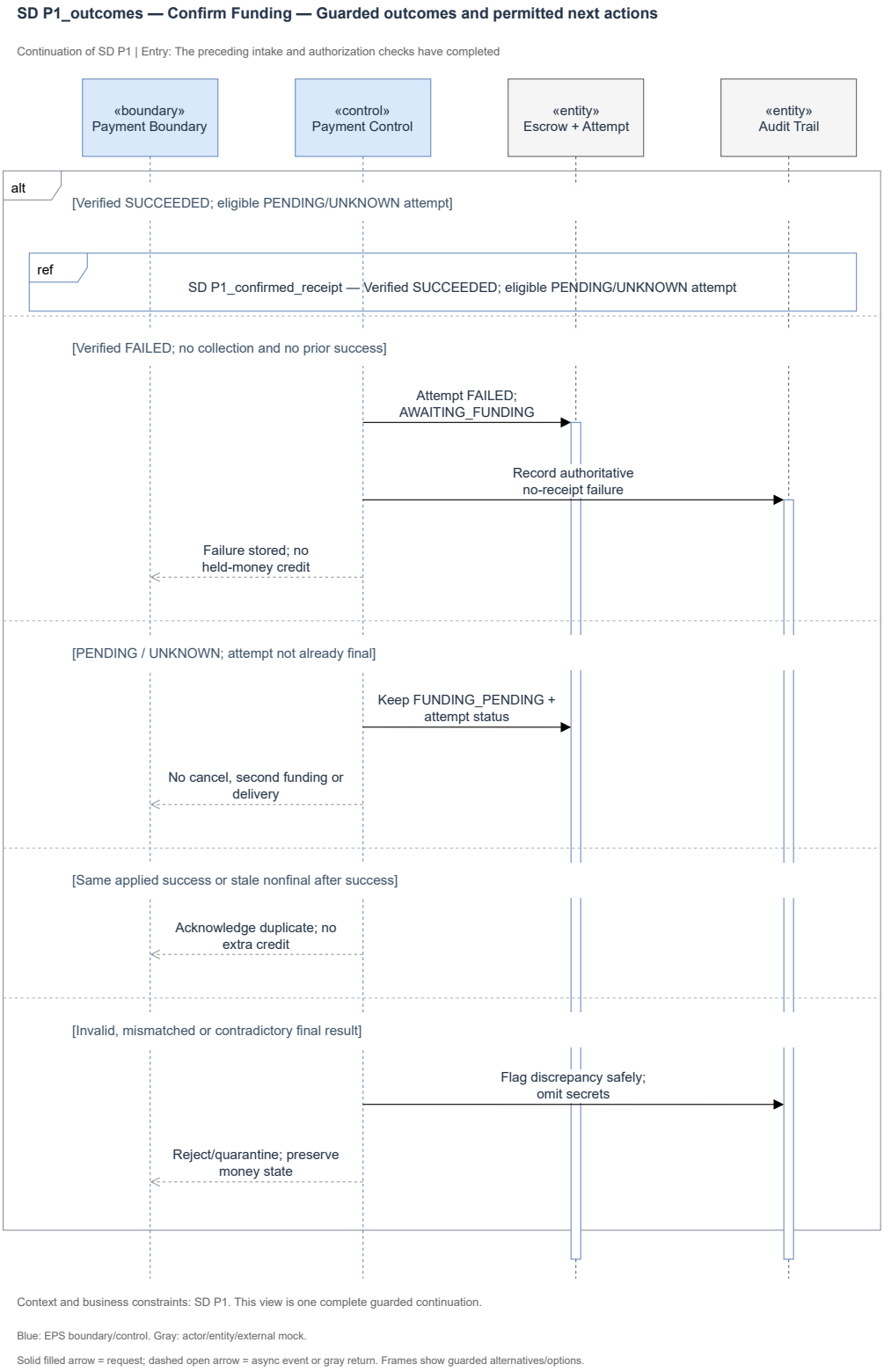


Figure 3.78: P1: Confirm Funding (continued: P1_outcomes)

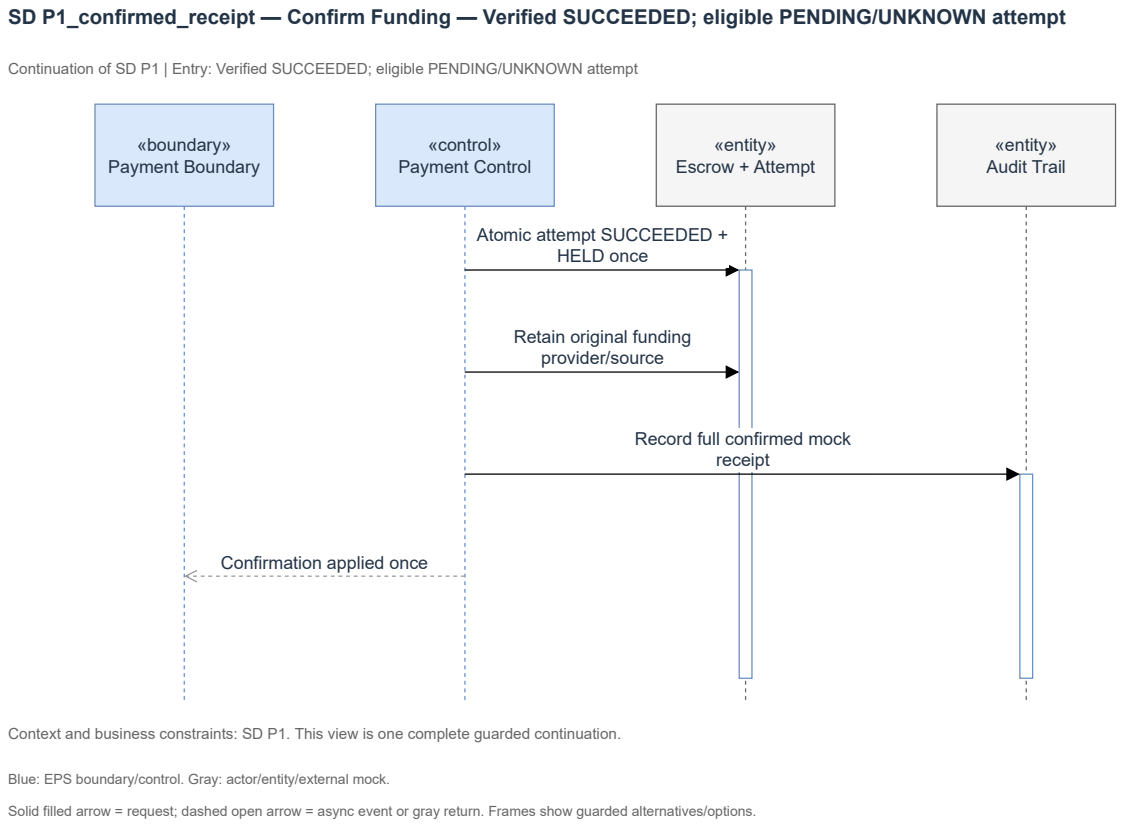


Figure 3.79: P1: Confirm Funding (continued: P1_confirmed_receipt)

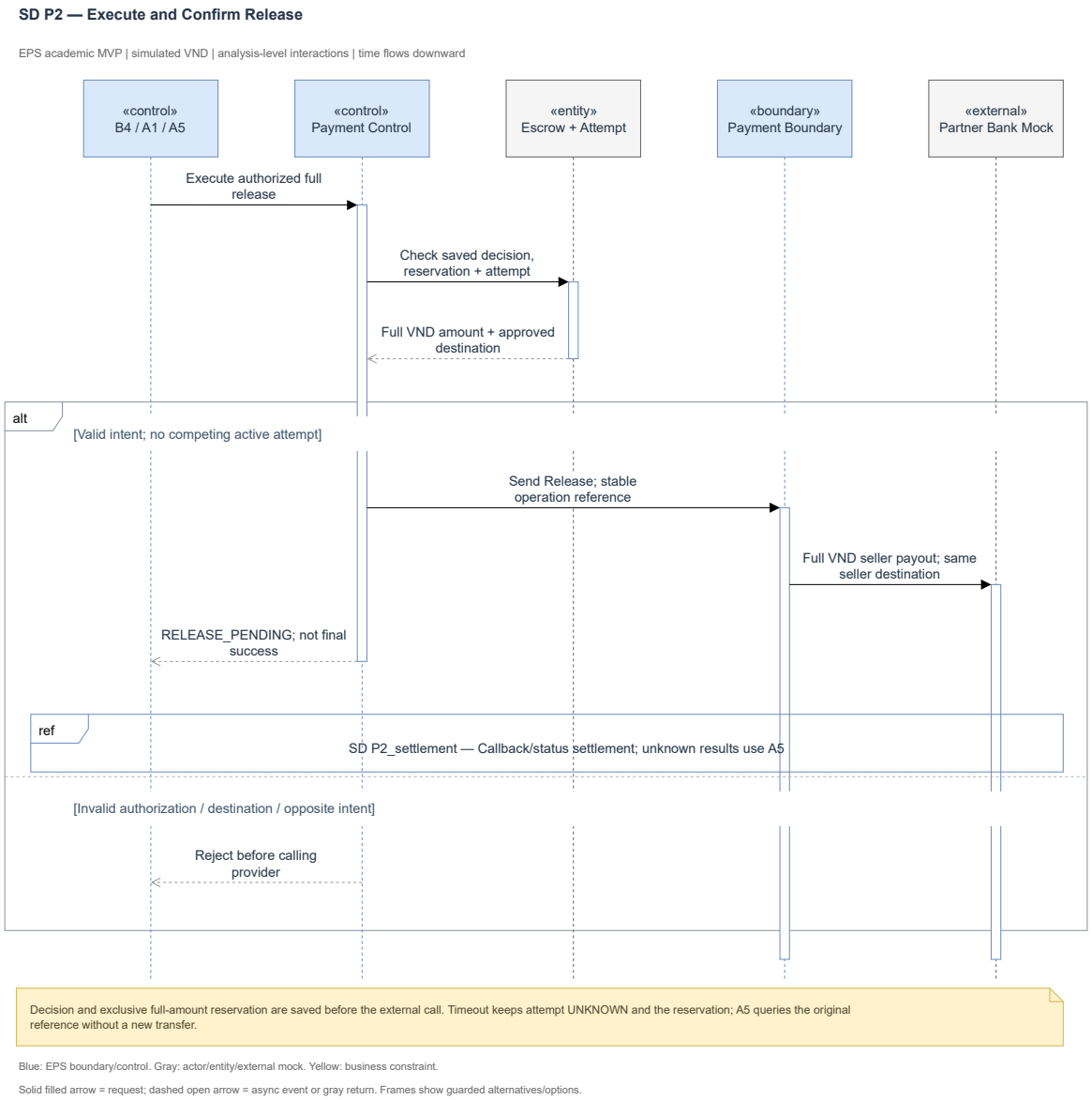


Figure 3.80: P2: Execute and Confirm Release

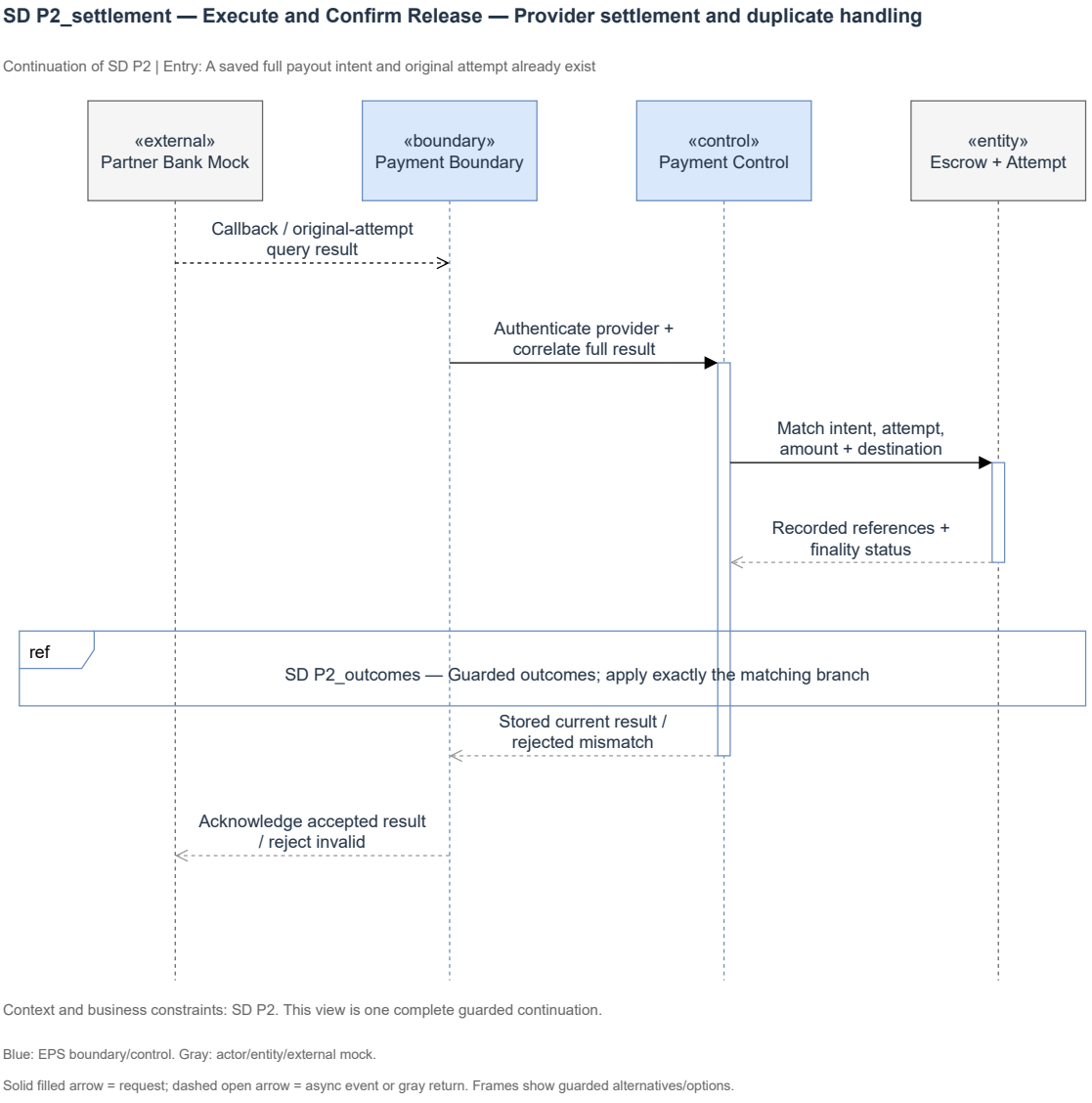


Figure 3.81: P2: Execute and Confirm Release (continued: P2_settlement)

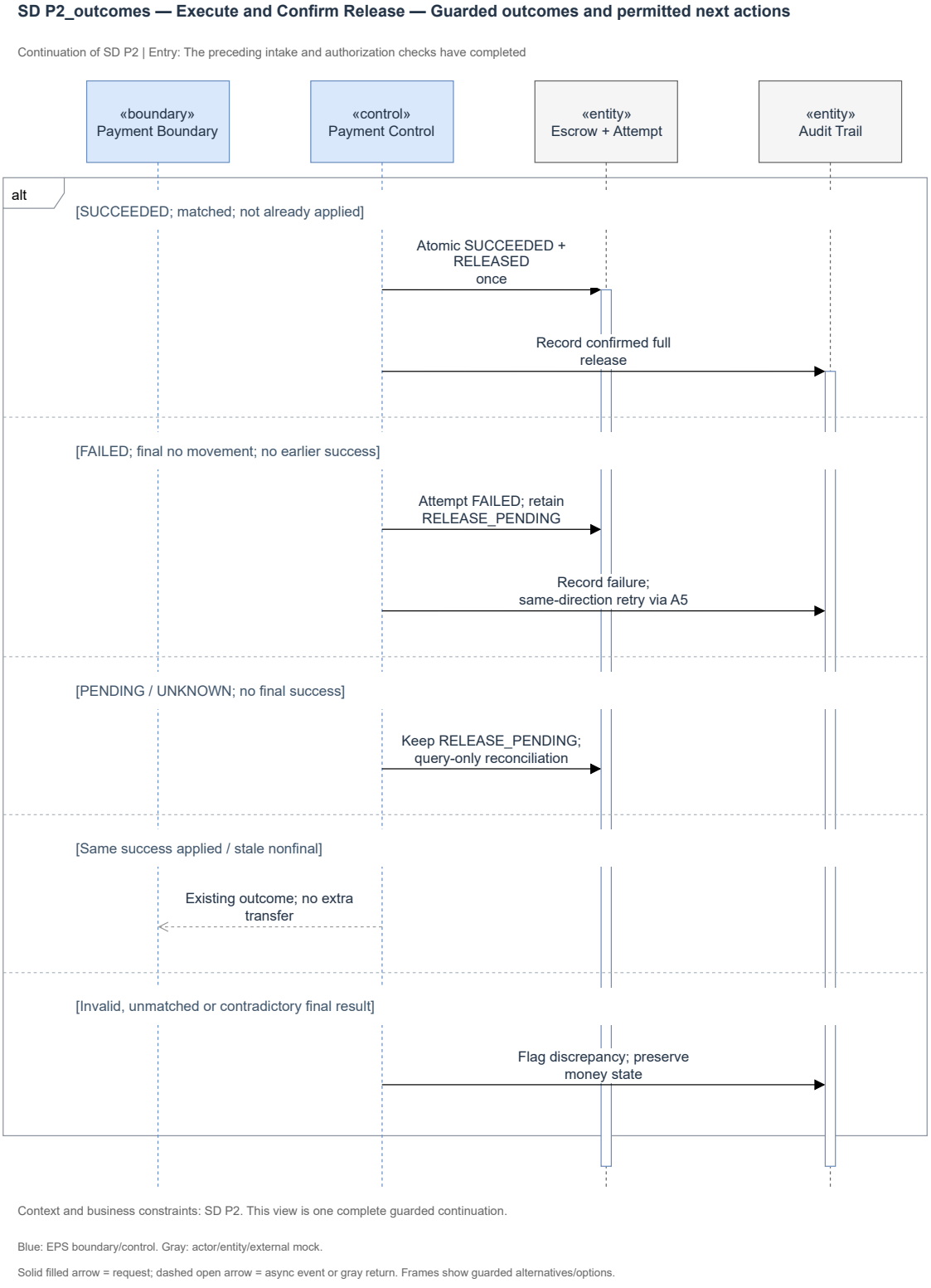


Figure 3.82: P2: Execute and Confirm Release (continued: P2_outcomes)

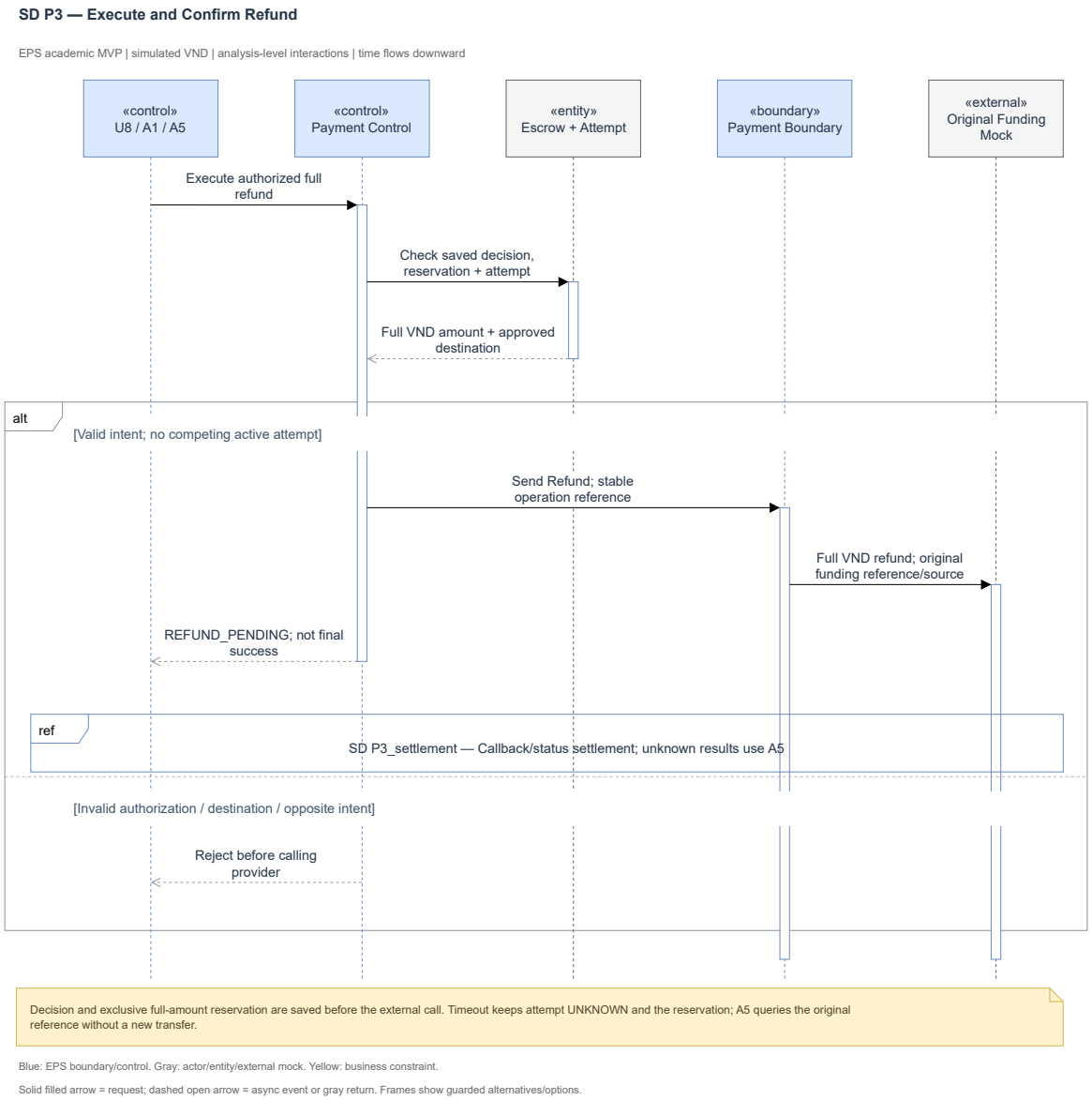


Figure 3.83: P3: Execute and Confirm Refund

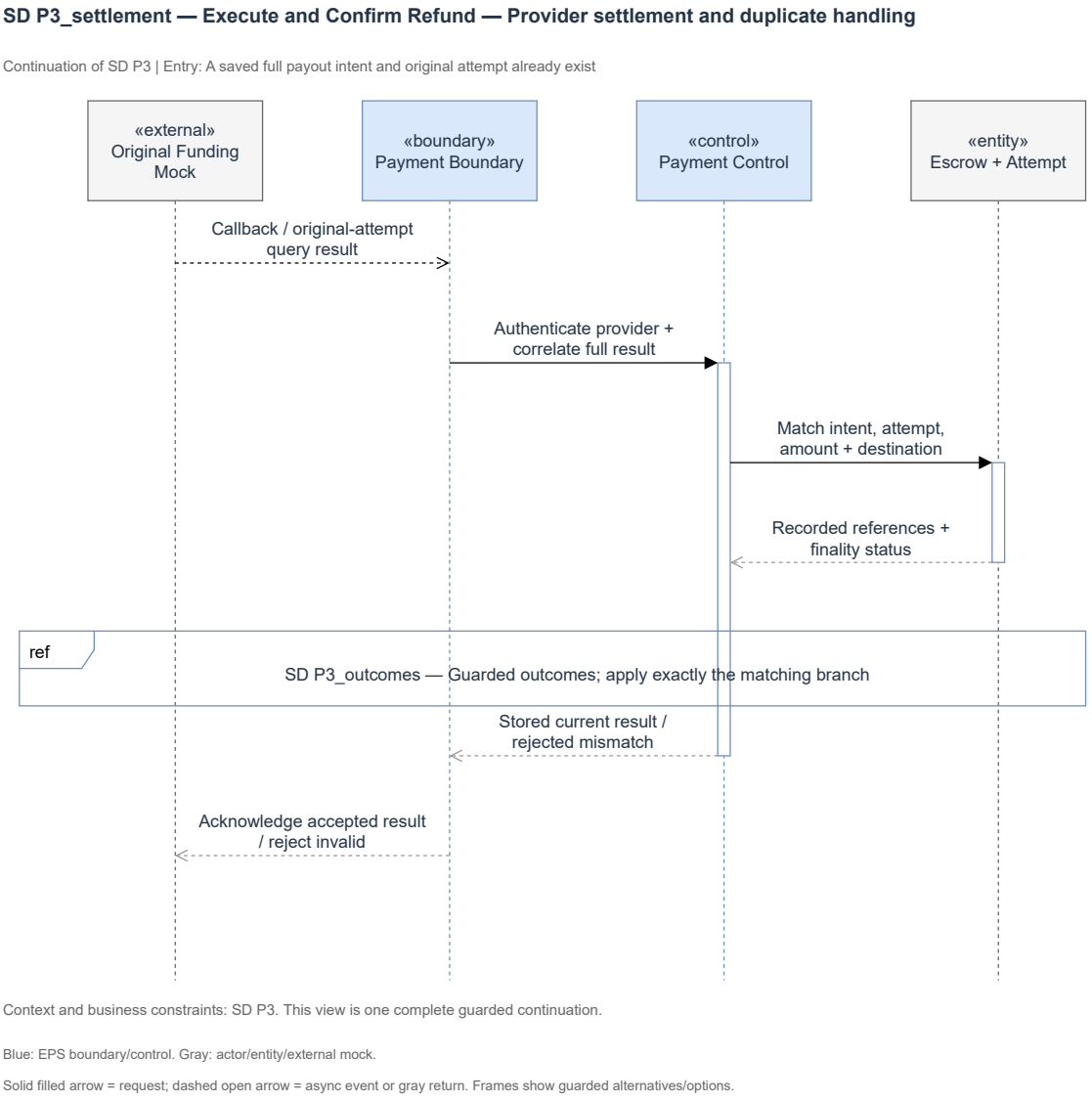


Figure 3.84: P3: Execute and Confirm Refund (continued: P3_settlement)

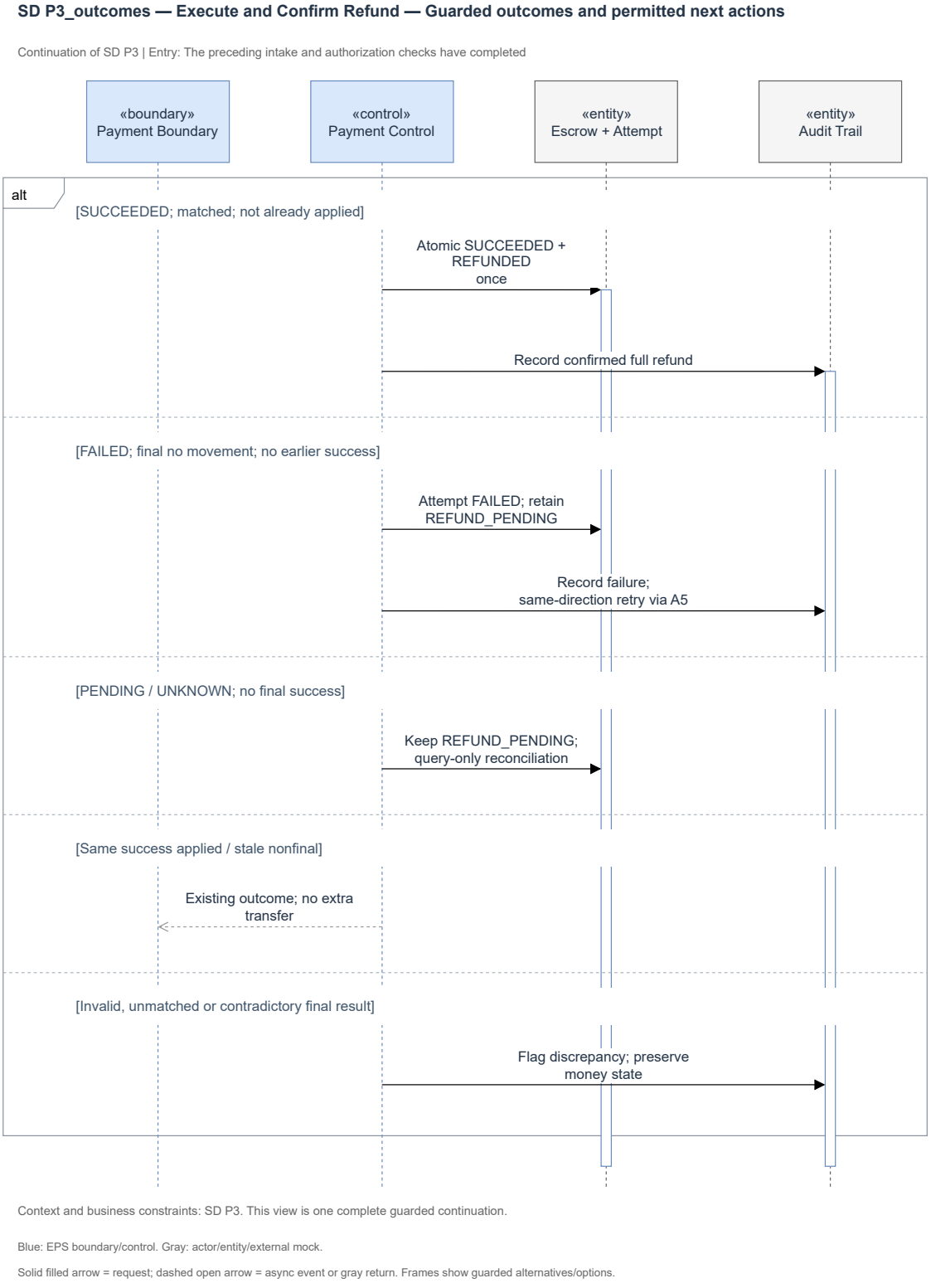


Figure 3.85: P3: Execute and Confirm Refund (continued: P3_outcomes)

3.5 Non-functional Requirements

The system must also meet the following non-functional requirements.

Performance Requirements:

- Target p95 response time is at most 2 seconds for internal create/query operations with 20 concurrent sessions and 1,000 simulated escrows, excluding user/provider waiting. This target has not been measured.
- Accepted operations must return a pending status and query reference when provider settlement is unresolved.

Operational Requirements:

- Preserve pending operations, policy snapshots and provider references after interruption so reconciliation can resume without duplicate transfers.
- Distinguish unknown outcomes from confirmed failures. Retry a payout only after confirmed failure with no money moved, preserving its direction and beneficiary.
- Keep audit records and dashboard counts consistent with recorded escrow states.

Security Requirements:

- Require MFA for human accounts and invalidate the session on logout. Authenticate Merchant clients separately.
- Check role, participant access and Merchant ownership on every protected operation.
- Use TLS and authenticate provider confirmations; reject mismatched amount, currency, destination or operation references.
- Keep the audit trail read-only and exclude passwords, MFA secrets and access tokens from logs.
- Process repeated or concurrent money decisions once, allowing only one final release or refund.

Cultural and Political Requirements:

- Use whole VND amounts and clear delivery, acceptance and evidence terms for both goods and services.
- Use synthetic data, label funds/providers as simulated and restrict personal and dispute evidence to authorised users.
- Legal claims in Planning remain unverified for real deployment; the MVP makes no claim of real-money custody or regulatory approval.